

Computational Companion to “Generalized Quasi-Symmetric Nets: A Constructive Approach to the Equimodular Elliptic Type of $n \times m$ Kokotsakis Polyhedra” — Helper

A. Nurmatov, M. Skopenkov, F. Rist, J. Klein, D. L. Michels
Tested on: Mathematica 14.0

Section 1

Results of Section 1 are used in the proof of Lemmas 1-2

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In[*]:= (* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[α, β, γ, δ, σ, a, b, c, d, r, s, f,
  M, assumptions, isTrueQ, statusChip, row, identities];
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α]; b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ]; d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain assumptions:the flat angles lie in (0,Pi)*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi, 0 < δ < Pi};
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
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statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];

(* =====*)
(*PART 2:THE SIX RATIO IDENTITIES*)
(* =====*)
identities = {"Identity 1", HoldForm[Sin[α] Sin[δ] / (Sin[β] Sin[γ])],
  HoldForm[r (r - M) / (M (r - 1))], isTrueQ},
{"Identity 2", HoldForm[Sin[σ - α] Sin[σ - δ] / (Sin[σ - β] Sin[σ - γ])],
  HoldForm[(r - M) / (r (r - 1))], isTrueQ},
{"Identity 3", HoldForm[Sin[γ] Sin[δ] / (Sin[α] Sin[β])],
  HoldForm[s (s - M) / (M (s - 1))], isTrueQ},
{"Identity 4", HoldForm[Sin[σ - γ] Sin[σ - δ] / (Sin[σ - α] Sin[σ - β])],
  HoldForm[(s - M) / (s (s - 1))], isTrueQ},
{"Identity 5", HoldForm[Sin[α] Sin[γ] / (Sin[β] Sin[δ])],
  HoldForm[f (f - M) / (M (f - 1))], isTrueQ},
{"Identity 6", HoldForm[Sin[σ - α] Sin[σ - γ] / (Sin[σ - β] Sin[σ - δ])],
  HoldForm[(f - M) / (f (f - 1))], isTrueQ}};

(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment → Center],
  Item[Style[TraditionalForm[rhs], 16], Alignment → Center],
  statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]};
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
    Style["Right Hand Side (RHS)", Bold, 14],
    Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame → All, FrameStyle → GrayLevel[0.75],
  Background → {None, {1 → RGBColor[0.99, 0.96, 0.88]}},
  Spacings → {2.2, 2.8},
  Alignment → {{Left, Center, Center, Center}, Center},
  ItemStyle → Directive[FontFamily → "Helvetica", 13]]

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Out[]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Identity 1	$\frac{\sin(\alpha) \sin(\delta)}{\sin(\beta) \sin(\gamma)}$	$\frac{r(r-M)}{M(r-1)}$	Verified
Identity 2	$\frac{\sin(\sigma-\alpha) \sin(\sigma-\delta)}{\sin(\sigma-\beta) \sin(\sigma-\gamma)}$	$\frac{r-M}{r(r-1)}$	Verified
Identity 3	$\frac{\sin(\gamma) \sin(\delta)}{\sin(\alpha) \sin(\beta)}$	$\frac{s(s-M)}{M(s-1)}$	Verified
Identity 4	$\frac{\sin(\sigma-\gamma) \sin(\sigma-\delta)}{\sin(\sigma-\alpha) \sin(\sigma-\beta)}$	$\frac{s-M}{s(s-1)}$	Verified
Identity 5	$\frac{\sin(\alpha) \sin(\gamma)}{\sin(\beta) \sin(\delta)}$	$\frac{f(f-M)}{M(f-1)}$	Verified
Identity 6	$\frac{\sin(\sigma-\alpha) \sin(\sigma-\gamma)}{\sin(\sigma-\beta) \sin(\sigma-\delta)}$	$\frac{f-M}{f(f-1)}$	Verified

Section 2

Results of Section 2 are used in the proof of Lemma 2

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In[ ]:= (* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[f, l, m, M, isTrueQ, statusChip, row, identities];
(*The identities below are identities of
  rational functions in Subscript[f,l],Subscript[f,m],M;
the subscripted symbols remain inert,so no domain assumptions are needed*)
isTrueQ[lhs_, rhs_] := TrueQ[Simplify[lhs == rhs]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background -> RGBColor[0.13, 0.55, 0.13], FrameStyle -> None,
  RoundingRadius -> 7, FrameMargins -> {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background -> RGBColor[0.82, 0.10, 0.10], FrameStyle -> None,
  RoundingRadius -> 7, FrameMargins -> {{12, 12}, {4, 4}}];
(* =====*)
(*PART 2:THE FOUR FACTORIZATION IDENTITIES*)
(* =====*)
identities = {"Factorization 1",

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HoldForm[Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1)) -
  Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))],
HoldForm[(Subscript[f, l] - Subscript[f, m])
  (Subscript[f, l] × Subscript[f, m] - (Subscript[f, l] + Subscript[f, m]) + M) /
  (M (Subscript[f, l] - 1) (Subscript[f, m] - 1))],
isTrueQ}, {"Factorization 2", HoldForm[
  (Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1))) *
  (Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))) - 1],
HoldForm[(Subscript[f, l] × Subscript[f, m] - M) (Subscript[f, l] ×
  Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
  (M^2 (Subscript[f, l] - 1) (Subscript[f, m] - 1))], isTrueQ},
{"Factorization 3", HoldForm[(Subscript[f, l] - M) /
  (Subscript[f, l] (Subscript[f, l] - 1)) -
  (Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))],
HoldForm[(Subscript[f, m] - Subscript[f, l]) (Subscript[f, l] ×
  Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
  (Subscript[f, l] × Subscript[f, m] (Subscript[f, l] - 1)
  (Subscript[f, m] - 1))], isTrueQ}, {"Factorization 4",
HoldForm[((Subscript[f, l] - M) / (Subscript[f, l] (Subscript[f, l] - 1))) *
  ((Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))) - 1],
HoldForm[(Subscript[f, l] × Subscript[f, m] - M)
  (Subscript[f, l] + Subscript[f, m] - Subscript[f, l] × Subscript[f, m] - M) /
  (Subscript[f, l] × Subscript[f, m] (Subscript[f, l] - 1)
  (Subscript[f, m] - 1))], isTrueQ}}];

(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment → Center],
   Item[Style[TraditionalForm[rhs], 16], Alignment → Center],
   statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]];
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
    Style["Right Hand Side (RHS)", Bold, 14],
    Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame → All, FrameStyle → GrayLevel[0.75],
  Background → {None, {1 → RGBColor[0.99, 0.96, 0.88]}},
  Spacings → {2.2, 2.8},
  Alignment → {{Left, Center, Center, Center}, Center},
  ItemStyle → Directive[FontFamily → "Helvetica", 13]]

```

Out[]:=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Factorization 1	$\frac{f_l(f_l-M)}{M(f_l-1)} - \frac{f_m(f_m-M)}{M(f_m-1)}$	$\frac{(f_l-f_m)(f_l f_m - (f_l+f_m)+M)}{M(f_l-1)(f_m-1)}$	Verified
Factorization 2	$\frac{(f_l(f_l-M))(f_m(f_m-M))}{(M(f_l-1))(M(f_m-1))} - 1$	$\frac{(f_l f_m - M)(f_l f_m - M(f_l+f_m-1))}{M^2(f_l-1)(f_m-1)}$	Verified
Factorization 3	$\frac{f_l-M}{f_l(f_l-1)} - \frac{f_m-M}{f_m(f_m-1)}$	$\frac{(f_m-f_l)(f_l f_m - M(f_l+f_m-1))}{f_l f_m(f_l-1)(f_m-1)}$	Verified
Factorization 4	$\frac{(f_l-M)(f_m-M)}{(f_l(f_l-1))(f_m(f_m-1))} - 1$	$\frac{(f_l f_m - M)(f_l+f_m-f_l f_m - M)}{f_l f_m(f_l-1)(f_m-1)}$	Verified

Section 3

Results of Section 3 are used in the proof of Lemma 4

```

In[ ]:= (* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[α, β, γ, δ, σ, ε, a, b, c, d, r, s, f, M, rad, sq,
  assumptions, isTrueQ, isTrueQeps, statusChip, row, identities];
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α]; b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ]; d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain
  assumptions:the flat angles and the sigma-shifted angles lie in (0,Pi);
these hold under the hypotheses of the lemma
  indicated in the subsection above (see paper)*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi,
  0 < δ < Pi, 0 < σ - α < Pi, 0 < σ - β < Pi, 0 < σ - γ < Pi, 0 < σ - δ < Pi};
rad = (f - M) (r - 1) (s - 1) / (f (1 - M));
sq = (Sin[σ] / Sin[σ - δ]) ^ 2;
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
(*Two-stage verification,
in the order of the table:STAGE 1 (the radicand row) identifies rad with the
  perfect square sq.STAGE 2 (the identity row) uses Stage 1:for t=
  Sin[sigma]/Sin[sigma-delta],the eps-split assumptions
  give t the sign eps (u/v checks),

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so Sqrt[rad]=Sqrt[t^2]=eps*t (abstract root law),
and eps*Sqrt[rad]*Sin[sigma-delta]=
  eps^2*Sin[sigma]=Sin[sigma] (cancellation).No Sqrt
  over trigonometric arguments reaches the kernel.*)
isTrueQeps[lhs_, rhs_] :=
  Module[{t, u, v}, TrueQ[FullSimplify[rad == sq, Assumptions → assumptions]] &&
    TrueQ[Assuming[u > 0 && v > 0, Simplify[u / v > 0]]] &&
    TrueQ[Assuming[u < 0 && v > 0, Simplify[u / v < 0]]] &&
    TrueQ[Assuming[t > 0, Simplify[Sqrt[t^2] == t]]] &&
    TrueQ[Assuming[t < 0, Simplify[Sqrt[t^2] == -t]]] &&
    TrueQ[Simplify[(Sin[σ] / Sin[σ - δ]) Sin[σ - δ] == Sin[σ]]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
(* =====*)
(*PART 2:THE RADICAND AND THE IDENTITY*)
(* =====*)
identities = {"Radicand equals a perfect square", HoldForm[
  (f - M) (r - 1) (s - 1) / (f (1 - M))], HoldForm[(Sin[σ] / Sin[σ - δ]) ^ 2], isTrueQ},
  {"Identity for sin(σ) (via the radicand row)", HoldForm[Sin[σ]],
  HoldForm[ε Sqrt[(f - M) (r - 1) (s - 1) / (f (1 - M))] Sin[σ - δ]], isTrueQeps}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment → Center],
  Item[Style[TraditionalForm[rhs], 16], Alignment → Center],
  statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]};
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
  Style["Right Hand Side (RHS)", Bold, 14],
  Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame → All, FrameStyle → GrayLevel[0.75],
  Background → {None, {1 → RGBColor[0.99, 0.96, 0.88]}},
  Spacings → {2.2, 2.8},
  Alignment → {{Left, Center, Center, Center}, Center},
  ItemStyle → Directive[FontFamily → "Helvetica", 13]]

```

Out[]:=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Radicaud equals a perfect square	$\frac{(f-M)(r-1)(s-1)}{f(1-M)}$	$\left(\frac{\sin(\sigma)}{\sin(\sigma-\delta)}\right)^2$	Verified
Identity for $\sin(\sigma)$ (via the radicaud row)	$\sin(\sigma)$	$\varepsilon \sqrt{\frac{(f-M)(r-1)(s-1)}{f(1-M)}} \sin(\sigma - \delta)$	Verified

Section 4

Results of Section 4 are used in the proof of Theorem 1

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In[ ]:= (* =====*)
(*PART 1:INITIALIZATION& STANDING HYPOTHESES*)
(* =====*)
ClearAll[f, i, j, l, m, c, s, si, sl, fi, fj, fl, fm,
  M, chi, x, display, propEq, standingOf, assumptions, isTrueQ,
  statusChip, caseCheck, results, starRef, fj3, fm3, fj4, fm4];
Format[fi, TraditionalForm] = Subscript[f, i];
Format[fl, TraditionalForm] = Subscript[f, l];
Format[chi, TraditionalForm] = x;
(*Standing hypotheses:M and all four f's are positive,
distinct from 1 and from each other's modulus pairing f≠M,
and each pair-product (f-1)(f-M) is positive;
these hold under the hypotheses indicated
  in the paper.The verification below is branch-wise:
  sign data entering through chi and the regime of M is enumerated explicitly,
so every square root reaching the kernel has a sign-determined
  radicaud.Parametrization:the first term of the identity carries zeta1 alone;
in every case the verifying choice is zeta1=chi (the value of chi
  on the branch).All display-only rows are wrapped in HoldForm,
so they cannot evaluate through symbols defined in other sections.*)
standingOf[fjE_, fmE_] := M > 0 && M ≠ 1 && And@@ (# > 0 & /@ {fi, fjE, fl, fmE}) &&
  And@@ (# ≠ 1 & /@ {fi, fjE, fl, fmE}) && And@@ (# ≠ M & /@ {fi, fjE, fl, fmE}) &&
  And@@ ((# - 1) (# - M) > 0 & /@ {fi, fjE, fl, fmE});
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
display[zeta1_, zeta2_, chiv_, fjE_, fmE_] :=
  zeta1 M Sqrt[(fi - 1) (fjE - 1) (fl - 1) (fmE - 1)] + zeta2 (1 - M) Sqrt[fi fjE fl fmE] -

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zeta2 chiv Sqrt[(fi - M) (fjE - M) (fl - M) (fmE - M)];
propEq[zeta1_, zeta2_, chiv_, fjE_, fmE_] := Style[TraditionalForm[
  HoldForm[#] &@ (display[zeta1, zeta2, chiv, fjE, fmE] == M (1 - M))], 16];
starRef = Style["(★)", Bold, 16, RGBColor[0.72, 0.45, 0.02]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
(*Two-stage verification per
  case:STAGE 1 (the collapse row) identifies the instantiated left-hand side
    of (star) with its square-root-free form under the branch assumptions;
  STAGE 2 (the closure row) verifies that the collapsed form equals
    M(1-M).The chip is Verified only if both stages hold.*)
caseCheck[title_, col_, condRow_, zeta1_, zeta2_, branchRow_, chiv_,
  fjE_, fmE_, extraAssum_, extraRow_, collapsedHeld_] := Module[
  {okCollapse, okZero, ok}, assumptions = standingOf[fjE, fmE] && extraAssum;
  okCollapse =
    isTrueQ[display[zeta1, zeta2, chiv, fjE, fmE], ReleaseHold[collapsedHeld]];
  okZero = isTrueQ[ReleaseHold[collapsedHeld], M (1 - M)];
  ok = okCollapse && okZero;
  Print[Panel[Column[
    {Row[{Style[title, Bold, 15, col], Spacer[16], Style[condRow, 13, col]}],
      Row[{Style["testing ", 13], starRef, Style[Row[{" with ",
        Subscript["g", 1], " = ", TraditionalForm[zeta1], " ",
        Subscript["g", 2], " = ", TraditionalForm[zeta2]}], Bold, 13],
        Spacer[24], branchRow}], propEq[zeta1, zeta2, chiv, fjE, fmE], Row[
        {Style["expanding the square roots: ", GrayLevel[0.35], 12], Style[
          TraditionalForm[HoldForm[#] &@ (collapsedHeld == M (1 - M))], 13}}],
        Row[{Style["→ ", Bold, 14], Style[TraditionalForm[HoldForm[0 == 0]],
          14], Spacer[26], statusChip[ok]}],
        Style[Row[{"additionally used: ", extraRow}], GrayLevel[0.35], 12}],
        Spacings → 1], Background → Lighter[col, 0.93], FrameMargins → 12]];
  ok];
(* =====*)
(*PART 2:STANDING HYPOTHESES AND THE IDENTITY (star)*)
(* =====*)
Print[Panel[Column[{Style["Standing Hypotheses:", Bold, 14],
  Style[Row[{TraditionalForm[HoldForm[Subscript[f, i]]], " ",
    TraditionalForm[HoldForm[M]], " > 0, ", TraditionalForm[HoldForm[
      Subscript[f, i]]], " ", TraditionalForm[HoldForm[M]], " ≠ 1, ",
      TraditionalForm[HoldForm[Subscript[f, i] ≠ M]], " ", TraditionalForm[
        HoldForm[(Subscript[f, i] - 1) (Subscript[f, i] - M) > 0]],
        " for each ", TraditionalForm[HoldForm[i]], " = 1, ..., 4, ",
        TraditionalForm[HoldForm[χ ∈ {±1}]]}], 13], "", Style[
      Row[{"The identity we want to verify for all cases below (for some ",

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Subscript["g", 1], " ", " ", Subscript["g", 2], " ∈ {±1}:"}], Bold, 14],
Row[{Style[TraditionalForm[HoldForm[Subscript[g, 1] M
Sqrt[Product[Subscript[f, i] - 1, {i, 1, 4}]] + Subscript[g, 2]
(1 - M) Sqrt[Product[Subscript[f, i], {i, 1, 4}]] - Subscript[g, 2]
χ Sqrt[Product[Subscript[f, i] - M, {i, 1, 4}]] = M (1 - M)]], 16],
Spacer[28], Style["(*)", Bold, 16, RGBColor[0.72, 0.45, 0.02]]}],
Spacings → 1.1], Background → RGBColor[0.99, 0.96, 0.88],
FrameMargins → 14]]];
(* =====*)
(*PART 3:CASE-BY-CASE VERIFICATION*)
(* =====*)
results = {};
(*Condition (i):fj=fi,fm=fl;
chi (fi-1)(fl-1)>0;
zeta1=chi=si*sl per branch,
zeta2=1.The four branches enumerate the signs of fi-1 and fl-1;
by the standing pair-products,fi-M and fi-1 have the same sign,
and similarly at l,so chi=si*sl.*)
Do[AppendTo[results, caseCheck["Condition (i):", RGBColor[0.10, 0.30, 0.70],
Row[{TraditionalForm[HoldForm[Subscript[f, j] = Subscript[f, i]]],
", ", TraditionalForm[HoldForm[Subscript[f, m] = Subscript[f, l]]],
", ", TraditionalForm[
HoldForm[χ (Subscript[f, i] - 1) (Subscript[f, l] - 1) > 0]]], si sl,
1, Style[Row[{"branch: sign(", TraditionalForm[HoldForm[fi - 1]],
") = ", si, ", sign(", TraditionalForm[HoldForm[fl - 1]], ") = ", sl,
" → χ = ", si sl, ", so \!\(\*SubscriptBox[\(g\), \{1\}]\) = χ"}],
13, GrayLevel[0.3]], si sl, fi, fl, si (fi - 1) > 0 &&
si (fi - M) > 0 && sl (fl - 1) > 0 && sl (fl - M) > 0,
Row[{TraditionalForm[HoldForm[#] &@ (si (fi - 1) > 0)], " ",
TraditionalForm[HoldForm[#] &@ (si (fi - M) > 0)],
", ", TraditionalForm[HoldForm[#] &@ (sl (fl - 1) > 0)],
", ", TraditionalForm[HoldForm[#] &@ (sl (fl - M) > 0)],
" (same-sign facts by the standing hypotheses)"}],
HoldForm[M (fi - 1) (fl - 1) + (1 - M) fi fl - (fi - M) (fl - M)]],
{si, {1, -1}}, {sl, {1, -1}}]];
(*Condition (ii):fi*fj=M,fl*fm=M,that is,fj=M/fi,fm=M/fl;
zeta1=chi(symbolic),zeta2=1;
no sign condition is consumed and the two chi-
dressed terms cancel syntactically.*)
AppendTo[results, caseCheck["Condition (ii):", RGBColor[0.55, 0.25, 0.70],
Row[{TraditionalForm[HoldForm[Subscript[f, i] × Subscript[f, j] = M]],
", ", TraditionalForm[
HoldForm[Subscript[f, l] × Subscript[f, m] = M]]], chi, 1, Style[
Row[{"single symbolic branch, \!\(\*SubscriptBox[\(g\), \{1\}]\) = χ"}],
13, GrayLevel[0.3]], chi, M / fi, M / fl, True,
"none (χ remains symbolic; no sign condition is consumed)",
HoldForm[chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)] +

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(1 - M) M - chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)]]];
(*Condition (iii): fi+fj-fi*fj=M, fl+fm-fl*fm=M,
that is, fj=(M-fi)/(1-fi), fm=(M-fl)/(1-fl);
chi (1-M)>0;
zeta1=chi=c per branch, zeta2=1. Two branches: chi=+1 with M<1,
and chi=-1 with M>1.*)
fj3 = (M - fi) / (1 - fi); fm3 = (M - fl) / (1 - fl);
Do[AppendTo[results, caseCheck["Condition (iii):",
  RGBColor[0.00, 0.50, 0.50], Row[{TraditionalForm[HoldForm[
    Subscript[f, i] + Subscript[f, j] - Subscript[f, i] × Subscript[f, j] = M]],
    ", ", TraditionalForm[HoldForm[
    Subscript[f, l] + Subscript[f, m] - Subscript[f, l] × Subscript[f, m] = M]],
    ", ", TraditionalForm[HoldForm[χ (1 - M) > 0]]], c, 1,
  Style[Row[{"branch: χ = ", c, ", ", " ", If[c == 1, "M < 1", "M > 1"],
    ", so \!\(\*SubscriptBox[\(\xi\), \(\eta\)]\) = χ"}],
    13, GrayLevel[0.3]], c, fj3, fm3, If[c == 1, M < 1, M > 1],
  Row[{TraditionalForm[HoldForm[#] &@If[c == 1, M < 1, M > 1]],
    " (this branch of ", TraditionalForm[HoldForm[χ (1 - M) > 0]], ")"}],
  HoldForm[M (1 - M) + (1 - M) Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))] -
    (1 - M) Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))]]], {c, {1, -1}}];
(*Condition (iv): M(fi+fj-1)=fi*fj, M(fl+fm-1)=fl*fm,
that is, fj=M(fi-1)/(fi-M), fm=M(fl-1)/(fl-M);
zeta1=chi=c per branch, and zeta2=-1 if chi(1-M)>0,
zeta2=+1 if chi(1-M)<0, encoded as zeta2=
-c*s with s=sign(1-M). Four branches over chi and the regime of M;
the collapse row uses per-branch literal templates so
the cancelling pair of square-root terms is shown.*)
fj4 = M (fi - 1) / (fi - M); fm4 = M (fl - 1) / (fl - M);
Do[AppendTo[results, caseCheck["Condition (iv):", RGBColor[0.75, 0.35, 0.05],
  Row[{TraditionalForm[HoldForm[M (Subscript[f, i] + Subscript[f, j] - 1) =
    Subscript[f, i] × Subscript[f, j]]], ", ", " ",
    TraditionalForm[HoldForm[M (Subscript[f, l] + Subscript[f, m] - 1) =
    Subscript[f, l] × Subscript[f, m]]]], c, -c s,
  Style[Row[{"branch: χ = ", c, ", ", " ", If[s == 1, "M < 1", "M > 1"],
    " → ", If[c s == 1, "χ(1-M) > 0", "χ(1-M) < 0"],
    ", \!\(\*SubscriptBox[\(\xi\), \(\eta\)]\) = χ"}],
    13, GrayLevel[0.3]], c, fj4, fm4, If[s == 1, M < 1, M > 1],
  Row[{TraditionalForm[HoldForm[#] &@If[s == 1, M < 1, M > 1]],
    " (this branch of the rule: ", Subscript["ξ", 2], " = -1 if ",
    TraditionalForm[HoldForm[χ (1 - M) > 0]], ", ", Subscript["ξ", 2],
    " = +1 if ", TraditionalForm[HoldForm[χ (1 - M) < 0]], ")"}], If[c s == 1,
  HoldForm[(1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] -
    (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] + M (1 - M)],
  HoldForm[-(1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +
    (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +
    M (1 - M)]]], {c, {1, -1}}, {s, {1, -1}}];
(* =====*)

```

(*PART 4:SUMMARY*)

(* =====*)

```
Print[
  Framed[Style[If[And@@results, "ALL CASES VERIFIED", "SOME CASES FAILED"],
    Bold, White, 16], Background →
    If[And@@results, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}];
```

Standing Hypotheses:

$f_i, M > 0, f_i, M \neq 1, f_i \neq M, (f_i - 1)(f_i - M) > 0$ for each $i = 1, \dots, 4, \chi \in \{\pm 1\}$

The identity we want to verify for all cases below (for some $\zeta_1, \zeta_2 \in \{\pm 1\}$):

$$\zeta_1 M \sqrt{\prod_{i=1}^4 (f_i - 1)} + \zeta_2 (1 - M) \sqrt{\prod_{i=1}^4 f_i} - \zeta_2 \chi \sqrt{\prod_{i=1}^4 (f_i - M)} = M(1 - M) \quad (\star)$$

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

testing $(\star)$ with $\zeta_1 = 1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = 1, \text{sign}(f_l - 1) = 1 \rightarrow \chi = 1, \text{ so } \zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) - \sqrt{(f_i - M)^2 (f_l - M)^2} + \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M)M$

$\rightarrow 0 = 0$

Verified

additionally used:

$-1 + f_i > 0, f_i - M > 0, -1 + f_l > 0, f_l - M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

testing $(\star)$ with $\zeta_1 = -1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = 1, \text{sign}(f_l - 1) = -1 \rightarrow \chi = -1, \text{ so } \zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) + \sqrt{(f_i - M)^2 (f_l - M)^2} - \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M)M$

$\rightarrow 0 = 0$

Verified

additionally used:

$-1 + f_i > 0, f_i - M > 0, 1 - f_l > 0, -f_l + M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi(f_i - 1)(f_l - 1) > 0$

testing (★) with $\zeta_1 = -1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = -1, \text{sign}(f_l - 1) = 1 \rightarrow \chi = -1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) + \sqrt{(f_i - M)^2 (f_l - M)^2} - \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M$

$$\rightarrow 0 = 0$$

Verified

additionally used:

$1 - f_i > 0, -f_i + M > 0, -1 + f_i > 0, f_l - M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi(f_i - 1)(f_l - 1) > 0$

testing (★) with $\zeta_1 = 1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = -1, \text{sign}(f_l - 1) = -1 \rightarrow \chi = 1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) - \sqrt{(f_i - M)^2 (f_l - M)^2} + \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M$

$$\rightarrow 0 = 0$$

Verified

additionally used:

$1 - f_i > 0, -f_i + M > 0, 1 - f_l > 0, -f_l + M > 0$ (same-sign facts by the standing hypotheses)

Condition (ii): $f_i f_j = M, f_l f_m = M$

testing (★) with $\zeta_1 = \chi, \zeta_2 = 1$ single symbolic branch, $\zeta_1 = \chi$

$$(1 - M) \sqrt{M^2} + \chi M \sqrt{(-1 + f_i)(-1 + f_l) \left(-1 + \frac{M}{f_i}\right) \left(-1 + \frac{M}{f_l}\right)} -$$

$$\chi \sqrt{(f_i - M)(f_l - M) \left(-M + \frac{M}{f_i}\right) \left(-M + \frac{M}{f_l}\right)} = (1 - M) M$$

expanding the square roots: $\chi M \sqrt{(f_i - 1) \left(\frac{M}{f_i} - 1\right) (f_l - 1) \left(\frac{M}{f_l} - 1\right)} +$

$$(1 - M) M - \chi M \sqrt{(f_i - 1) \left(\frac{M}{f_i} - 1\right) (f_l - 1) \left(\frac{M}{f_l} - 1\right)} = (1 - M) M$$

$$\rightarrow 0 = 0$$

Verified

additionally used: none (χ remains symbolic; no sign condition is consumed)

Condition (iii): $f_i + f_j - f_i f_j = M$, $f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M < 1$, so $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{f_i f_l (-f_i+M)(-f_l+M)}{(1-f_i)(1-f_l)}} + M \sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{-f_i+M}{1-f_i}\right)\left(-1+\frac{-f_l+M}{1-f_l}\right)} - \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{-f_i+M}{1-f_i}\right)\left(-M+\frac{-f_l+M}{1-f_l}\right)} = (1-M)M$$

expanding the square roots:

$$M(1-M) + (1-M) \sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} - (1-M) \sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} = (1-M)M$$

→ 0 = 0

Verified

additionally used: $M < 1$ (this branch of $\chi(1-M) > 0$)

Condition (iii): $f_i + f_j - f_i f_j = M$, $f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$

testing (★) with $\zeta_1 = -1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M > 1$, so $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{f_i f_l (-f_i+M)(-f_l+M)}{(1-f_i)(1-f_l)}} - M \sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{-f_i+M}{1-f_i}\right)\left(-1+\frac{-f_l+M}{1-f_l}\right)} + \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{-f_i+M}{1-f_i}\right)\left(-M+\frac{-f_l+M}{1-f_l}\right)} = (1-M)M$$

expanding the square roots:

$$M(1-M) + (1-M) \sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} - (1-M) \sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} = (1-M)M$$

→ 0 = 0

Verified

additionally used: $M > 1$ (this branch of $\chi(1-M) > 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = -1$ branch: $\chi = 1$, $M < 1 \rightarrow \chi(1-M) > 0$, $\zeta_1 = \chi$

$$-\left((1-M) \sqrt{\frac{(-1+f_i)f_i(-1+f_l)f_l M^2}{(f_i-M)(f_l-M)}}\right) + M \sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{(-1+f_i)M}{f_i-M}\right)\left(-1+\frac{(-1+f_l)M}{f_l-M}\right)} + \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{(-1+f_i)M}{f_i-M}\right)\left(-M+\frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$

expanding the square roots:

$$(1-M) \sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} - (1-M) \sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$

→ 0 = 0

Verified

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M > 1 \rightarrow \chi(1-M) < 0$, $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} +$$

$$M \sqrt{(-1+f_i)(-1+f_l) \left(-1 + \frac{(-1+f_i)M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l)M}{f_l-M}\right)} -$$

$$\sqrt{(f_i-M)(f_l-M) \left(-M + \frac{(-1+f_i)M}{f_i-M}\right) \left(-M + \frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$

expanding the square roots:

$$-(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$

$\rightarrow 0 = 0$

Verified

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = -1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M < 1 \rightarrow \chi(1-M) < 0$, $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} -$$

$$M \sqrt{(-1+f_i)(-1+f_l) \left(-1 + \frac{(-1+f_i)M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l)M}{f_l-M}\right)} +$$

$$\sqrt{(f_i-M)(f_l-M) \left(-M + \frac{(-1+f_i)M}{f_i-M}\right) \left(-M + \frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$

expanding the square roots:

$$-(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$

$\rightarrow 0 = 0$

Verified

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = -1$, $\zeta_2 = -1$ branch: $\chi = -1$, $M > 1 \rightarrow \chi(1-M) > 0$, $\zeta_1 = \chi$

$$-\left((1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}}\right) -$$

$$M \sqrt{(-1+f_i)(-1+f_l) \left(-1 + \frac{(-1+f_i)M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l)M}{f_l-M}\right)} -$$

$$\sqrt{(f_i-M)(f_l-M) \left(-M + \frac{(-1+f_i)M}{f_i-M}\right) \left(-M + \frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$

expanding the square roots:

$$(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} - (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$

$\rightarrow 0 = 0$

Verified

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

ALL CASES VERIFIED

Section 5

Results of Section 5 are used in the proof of Remark 5

```
In[*]:= (* ===== *)
(*PART 1:INITIALIZATION*)
(* ===== *)
ClearAll[gi, gj, k, kp, ei, ej, xi, h1of, h2of, h3of, baseAssum,
wConfigs, ePairs, dirForward, dirBackward, branchOK, smallChip,
statusChip, panelH, resultsAll, condOf, vars, hsym, stmtOf];
Format[gi, TraditionalForm] = Subscript[g, i];
Format[gj, TraditionalForm] = Subscript[g, j];
Format[kp, TraditionalForm] = k';
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
vars = {gi, gj, k, kp};
(*The three h's in the case k^2≠(gi^2-1)(gj^2-1),
with xi a numeric sign per branch.*)
```

```

h1of[e1_, e2_, x_] :=
  (e1 gj Sqrt[(gi^2 - 1) (gj^2 - kp^2) x] + e2 gi Sqrt[(gj^2 - 1) (gi^2 - kp^2) x]) /
  (k^2 - (gi^2 - 1) (gj^2 - 1));
h2of[e1_, e2_, x_] :=
  (Sqrt[(gi^2 - kp^2) (gj^2 - kp^2) x] + e1 e2 gi gj Sqrt[(gi^2 - 1) (gj^2 - 1) x]) /
  (k^2 - (gi^2 - 1) (gj^2 - 1));
h3of[e1_, e2_, x_] :=
  (k^2 gi gj + e1 e2 x Sqrt[(gi^2 - 1) (gi^2 - kp^2) (gj^2 - 1) (gj^2 - kp^2)]) /
  (k^2 - (gi^2 - 1) (gj^2 - 1));
(*Standing assumptions:k,k' with k'^2=1-k^2;
the case guard;
window assumptions added per branch.*)
baseAssum = 0 < k < 1 && kp > 0 && kp^2 == 1 - k^2 && k^2 ≠ (gi^2 - 1) (gj^2 - 1);
wConfigs =
  {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\), \!\(\(*SubscriptBox[\(g\), \(\j\)]\)\)
    ∈ (1, ∞)", gi > 1 && gj > 1, 1},
  {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\) ∈ (1, ∞), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) ∈ (0, k')", gi > 1 && 0 < gj < kp, -1},
  {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\) ∈ (0, k'), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) ∈ (1, ∞)", 0 < gi < kp && gj > 1, -1},
  {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) ∈ (0, k')", 0 < gi < kp && 0 < gj < kp, 1}};
ePairs = {{1, 1}, {1, -1}, {-1, 1}, {-1, -1}};
(*Two-directional branch check:no zero off the condition locus,
and no nonzero value on it.A condition False
means the claim on this branch is pure nonvanishing.*)
dirForward[h_, cond_, assum_] :=
  Reduce[assum && h == 0 && ! cond, vars, Reals] === False;
dirBackward[h_, cond_, assum_] :=
  If[cond === False, True, Reduce[assum && cond && h ≠ 0, vars, Reals] === False];
branchOK[h_, cond_, assum_] :=
  dirForward[h, cond, assum] && dirBackward[h, cond, assum];
(*Per-branch condition of each claim;
False encodes "never vanishes".*)
condOf[1, e1_, e2_] := If[e1 == -e2, gi == gj, False];
condOf[2, e1_, e2_] := If[e1 == -e2, gi gj == kp, False];
condOf[3, e1_, e2_] :=
  If[e1 == e2, k^2 (gi^2 + gj^2) == k^2 - (gi^2 - 1) (gj^2 - 1), False];
hsym[n_] := Subscript["h", "ij" <> ToString[n]];
stmtOf[n_, e1_, e2_] := Module[{c = condOf[n, e1, e2]}, If[c === False,
  Row[{hsym[n], " ≠ 0"}], Row[{hsym[n], " = 0 ⇔ ", TraditionalForm[c]}]]];
(* =====*)
(*PART 2:THE QUANTITIES AND THE CLAIMS TO TEST*)
(* =====*)
Print[Panel[Column[{Style[
  "The quantities under test, in the case \!\(\(*SuperscriptBox[\(k\),
    \(\j\)]\)\) ≠ \!\(\(*SubsuperscriptBox[\(g\), \(\i\), \(\j\)]\)\)

```

```

- 1) (\!\(\*SubsuperscriptBox[\(g\), \(\j\), \(\j\)] - 1):",
Bold, 14], Style[TraditionalForm[HoldForm[Subscript[h, ij1] ==
(Subscript[e, i] × Subscript[g, j] Sqrt[(Subscript[g, i]^2 - 1)
(Subscript[g, j]^2 - k'^2) Subscript[\(\xi\), ij]] +
Subscript[e, j] × Subscript[g, i] Sqrt[(Subscript[g, j]^2 - 1)
(Subscript[g, i]^2 - k'^2) Subscript[\(\xi\), ij]]) /
(k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))]], 14],
Style[TraditionalForm[HoldForm[Subscript[h, ij2] ==
(Sqrt[(Subscript[g, i]^2 - k'^2) (Subscript[g, j]^2 - k'^2)
Subscript[\(\xi\), ij]] + Subscript[e, i] ×
Subscript[e, j] × Subscript[g, i] × Subscript[g, j] Sqrt[
(Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1) Subscript[\(\xi\), ij]]) /
(k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))]], 14],
Style[TraditionalForm[HoldForm[Subscript[h, ij3] ==
(k^2 Subscript[g, i] × Subscript[g, j] +
Subscript[e, i] × Subscript[e, j] × Subscript[\(\xi\), ij]
Sqrt[(Subscript[g, i]^2 - 1) (Subscript[g, i]^2 - k'^2)
(Subscript[g, j]^2 - 1) (Subscript[g, j]^2 - k'^2)]] /
(k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))]], 14],
Style["and, in the case \!\(\*SuperscriptBox[\(k\), \(\j\)]\)",
= (\!\(\*SubsuperscriptBox[\(g\), \(\i\), \(\j\)] -
1) (\!\(\*SubsuperscriptBox[\(g\), \(\j\), \(\j\)] -
1) with \!\(\*SubscriptBox[\(e\), \(\i\)]\) =
-\!\(\*SubscriptBox[\(e\), \(\j\)]\):", Bold, 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] ==
Subscript[e, i] (Subscript[g, i]^2 - Subscript[g, j]^2) /
(2 k Subscript[g, i] × Subscript[g, j])]],
" ", TraditionalForm[HoldForm[Subscript[h, ij2] ==
(Subscript[g, i]^2 × Subscript[g, j]^2 - k'^2) /
(2 k Subscript[g, i] × Subscript[g, j])]], " ", TraditionalForm[
HoldForm[Subscript[h, ij3] == (Subscript[g, i]^2 + Subscript[g, j]^2) /
(2 Subscript[g, i] × Subscript[g, j])]]], 14],
"", Style["Standing Hypotheses:", Bold, 14], Style[
Row[{"k ∈ (0, 1), ", TraditionalForm[HoldForm[k'^2 == 1 - k^2]],
", ", TraditionalForm[HoldForm[Subscript[g, i]]],
", ", TraditionalForm[HoldForm[Subscript[g, j]]],
" ∈ (0, k') ∪ (1, ∞), ",
TraditionalForm[HoldForm[Subscript[e, i]]],
", ", TraditionalForm[HoldForm[Subscript[e, j]]],
" ∈ {±1}, ", TraditionalForm[HoldForm[Subscript[\(\xi\), ij] ==
Sign[(Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1)]]]], 13], "",
Style["The claims we test (over all window configurations of
\!\(\*SubscriptBox[\(g\), \(\i\)]\), \!\(\*SubscriptBox[\(g\),
\(\j\)]\) and all \!\(\*SubscriptBox[\(e\), \(\i\)]\),
\!\(\*SubscriptBox[\(e\), \(\j\)]\), in both directions):",
Bold, 14], Style[TraditionalForm[HoldForm[Subscript[h, ij1] == 0 ⇔
Subscript[g, i] = Subscript[g, j] &&

```

```

Subscript[e, i] == -Subscript[e, j]], 15],
Style[TraditionalForm[HoldForm[Subscript[h, ij2] == 0 ⇔
Subscript[g, i] × Subscript[g, j] == k' &&
Subscript[e, i] == -Subscript[e, j]], 15],
Style[TraditionalForm[HoldForm[Subscript[h, ij3] == 0 ⇔
k^2 (Subscript[g, i]^2 + Subscript[g, j]^2) ==
k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1) &&
Subscript[e, i] == Subscript[e, j]], 15]],
Spacings → 1.1], Background → RGBColor[0.99,
0.96,
0.88], FrameMargins → 14]]];
(* =====*)
(*PART 3:CASE k^2≠(gi^2-1)(gj^2-1)*)
(* =====*)
resultsAll = {};
panelH[n_, hfun_, col_, title_] := Module[{grid, ok, headRow, stmtRow},
grid = Table[branchOK[hfun[ep[[1]], ep[[2]], w[[3]], condOf[n, ep[[1]], ep[[2]],
baseAssum&&w[[2]], {w, wConfigs}, {ep, ePairs}]],
ok = And@@ Flatten[grid];
AppendTo[resultsAll, ok];
headRow = Prepend[
Table[Style[Row[{"(", ep[[1]], ", ", ep[[2]], ")"}], 12, Bold], {ep, ePairs}],
Style["(!\\(\\*SubscriptBox[(e), (i)]\\), \\!\\(\\*SubscriptBox[(e),
(j)]\\)) →", 12, GrayLevel[0.3]]];
stmtRow = Prepend[Table[Style[stmtOf[n, ep[[1]], ep[[2]], 11, GrayLevel[0.25]],
{ep, ePairs}], Style["tested →", 11, GrayLevel[0.3]]];
Print[Panel[
Column[{Row[{Style[title, Bold, 15, col], Spacer[20], statusChip[ok]}],
Grid[Join[{headRow}, {stmtRow}, Table[Prepend[smallChip/@grid[[iw]],
Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]]], {iw, 4}]],
Alignment → Left, Spacings → {2, 0.7}]], Spacings → 1],
Background → Lighter[col, 0.93], FrameMargins → 12]]];
panelH[1, h1of, RGBColor[0.10, 0.30, 0.70],
"First claim (case \\(\\*SuperscriptBox[(k),
(2)]\\) ≠ (\\(\\*SubsuperscriptBox[(g), (i),
(2)]\\)-1) (\\(\\*SubsuperscriptBox[(g), (j), (2)]\\)-1)):"];
panelH[2, h2of, RGBColor[0.55, 0.25, 0.70],
"Second claim (case \\(\\*SuperscriptBox[(k),
(2)]\\) ≠ (\\(\\*SubsuperscriptBox[(g), (i),
(2)]\\)-1) (\\(\\*SubsuperscriptBox[(g), (j), (2)]\\)-1)):"];
panelH[3, h3of, RGBColor[0.00, 0.50, 0.50],
"Third claim (case \\(\\*SuperscriptBox[(k),
(2)]\\) ≠ (\\(\\*SubsuperscriptBox[(g), (i),
(2)]\\)-1) (\\(\\*SubsuperscriptBox[(g), (j), (2)]\\)-1)):"];
(* =====*)
(*PART 4:CASE k^2=(gi^2-1)(gj^2-1) WITH ei=-ej*)
(* =====*)

```

```

Module[{assum7, ok1, ok2, ok3, ok4, ok},
  assum7 = 0 < k < 1 && kp > 0 && kp^2 == 1 - k^2 && k^2 == (gi^2 - 1) (gj^2 - 1) &&
    ((gi > 1 && gj > 1) || (0 < gi < kp && 0 < gj < kp));
  ok1 = And @@ Table[
    branchOK[e1 (gi^2 - gj^2) / (2 k gi gj), gi == gj, assum7], {e1, {1, -1}}];
  ok2 = branchOK[(gi^2 gj^2 - kp^2) / (2 k gi gj), gi gj == kp, assum7];
  ok3 = Reduce[assum7 && (gi^2 + gj^2) / (2 gi gj) == 0, vars, Reals] === False;
  ok4 = Reduce[assum7 && k^2 (gi^2 + gj^2) == k^2 - (gi^2 - 1) (gj^2 - 1),
    vars, Reals] === False;
  ok = ok1 && ok2 && ok3 && ok4;
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["Case \!\(\(*SuperscriptBox[\(k\), \((2\))]\) =
      (\!\(\(*SubsuperscriptBox[\(g\), \((i\),
      \((2\))]\)-1) (\!\(\(*SubsuperscriptBox[\(g\), \((j\),
      \((2\))]\)-1), with \!\(\(*SubscriptBox[\(e\), \((i\))]\)
      = -\!\(\(*SubscriptBox[\(e\), \((j\))]\):", Bold, 15,
      RGBColor[0.75, 0.35, 0.05]], Spacer[20], statusChip[ok]]},
    Column[{Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] ==
      Subscript[e, i] (Subscript[g, i]^2 - Subscript[g, j]^2) /
      (2 k Subscript[g, i] × Subscript[g, j])]],
      ": ", hsym[1], " = 0 ⇔ ", TraditionalForm[HoldForm[
      Subscript[g, i] == Subscript[g, j]]], " ", smallChip[ok1]]}, 13],
    Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij2] ==
      (Subscript[g, i]^2 × Subscript[g, j]^2 - k'^2) / (2 k Subscript[
      g, i] × Subscript[g, j])]], ": ", hsym[2], " = 0 ⇔ ",
      TraditionalForm[HoldForm[Subscript[g, i] × Subscript[g, j] == k']],
      " ", smallChip[ok2]]}, 13], Style[
      Row[{TraditionalForm[HoldForm[Subscript[h, ij3] == (Subscript[g, i]^2 +
      Subscript[g, j]^2) / (2 Subscript[g, i] × Subscript[g, j])]],
      " ≥ 1, so ", hsym[3], " ≠ 0 ", smallChip[ok3]]}, 13],
    Style[Row[{"condition of the third claim unsatisfiable here ",
      smallChip[ok4]}], 13]], Spacings → 0.8}},
    Spacings → 1], Background → Lighter[RGBColor[0.75, 0.35,
      0.05],
      0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 5:SUMMARY*)
(* =====*)
Print[Framed[Style[If[And @@ resultsAll, "ALL CLAIMS VERIFIED",
  "SOME CLAIMS FAILED"], Bold, White, 16], Background →
  If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];

```

The quantities under test, in the case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$:

$$h_{ij1} = \frac{e_i g_j \sqrt{(g_i^2 - 1)(g_j^2 - (k')^2)} \xi_{ij} + e_j g_i \sqrt{(g_i^2 - 1)(g_j^2 - (k')^2)} \xi_{ij}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$h_{ij2} = \frac{\sqrt{(g_i^2 - (k')^2)(g_j^2 - (k')^2)} \xi_{ij} + e_i e_j g_i g_j \sqrt{(g_i^2 - 1)(g_j^2 - 1)} \xi_{ij}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$h_{ij3} = \frac{k^2 g_i g_j + e_i e_j \xi_{ij} \sqrt{(g_i^2 - 1)(g_j^2 - (k')^2)(g_j^2 - 1)(g_i^2 - (k')^2)}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

and, in the case $k^2 = (g_i^2 - 1)(g_j^2 - 1)$ with $e_i = -e_j$:

$$h_{ij1} = \frac{e_i (g_i^2 - g_j^2)}{2 k g_i g_j} \quad h_{ij2} = \frac{g_i^2 g_j^2 - (k')^2}{2 k g_i g_j} \quad h_{ij3} = \frac{g_i^2 + g_j^2}{2 g_i g_j}$$

Standing Hypotheses:

$$k \in (0, 1), \quad (k')^2 = 1 - k^2, \quad g_i, g_j$$

$$\in (0, k') \cup (1, \infty), \quad e_i, e_j \in \{\pm 1\}, \quad \xi_{ij} = \text{sgn}((g_i^2 - 1)(g_j^2 - 1))$$

The claims we test (over all window

configurations of g_i, g_j and all e_i, e_j , in both directions):

$$h_{ij1} = 0 \Leftrightarrow g_i = g_j \wedge e_i = -e_j$$

$$h_{ij2} = 0 \Leftrightarrow g_i g_j = k' \wedge e_i = -e_j$$

$$h_{ij3} = 0 \Leftrightarrow k^2 (g_i^2 + g_j^2) = k^2 - (g_i^2 - 1)(g_j^2 - 1) \wedge e_i = e_j$$

First claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):

Verified

$(e_i, e_j) \rightarrow$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
tested $\rightarrow$	$h_{ij1} \neq 0$	$h_{ij1} = 0 \Leftrightarrow g_i = g_j$	$h_{ij1} = 0 \Leftrightarrow g_i = g_j$	$h_{ij1} \neq 0$
$g_i, g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i \in (1, \infty), g_j \in (0, k')$	✓	✓	✓	✓
$g_i \in (0, k'), g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i, g_j \in (0, k')$	✓	✓	✓	✓

Second claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):

Verified

$(e_i, e_j) \rightarrow$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
tested $\rightarrow$	$h_{ij2} \neq 0$	$h_{ij2} = 0 \Leftrightarrow g_i g_j = k'$	$h_{ij2} = 0 \Leftrightarrow g_i g_j = k'$	$h_{ij2} \neq 0$
$g_i, g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i \in (1, \infty), g_j \in (0, k')$	✓	✓	✓	✓
$g_i \in (0, k'), g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i, g_j \in (0, k')$	✓	✓	✓	✓

Third claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):

Verified

$(e_i, e_j) \rightarrow$	$(1, 1)$	$(1, -1)$	$(-1, 1)$	$(-1, -1)$
tested $\rightarrow$	$h_{ij3} = 0 \Leftrightarrow k^2 (g_i^2 + g_j^2) = k^2 - (g_i^2 - 1)(g_j^2 - 1)$	$h_{ij3} \neq 0$	$h_{ij3} \neq 0$	$h_{ij3} = 0 \Leftrightarrow k^2 (g_i^2 + g_j^2) = k^2 - (g_i^2 - 1)(g_j^2 - 1)$
$g_i, g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i \in (1, \infty), g_j \in (0, k')$	✓	✓	✓	✓
$g_i \in (0, k'), g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i, g_j \in (0, k')$	✓	✓	✓	✓

Case $k^2 = (g_i^2 - 1)(g_j^2 - 1)$, with $e_i = -e_j$:

Verified

$$h_{ij1} = \frac{e_i(g_i^2 - g_j^2)}{2k g_i g_j}: h_{ij1} = 0 \Leftrightarrow g_i = g_j \quad \checkmark$$

$$h_{ij2} = \frac{g_i^2 g_j^2 - (k')^2}{2k g_i g_j}: h_{ij2} = 0 \Leftrightarrow g_i g_j = k' \quad \checkmark$$

$$h_{ij3} = \frac{g_i^2 + g_j^2}{2g_i g_j} \geq 1, \text{ so } h_{ij3} \neq 0 \quad \checkmark$$

condition of the third claim unsatisfiable here $\checkmark$

ALL CLAIMS VERIFIED

Section 6

Results of Section 6 are used in the proof of Lemma 12:
Case 1

```

In[*]:= (* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[UU, XX, YY, ZZ, tt, wConfigs, sTriples, dataOf, lhsOf, derOf,
  assumOf, derCheck, resultsAll, statusChip, smallChip, lblZ, lblY, lblX];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background -> RGBColor[0.13, 0.55, 0.13], FrameStyle -> None,
  RoundingRadius -> 7, FrameMargins -> {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background -> RGBColor[0.82, 0.10, 0.10], FrameStyle -> None,
  RoundingRadius -> 7, FrameMargins -> {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
(*Sign regions:
  two z_l-windows (from g_l in (0,k') or (1,infinity)) and the sign of y_l;

```

```

the sign of  $x_1$  is then forced by  $u x_1 y_1 z_1 > 0$ . The window  $z_1 < -1$  carries  $1 + u z_1 > 0$ . On each region,  $p_1$  and  $q_1$  take their explicit radical values and the nested square roots collapse;
the identity is checked there for all 8 sign combinations of  $e', e_0, \epsilon_{s_l}.$ 
lblZ[1] = TraditionalForm[HoldForm[Subscript[z, 1] > 0]];
lblZ[2] = TraditionalForm[HoldForm[Subscript[z, 1] < -1]];
lblY[1] = TraditionalForm[HoldForm[Subscript[y, 1] > 0]];
lblY[-1] = TraditionalForm[HoldForm[Subscript[y, 1] < 0]];
lblX[1] = TraditionalForm[HoldForm[Subscript[x, 1] > 0]];
lblX[-1] = TraditionalForm[HoldForm[Subscript[x, 1] < 0]];
wConfigs = {{Row[{lblZ[1], " ", lblY[1], " hence ", lblX[1]}], ZZ, 1, 1},
  {Row[{lblZ[1], " ", lblY[-1], " hence ", lblX[-1]}], ZZ, -1, -1},
  {Row[{lblZ[2], " ", lblY[1], " hence ", lblX[-1]}], -1 - ZZ, 1, -1},
  {Row[{lblZ[2], " ", lblY[-1], " hence ", lblX[1]}], -1 - ZZ, -1, 1}};
sTriples = Tuples[{1, -1}, 3];
dataOf[{lab_, z1_, sy_, sx_}] := Module[{yl = sy YY, x1 = sx XX,
  kk = Sqrt[UU], kp = Sqrt[1 - UU], gl, p1, ql, Dd}, gl = Sqrt[1 + 1 / z1];
  p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
  ql = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
  Dd = -(1 - x1 tt^2) (1 - UU x1 tt^2);
  {z1, yl, x1, kk, kp, gl, p1, ql, Dd}];
lhsOf[{z1_, yl_, x1_, kk_, kp_, gl_, p1_, ql_, Dd_}, ep_, e0_, eps_] :=
  Module[{snl, cnl, RR = Sqrt[x1 Dd]},
    snl = I eps Sqrt[(gl^2 - 1) p1^2 ql^2] / (kk p1 ql);
    cnl = eps Sqrt[(gl^2 - kp^2) p1^2 ql^2] / (kk p1 ql);
    (z1 ql cnl gl (tt / p1) + ep z1 ql snl (e0 I RR / (XX p1))) / (z1 + x1 tt^2)];
derOf[{z1_, yl_, x1_, kk_, kp_, gl_, p1_, ql_, Dd_}, ep_, e0_, eps_] :=
  eps (tt Sqrt[UU x1 yl (1 + z1) (1 + UU z1)] - ep e0 Sqrt[UU yl z1 Dd]) /
    (yl UU (z1 + x1 tt^2));
assumOf[{lab_, z1_, sy_, sx_}] := 0 < UU < 1 && XX > 0 && YY > 0 && ZZ > 0 &&
  Element[tt, Reals] && sx XX (- (1 - sx XX tt^2) (1 - UU sx XX tt^2)) >= 0 &&
  1 + UU z1 > 0 && z1 + sx XX tt^2 != 0;
derCheck[w_, {ep_, e0_, eps_}] := Module[{d = dataOf[w]},
  TimeConstrained[FullSimplify[lhsOf[d, ep, e0, eps] - derOf[d, ep, e0, eps] == 0,
    assumOf[w]], 180, False] == True];
(* =====*)
(*PART 2:THE GIVEN HYPOTHESES AND THE FORMULAS TO TEST*)
(* =====*)
Print[Panel[Column[{Style[Row[{"Given hypotheses, Case 1  ("},
  TraditionalForm[HoldForm[u > 0]], ")] :"}], Bold, 14],
  Style[Row[{TraditionalForm[HoldForm[0 < u < 1]], " ",
    TraditionalForm[HoldForm[k = Sqrt[u]]], " ", TraditionalForm[
      HoldForm[k' = Sqrt[1 - u]]], " ", TraditionalForm[HoldForm[k]],
      " ", TraditionalForm[HoldForm[k']], " \in (0, 1)"}], 14],
  Style[Row[{"For ", TraditionalForm[HoldForm[i]], " = 1, 2, 3, 4 :"}],
    Bold, 14], Style[Row[{TraditionalForm[HoldForm[Subscript[x, i]]],
      " ", TraditionalForm[HoldForm[Subscript[y, i]]], " ",

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TraditionalForm[HoldForm[Subscript[z, i]], " ∈ ℝ",
", ", TraditionalForm[HoldForm[Subscript[x, i]],
", ", TraditionalForm[HoldForm[Subscript[y, i]],
", ", TraditionalForm[HoldForm[Subscript[z, i]],
" ≠ -1, 0, ", TraditionalForm[HoldForm[Subscript[g, i] =
dn[Subscript[t, i] = Sqrt[1 + 1 / Subscript[z, i]]]], 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, i]], " ",
TraditionalForm[HoldForm[Subscript[q, i]], " ∈ ",
Subscript["ℝ", "> 0"], " ∪ ℐ ", Subscript["ℝ", "> 0"],
", ", TraditionalForm[HoldForm[Subscript[e, 0]],
", ", TraditionalForm[HoldForm[Derivative[1][e]],
", ", TraditionalForm[HoldForm[Subscript[e, i]],
", ", TraditionalForm[HoldForm[Subscript[ε, i]], " ",
TraditionalForm[HoldForm[Subscript[ξ, 23]]], " ∈ {±1}"]], 14],
Style[Row[{TraditionalForm[HoldForm[u Subscript[x, i] ×
Subscript[y, i] × Subscript[z, i] > 0]], " ",
TraditionalForm[HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]],
", ", TraditionalForm[HoldForm[1 + u Subscript[z, i] > 0]]], 14],
Style[Row[{TraditionalForm[HoldForm[∂[Subscript[x, 1], t] =
- (1 - Subscript[x, 1] t^2) (1 - u Subscript[x, 1] t^2)]],
", ", TraditionalForm[HoldForm[R = Sqrt[Subscript[x, 1] ×
∂[Subscript[x, 1], t]]]], " ", TraditionalForm[
HoldForm[Subscript[x, 1] × ∂[Subscript[x, 1], t] ≥ 0]]],
14], ""], Style["Formulas:", Bold, 14], Style[
Row[{TraditionalForm[HoldForm[sn[v + w] = (sn[v] × cn[w] × dn[w] +
sn[w] × cn[v] × dn[v]) / (1 - k^2 sn[v]^2 × sn[w]^2)]],
" ", Style["the first addition formula", Italic, 13,
RGBColor[0.10, 0.30, 0.70]]], 14], Style[
Row[{TraditionalForm[HoldForm[{sn, cn, dn}[-v] = {-sn[v], cn[v], dn[v]}]],
" ", Style["the parity formulas", Italic,
13, RGBColor[0.75, 0.35, 0.05]]], 14], ""],
Style[Row[{"We want to test these formulas ", Style["*", 14], " :"}],
Bold, 14],
Style[TraditionalForm[HoldForm[Subscript[W, 2] = Subscript[ε, 2]
(t Sqrt[u Subscript[x, 1] × Subscript[y, 2] (1 + Subscript[z, 2])
(1 + u Subscript[z, 2])] + Subscript[e, 0] × Subscript[e, 2] Sqrt[
u Subscript[y, 2] × Subscript[z, 2] × ∂[Subscript[x, 1], t]]) /
(Subscript[y, 2] u (Subscript[z, 2] + Subscript[x, 1] t^2))]], 15],
Style[TraditionalForm[HoldForm[Subscript[W, 1] = Subscript[ε, 1]
(t Sqrt[u Subscript[x, 1] × Subscript[y, 1] (1 + Subscript[z, 1])
(1 + u Subscript[z, 1])] - Subscript[e, 0] × Subscript[e, 1] Sqrt[
u Subscript[y, 1] × Subscript[z, 1] × ∂[Subscript[x, 1], t]]) /
(Subscript[y, 1] u (Subscript[z, 1] + Subscript[x, 1] t^2))]], 15],
Style[Row[{TraditionalForm[HoldForm[U = Subscript[ε, 2] × Subscript[ε, 3]
Sign[u Subscript[z, 2]] (Abs[Subscript[x, 1]] Subscript[h, 232] ×
Subscript[h, 233] t + Subscript[e, 0] × Subscript[h, 231]
Sqrt[Subscript[x, 1] × ∂[Subscript[x, 1], t]]) / (Sqrt[Abs[

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Subscript[x, 1] × Subscript[x, 3]]] (1 + Subscript[ε, 23] k^2
Subscript[x, 1] × Subsuperscript[h, 231, 2] t^2))]]], " ",
Style["in cases (a) and (b)", Italic, 13, GrayLevel[0.35]]], 15],
Style[Row[{TraditionalForm[HoldForm[U = Subscript[ε, 2] ×
Subscript[ε, 3] Sign[u Subscript[z, 2]] Sign[Subscript[x, 1]] /
(k Sqrt[Abs[Subscript[x, 1] × Subscript[x, 3]]] t)]], " ",
Style["in case (c)", Italic, 13, GrayLevel[0.35]]], 15]],
Spacings → 1.1], Background → RGBColor[0.99, 0.96,
0.88], FrameMargins → 14]]];
(* =====*)
(*PART 3:THE DERIVATION AND ITS VERIFICATION*)
(* =====*)
resultsAll = {};
Module[{grid, regionOK, ok},
grid = Table[derCheck[w, s], {w, wConfigs}, {s, sTriples}];
regionOK = And@@@grid;
ok = And@@regionOK;
AppendTo[resultsAll, ok];
Print[
Panel[Column[{Row[{Style["The derivation: the formulas for ", Bold, 15,
RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[
HoldForm[Subscript[W, 2]]], Bold, 15, RGBColor[0.10, 0.30, 0.70]],
Style[" and ", Bold, 15, RGBColor[0.10, 0.30, 0.70]],
Style[TraditionalForm[HoldForm[Subscript[W, 1]]], Bold,
15, RGBColor[0.10, 0.30, 0.70]], Style[" for ", Bold, 15,
RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[u > 0]],
Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok]]],
Style["Given hypotheses and formulas, in addition to those above:",
Bold, 13], Style[
Row[{"For ", TraditionalForm[HoldForm[l]], " = 1, 2 :"}], Bold, 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[x, 1] ×
Subsuperscript[p, 1, 2] = 1]], " ", TraditionalForm[
HoldForm[Subscript[y, l] × Subsuperscript[q, l, 2] = 1]], " ",
TraditionalForm[HoldForm[Subscript[p, l] = Subscript[p, 1]]]], 13],
Style[TraditionalForm[HoldForm[sn[Subscript[t, l]] = I Subscript[ε, l]
Sqrt[(Subsuperscript[g, l, 2] - 1) Subsuperscript[p, l, 2] ×
Subsuperscript[q, l, 2]] / (k Subscript[p, l] × Subscript[q, l])]],
14], Style[TraditionalForm[HoldForm[cn[Subscript[t, l]] =
Subscript[ε, l] Sqrt[(Subsuperscript[g, l, 2] - k'^2)
Subsuperscript[p, l, 2] × Subsuperscript[q, l, 2]] /
(k Subscript[p, l] × Subscript[q, l])]], 14],
Style[Row[{TraditionalForm[HoldForm[sn[ε] = t / Subscript[p, 1]]],
", ", TraditionalForm[HoldForm[Subscript[p, 1] × cn[ε] × dn[ε] =
Subscript[e, 0] I R / Abs[Subscript[x, 1]]]]], 14],
Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[Derivative[1][e] ×
Subscript[t, l]] = {Derivative[1][e] × sn[Subscript[t, l]],
cn[Subscript[t, l]], dn[Subscript[t, l]]}]],

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", Style["by the parity formulas", Italic, 13,
  RGBColor[0.75, 0.35, 0.05]]], 14], Style[Row[{TraditionalForm[
  HoldForm[sn[Subscript[t, l]]^2 == (1 - Subsuperscript[g, l, 2]) / k^2 ==
    - (1 / (u Subscript[z, l]))]], ", ",
  TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, l]]^2 × sn[ϕ]^2 ==
    (Subscript[z, l] + Subscript[x, 1] t^2) / Subscript[z, l] ≠ 0]],
  ", by the choice of ", TraditionalForm[HoldForm[t]]}],
14], "", Style["Verification:", Bold, 13],
Style[Row[{"Since ", TraditionalForm[HoldForm[± Subscript[t, l]]],
  " and ", TraditionalForm[HoldForm[ϕ]], " are not poles and ",
  TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, l]]^2 × sn[ϕ]^2 ≠ 0]],
  ", ", Style["the first addition formula", Italic, 13, RGBColor[0.10,
    0.30, 0.70]], ", at ", TraditionalForm[HoldForm[v = ϕ]], ", ",
  TraditionalForm[HoldForm[w = Derivative[1][e] × Subscript[t, l]]],
  ", holds pointwise and, together with ",
  Style["the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]],
  ", gives"}], 13], Style[TraditionalForm[
  HoldForm[Subscript[q, l] × sn[ϕ + Derivative[1][e] × Subscript[t, l]] =
    (Subscript[z, l] × Subscript[q, l] × cn[Subscript[t, l]] ×
      dn[Subscript[t, l]] × sn[ϕ] + Derivative[1][e] × Subscript[z, l] ×
      Subscript[q, l] × sn[Subscript[t, l]] × cn[ϕ] × dn[ϕ]) /
    (Subscript[z, l] + Subscript[x, 1] t^2)], 14],
Style["Using the given formulas, we get", 13],
Style[Row[{TraditionalForm[
  HoldForm[Subscript[q, l] × sn[ϕ + Derivative[1][e] × Subscript[t, l]] =
    Subscript[ε, l] (t Sqrt[u Subscript[x, 1] × Subscript[y, l]
      (1 + Subscript[z, l]) (1 + u Subscript[z, l])) -
    Derivative[1][e] × Subscript[e, 0] Sqrt[u Subscript[
      y, l] × Subscript[z, l] × D[Subscript[x, 1], t]]) /
    (Subscript[y, l] u (Subscript[z, l] + Subscript[x, 1] t^2))}],
  " ", Style["★", 15, RGBColor[0.10, 0.30, 0.70]]], 14],
Style[Row[{"Setting ", TraditionalForm[HoldForm[l = 2]], ", ",
  TraditionalForm[HoldForm[Derivative[1][e] = -Subscript[e, 2]]],
  " in ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]],
  " gives the first formula of ", Style["★", 13], " ."}], 13],
Style[Row[{"Setting ", TraditionalForm[HoldForm[l = 1]], ", ",
  TraditionalForm[HoldForm[Derivative[1][e] = Subscript[e, 1]]],
  " in ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]],
  " gives the second formula of ", Style["★", 13], " ."}], 13], "",
Style[Row[{"The identity ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]],
  " is verified separately on each sign region:"}], 13],
Grid[Table[{Style[wConfigs[iw, 1], 12, GrayLevel[0.3]],
  Style[Row[{Style["★", 12, RGBColor[0.10, 0.30, 0.70]],
    " for all 8 sign combinations of ",
    TraditionalForm[HoldForm[Derivative[1][e]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, 0]]], ", ", TraditionalForm[
      HoldForm[Subscript[ε, l]]], " ∈ {±1} → 0 == 0"}],

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12, GrayLevel[0.3]], smallChip[regionOK[[iw]]], {iw, 4}],
Alignment → Left, Spacings → {2, 0.7}]], Spacings → 1],
Background → Lighter[RGBColor[0.10, 0.30, 0.70], 0.93],
FrameMargins →
12]]];];
(* =====*)
(*PART 4:U-FORMULA IN CASES (a) AND (b)*)
(* =====*)
ClearAll[X3, RR, h1, h2, h3, ee0, ep2, ep3, kk, w3Configs,
thAssum, thLhs, thRhs, thCheck, lblX3, lblY2, lblXi, lblUZ];
lblX3[1] = TraditionalForm[HoldForm[Subscript[x, 3] > 0]];
lblX3[-1] = TraditionalForm[HoldForm[Subscript[x, 3] < 0]];
lblY2[1] = TraditionalForm[HoldForm[Subscript[y, 2] > 0]];
lblY2[-1] = TraditionalForm[HoldForm[Subscript[y, 2] < 0]];
lblXi[1] = TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]];
lblXi[-1] = TraditionalForm[HoldForm[Subscript[ξ, 23] == -1]];
lblUZ[1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] == 1]];
lblUZ[-1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] == -1]];
w3Configs =
Table[{Row[{lblX[sx], ", ", lblX3[s3], ", ", lblY2[sy], ", hence ",
lblXi[sx s3], ", ", lblUZ[sx sy]]], sx, s3, sy},
{sx, {1, -1}}, {s3, {1, -1}}, {sy, {1, -1}}]];
w3Configs = Flatten[w3Configs, 2];
thCheck[{lab_, sx_, s3_, sy_}] :=
Module[{x1 = sx XX, x3 = s3 X3, xi = sx s3, suz = sx sy, p1, p3, q2, S23,
denom, lhs, rhs, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
denom = 1 + xi kk^2 x1 h1^2 tt^2;
lhs = p3 (S23 h2 h3 (tt / p1) - I S23 h1 (ee0 I RR / (XX p1))) / denom;
rhs = ep2 ep3 suz (XX h2 h3 tt + ee0 h1 RR) / (Sqrt[XX X3] denom);
assum = XX > 0 && X3 > 0 && YY > 0 && 0 < kk < 1 && RR ≥ 0 &&
Element[tt | h1 | h2 | h3 | ee0 | ep2 | ep3, Reals] && denom ≠ 0;
TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] == True];
Module[{grid, ok}, grid = thCheck/@w3Configs;
ok = And@@grid;
AppendTo[resultsAll, ok];
Print[
Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
RGBColor[0.55, 0.25, 0.70]], Style[TraditionalForm[HoldForm[U]],
Bold, 15, RGBColor[0.55, 0.25, 0.70]], Style[
" in cases (a) and (b) for ", Bold, 15, RGBColor[0.55, 0.25, 0.70]],
Style[TraditionalForm[HoldForm[u > 0]], Bold, 15,
RGBColor[0.55, 0.25, 0.70]], Spacer[20], statusChip[ok]]},
Style["Given hypotheses and formulas, in addition to those above:",
Bold, 13], Style[Row[{TraditionalForm[HoldForm[Subscript[x, 3] ×

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Subsuperscript[p, 3, 2] == 1]], ", ", TraditionalForm[
HoldForm[Subscript[y, 2] × Subsuperscript[q, 2, 2] == 1]], ", ",
TraditionalForm[HoldForm[Subscript[p, 2] == Subscript[p, 1]]],
", ", TraditionalForm[HoldForm[Subscript[q, 3] == Subscript[q, 2]]],
", ", TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233]]], " ∈ ℝ"}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, 23] ==
Sign[Subscript[x, 1] × Subscript[x, 3]]]],
", ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
Sign[Subscript[x, 1] × Subscript[y, 2]]]]}], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ς] == t / Subscript[p, 1]]],
", ", TraditionalForm[HoldForm[Subscript[p, 1] × cn[ς] × dn[ς] ==
Subscript[e, 0] I R / Abs[Subscript[x, 1]]]]}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[S, 23] == Subscript[ε, 2] ×
Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2] × Subsuperscript[
p, 3, 2] × Subsuperscript[q, 2, 4] × Subscript[ξ, 23]] /
(Subscript[p, 1] × Subscript[p, 3] × Subsuperscript[q, 2, 2])]],
", ", TraditionalForm[HoldForm[Subsuperscript[S, 23, 2] ==
Subscript[ξ, 23]]]]], 13],
Style[TraditionalForm[HoldForm[{sn, cn, dn}[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]] ==
{I Subscript[S, 23] × Subscript[h, 231], Subscript[S, 23] ×
Subscript[h, 232], Subscript[h, 233]}]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]] ==
I Subscript[S, 23] × Subscript[h, 231]]], ", ",
TraditionalForm[HoldForm[sn[Subscript[e, 2] × Subscript[t, 2] +
Subscript[e, 3] × Subscript[t, 3]]^2 ==
-Subscript[ξ, 23] × Subsuperscript[h, 231, 2]]], ", ",
TraditionalForm[HoldForm[cn[Subscript[e, 2] × Subscript[t, 2] +
Subscript[e, 3] × Subscript[t, 3]] × dn[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]] ==
Subscript[S, 23] × Subscript[h, 232] × Subscript[h, 233]]]]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ς]^2 == Subscript[x, 1] t^2]],
", ", TraditionalForm[
HoldForm[1 - k^2 sn[ς]^2 × sn[Subscript[e, 2] × Subscript[t, 2] +
Subscript[e, 3] × Subscript[t, 3]]^2 == 1 + Subscript[ξ, 23]
k^2 Subscript[x, 1] × Subsuperscript[h, 231, 2] t^2 ≠ 0]],
", by the choice of ", TraditionalForm[HoldForm[t]]}],
13], "", Style["Verification:", Bold, 13],
Style[Row[{"Since ", TraditionalForm[HoldForm[± (Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3])]], " and ",
TraditionalForm[HoldForm[ς]], " are not poles and ",
TraditionalForm[HoldForm[1 - k^2 sn[ς]^2 × sn[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]]^2 ≠ 0]],
", ", Style["the first addition formula", Italic, 13,

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RGBColor[0.10, 0.30, 0.70]],
", at ", TraditionalForm[HoldForm[v =  $\varsigma$ ]], ", ",
TraditionalForm[HoldForm[w = -Subscript[e, 2]  $\times$  Subscript[t, 2] -
Subscript[e, 3]  $\times$  Subscript[t, 3]]],
", holds pointwise and, together with ", Style[
"the parity formulas", Italic, 13,
RGBColor[0.75, 0.35, 0.05]], ", gives"}], 13],
Style[TraditionalForm[HoldForm[Subscript[p, 3]  $\times$  sn[ $\varsigma$  - Subscript[e, 2]  $\times$ 
Subscript[t, 2] - Subscript[e, 3]  $\times$  Subscript[t, 3]] =
(Subscript[p, 3] (cn[Subscript[e, 2]  $\times$  Subscript[t, 2] +
Subscript[e, 3]  $\times$  Subscript[t, 3]]  $\times$  dn[Subscript[e, 2]  $\times$ 
Subscript[t, 2] + Subscript[e, 3]  $\times$  Subscript[t, 3]]  $\times$  sn[ $\varsigma$ ] -
sn[Subscript[e, 2]  $\times$  Subscript[t, 2] + Subscript[e, 3]  $\times$ 
Subscript[t, 3]]  $\times$  cn[ $\varsigma$ ]  $\times$  dn[ $\varsigma$ ])) / (1 + Subscript[ $\xi$ , 23]
k^2 Subscript[x, 1]  $\times$  Subsuperscript[h, 231, 2] t^2)], 14],
Style["Using the given hypotheses and formulas, we get", 13],
Style[
Row[{TraditionalForm[HoldForm[Subscript[p, 3]  $\times$  sn[ $\varsigma$  - Subscript[e, 2]  $\times$ 
Subscript[t, 2] - Subscript[e, 3]  $\times$  Subscript[t, 3]] =
Subscript[ $\varepsilon$ , 2]  $\times$  Subscript[ $\varepsilon$ , 3] Sign[u Subscript[z, 2]]
(Abs[Subscript[x, 1]] Subscript[h, 232]  $\times$  Subscript[h, 233] t +
Subscript[e, 0]  $\times$  Subscript[h, 231]
Sqrt[Subscript[x, 1]  $\times$   $\mathcal{D}$ [Subscript[x, 1], t]]) / (Sqrt[
Abs[Subscript[x, 1]  $\times$  Subscript[x, 3]] (1 + Subscript[ $\xi$ , 23]
k^2 Subscript[x, 1]  $\times$  Subsuperscript[h, 231, 2] t^2))}],
" ", Style["♦", 14, RGBColor[0.55, 0.25, 0.70]]}], 14],
Style[Row[{"which is the third formula of ", Style["*", 13], " ."}],
13], ""],
Style[Row[{"The identity ", Style["♦", 13, RGBColor[0.55, 0.25, 0.70]],
" is verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
", ", TraditionalForm[HoldForm[Subscript[h, 233]]],
", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
", ", TraditionalForm[HoldForm[Subscript[ $\varepsilon$ , 2]]],
", ", TraditionalForm[HoldForm[Subscript[ $\varepsilon$ , 3]]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"}], 13],
Style[Row[{"In case (a), both values of ", TraditionalForm[
HoldForm[Subscript[ $\xi$ , 23]]], " occur, because the sign of ",
TraditionalForm[HoldForm[Subscript[z, 2]  $\times$  Subscript[z, 3]]],
" is not constrained; in case (b), ",
TraditionalForm[HoldForm[Subscript[ $\xi$ , 23] = 1]], ", because ",
TraditionalForm[HoldForm[k^2 = (Subsuperscript[g, 2, 2] - 1)
(Subsuperscript[g, 3, 2] - 1) > 0]], " and ", TraditionalForm[
HoldForm[Subsuperscript[g, i, 2] - 1 = 1 / Subscript[z, i]]], ", so ",
TraditionalForm[HoldForm[Subscript[z, 2]  $\times$  Subscript[z, 3] > 0]]],

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" . The eight sign regions cover both cases.}],
12, GrayLevel[0.35]], Grid[Table[
  {Style[w3Configs[[iw, 1]], 12, GrayLevel[0.3]], Style[
    Row[{Style["♦", 12, RGBColor[0.55, 0.25, 0.70]], " → 0 == 0"}],
    12, GrayLevel[0.3]], smallChip[grid[[iw]]], {iw, 8}},
  Alignment → Left, Spacings → {2, 0.7}]], Spacings → 1],
Background → Lighter[RGBColor[0.55, 0.25, 0.70],
0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 5:U-FORMULA IN CASE (c) FOR u>0*)
(* =====*)
ClearAll[cCheck];
cCheck[{lab_, sx_, s3_, sy_}] :=
Module[{p1, p3, q2, xi = sx s3, suz = sx sy, S23, lhs, rhs, assum},
  p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
  p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
  q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
  S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
  lhs = S23 p1 p3 / (kk tt);
  rhs = suz ep2 ep3 sx / (kk Sqrt[XX X3] tt);
  assum =
    XX > 0 && X3 > 0 && YY > 0 && 0 < kk < 1 && Element[tt | ep2 | ep3, Reals] && tt ≠ 0;
  TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] == True];
Module[{cConfigs, grid, ok}, cConfigs = Select[w3Configs, #[[2]] × #[[3]] == 1 &];
grid = cCheck /@ cConfigs;
ok = And @@ grid;
AppendTo[resultsAll, ok];
Print[
  Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
    RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[U]], Bold,
    15, RGBColor[0.00, 0.50, 0.50]], Style[" in case (c) for ", Bold, 15,
    RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u > 0]],
    Bold, 15, RGBColor[0.00, 0.50, 0.50]], Spacer[20], statusChip[ok]}],
    Style["Given hypotheses and formulas, in addition to those above:",
    Bold, 13], Style[Row[{TraditionalForm[HoldForm[
      Subscript[ξ, 23] == Sign[Subscript[x, 1] × Subscript[x, 3]]],
      ", ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
        Sign[Subscript[x, 1] × Subscript[y, 2]]]]], 13],
    Style[TraditionalForm[HoldForm[Subscript[S, 23] == Subscript[ε, 2] ×
      Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2] × Subsuperscript[p, 3,
        2] × Subsuperscript[q, 2, 4] × Subscript[ξ, 23]] / (Subscript[
        p, 1] × Subscript[p, 3] × Subsuperscript[q, 2, 2])]], 13],
    Style[Row[{TraditionalForm[HoldForm[Subscript[t, 2] + Subscript[t, 3] ==
      2 m K + I K']], ", ", TraditionalForm[HoldForm[m ∈ ℤ]], ", ",
      TraditionalForm[HoldForm[(-1)^m == Subscript[S, 23]]], ", ",
      TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]]], 13],
    Style[Row[{TraditionalForm[HoldForm[sn[v + 2 m K] == (-1)^m sn[v]]],

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", " ", TraditionalForm[HoldForm[sn[v ± I K'] == 1 / (k sn[v])]],
" ", Style["the shift formulas", Italic,
13, RGBColor[0.00, 0.50, 0.50]]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ς] == t / Subscript[p, 1]]],
", " ", TraditionalForm[HoldForm[t ≠ 0]],
", by the choice of ", TraditionalForm[HoldForm[t]]], 13], ""],
Style["Verification:", Bold, 13], Style[Row[{"Since ", TraditionalForm[
HoldForm[Subscript[e, 2] == Subscript[e, 3]], " and ",
TraditionalForm[HoldForm[(-1)^(m Subscript[e, 2]) == (-1)^m]],
", ", Style["the shift formulas", Italic, 13,
RGBColor[0.00, 0.50, 0.50]], " give"}], 13],
Style[TraditionalForm[HoldForm[Subscript[p, 3] ×
sn[ς - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
(-1)^m Subscript[p, 3] / (k sn[ς]) == Subscript[S, 23] ×
Subscript[p, 1] × Subscript[p, 3] / (k t)], 14],
Style["Using the given hypotheses and formulas, we get", 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3] × sn[ς - Subscript[e,
2] (Subscript[t, 2] + Subscript[t, 3])] == Subscript[ε, 2] ×
Subscript[ε, 3] Sign[u Subscript[z, 2]] Sign[Subscript[x, 1]] /
(k Sqrt[Abs[Subscript[x, 1] × Subscript[x, 3]] t])],
" ", Style["▲", 14, RGBColor[0.00, 0.50, 0.50]]], 14],
Style[Row[{"which is the fourth formula of ", Style["★", 13], " ."}],
13], "",
Style[Row[{"The identity ", Style["▲", 13, RGBColor[0.00, 0.50, 0.50]],
" is verified separately on each of the four sign
regions realizable in case (c), with ",
TraditionalForm[HoldForm[Subscript[ε, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[ε, 3]]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"}], 13],
Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
", because ", TraditionalForm[HoldForm[k^2 ==
(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
" and ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
1 / Subscript[z, i]]], ", so ",
TraditionalForm[HoldForm[Subscript[z, 2] × Subscript[z, 3] > 0]],
" ."}], 12, GrayLevel[0.35]],
Grid[Table[{Style[cConfigs[iw, 1]], 12, GrayLevel[0.3]], Style[
Row[{Style["▲", 12, RGBColor[0.00, 0.50, 0.50]], " → 0 == 0"}],
12, GrayLevel[0.3]], smallChip[grid[iw]]], {iw, 4}],
Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.00, 0.50, 0.50], 0.93],
FrameMargins →
12]]];];
(* =====*)
(*PART 6:SUMMARY*)
(* =====*)

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Print[Framed[Style[If[And@@resultsAll, "CASE 1: ALL CLAIMS VERIFIED",
  "SOME CLAIMS FAILED"], Bold, White, 16], Background →
  If[And@@resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]]];
```

Given hypotheses, Case 1 ($u > 0$):

$$0 < u < 1, \quad k = \sqrt{u}, \quad k' = \sqrt{1-u}, \quad k, k' \in (0, 1)$$

For $i = 1, 2, 3, 4$:

$$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0, \quad g_i = \operatorname{dn}(t_i) = \sqrt{1 + \frac{1}{z_i}}$$

$$p_i, q_i \in \mathbb{R}_{>0} \cup i \mathbb{R}_{>0}, \quad e_0, e', e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$$

$$u x_i y_i z_i > 0, \quad z_i (1 + z_i) > 0, \quad 1 + u z_i > 0$$

$$\mathcal{D}(x_1, t) = -(1 - x_1 t^2)(1 - u x_1 t^2), \quad R = \sqrt{x_1 \mathcal{D}(x_1, t)}, \quad x_1 \mathcal{D}(x_1, t) \geq 0$$

Formulas:

$$\operatorname{sn}(v + w) = \frac{\operatorname{sn}(v) \operatorname{cn}(w) \operatorname{dn}(w) + \operatorname{sn}(w) \operatorname{cn}(v) \operatorname{dn}(v)}{1 - k^2 \operatorname{sn}(v)^2 \operatorname{sn}(w)^2} \quad \text{the first addition formula}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(-v) = \{-\operatorname{sn}(v), \operatorname{cn}(v), \operatorname{dn}(v)\} \quad \text{the parity formulas}$$

We want to test these formulas ★:

$$W_2 = \frac{\varepsilon_2 (t \sqrt{u x_1 y_2 (1+z_2) (1+u z_2)} + e_0 e_2 \sqrt{u y_2 z_2 \mathcal{D}(x_1, t)})}{y_2 u (z_2 + x_1 t^2)}$$

$$W_1 = \frac{\varepsilon_1 (t \sqrt{u x_1 y_1 (1+z_1) (1+u z_1)} - e_0 e_1 \sqrt{u y_1 z_1 \mathcal{D}(x_1, t)})}{y_1 u (z_1 + x_1 t^2)}$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) (|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{|x_1 x_3|} (1 + \xi_{23} k^2 x_1 h_{231}^2 t^2)} \quad \text{in cases (a) and (b)}$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) \operatorname{sgn}(x_1)}{k \sqrt{|x_1 x_3|} t} \quad \text{in case (c)}$$

The derivation: the formulas for W_2 and W_1 for $u > 0$

Verified

Given hypotheses and formulas, in addition to those above:

For $l = 1, 2$:

$$x_1 p_1^2 = 1, \quad y_l q_l^2 = 1, \quad p_l = p_1$$

$$\operatorname{sn}(t_l) = \frac{i \varepsilon_l \sqrt{(g_l^2 - 1) p_l^2 q_l^2}}{k p_l q_l}$$

$$\operatorname{cn}(t_l) = \frac{\varepsilon_l \sqrt{(g_l^2 - (k')^2) p_l^2 q_l^2}}{k p_l q_l}$$

$$\operatorname{sn}(\zeta) = \frac{t}{p_1}, \quad p_1 \operatorname{cn}(\zeta) \operatorname{dn}(\zeta) = \frac{e_0 i R}{|x_1|}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(e' t_l) = \{e' \operatorname{sn}(t_l), \operatorname{cn}(t_l), \operatorname{dn}(t_l)\} \quad \text{by the parity formulas}$$

$$\operatorname{sn}(t_l)^2 = \frac{1 - g_l^2}{k^2} = -\frac{1}{u z_l}, \quad 1 - k^2 \operatorname{sn}(t_l)^2 \operatorname{sn}(\zeta)^2 = \frac{z_l + x_1 t^2}{z_l} \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $\pm t_l$ and ζ are not poles and $1 - k^2 \operatorname{sn}(t_l)^2 \operatorname{sn}(\zeta)^2 \neq 0$, the first addition formula, at $v = \zeta$, $w = e' t_l$, holds pointwise and, together with the parity formulas, gives

$$q_l \operatorname{sn}(\zeta + e' t_l) = \frac{z_l q_l \operatorname{cn}(t_l) \operatorname{dn}(t_l) \operatorname{sn}(\zeta) + e' z_l q_l \operatorname{sn}(t_l) \operatorname{cn}(\zeta) \operatorname{dn}(\zeta)}{z_l + x_1 t^2}$$

Using the given formulas, we get

$$q_l \operatorname{sn}(\zeta + e' t_l) = \frac{\varepsilon_l (t \sqrt{u x_1 y_l (1 + z_l) (1 + u z_l)} - e' e_0 \sqrt{u y_l z_l \mathcal{D}(x_1, t)})}{y_l u (z_l + x_1 t^2)} \quad \star$$

Setting $l = 2$, $e' = -e_2$ in $\star$ gives the first formula of $\star$.

Setting $l = 1$, $e' = e_1$ in $\star$ gives the second formula of $\star$.

The identity $\star$ is verified separately on each sign region:

$z_l > 0, y_l > 0$, hence $x_1 > 0$	$\star$ for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\} \rightarrow 0 == 0$	✓
$z_l > 0, y_l < 0$, hence $x_1 < 0$	$\star$ for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\} \rightarrow 0 == 0$	✓
$z_l < -1, y_l > 0$, hence $x_1 < 0$	$\star$ for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\} \rightarrow 0 == 0$	✓
$z_l < -1, y_l < 0$, hence $x_1 > 0$	$\star$ for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\} \rightarrow 0 == 0$	✓

The derivation: the formula for U in cases (a) and (b) for $u > 0$

Verified

Given hypotheses and formulas, in addition to those above:

$$x_3 p_3^2 = 1, \quad y_2 q_2^2 = 1, \quad p_2 = p_1, \quad q_3 = q_2, \quad h_{231}, h_{232}, h_{233} \in \mathbb{R}$$

$$\xi_{23} = \text{sgn}(x_1 x_3), \quad \text{sgn}(u z_2) = \text{sgn}(x_1 y_2)$$

$$\text{sn}(\varsigma) = \frac{t}{p_1}, \quad p_1 \text{cn}(\varsigma) \text{dn}(\varsigma) = \frac{e_0 i R}{|x_1|}$$

$$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{p_1^2 p_3^2 q_2^4 \xi_{23}}}{p_1 p_3 q_2^2}, \quad S_{23}^2 = \xi_{23}$$

$$\{\text{sn}, \text{cn}, \text{dn}\}(e_2 t_2 + e_3 t_3) = \{i S_{23} h_{231}, S_{23} h_{232}, h_{233}\}$$

$$\text{sn}(e_2 t_2 + e_3 t_3) = i S_{23} h_{231}, \quad \text{sn}(e_2 t_2 + e_3 t_3)^2 = -\xi_{23} h_{231}^2$$

$$, \quad \text{cn}(e_2 t_2 + e_3 t_3) \text{dn}(e_2 t_2 + e_3 t_3) = S_{23} h_{232} h_{233}$$

$$\text{sn}(\varsigma)^2 = x_1 t^2, \quad 1 - k^2 \text{sn}(\varsigma)^2 \text{sn}(e_2 t_2 + e_3 t_3)^2 = 1 + \xi_{23} k^2 x_1 h_{231}^2 t^2 \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $\pm(e_2 t_2 + e_3 t_3)$ and ς are not poles and

$$1 - k^2 \text{sn}(\varsigma)^2 \text{sn}(e_2 t_2 + e_3 t_3)^2 \neq 0, \quad \text{the first addition formula, at } v = \varsigma,$$

$w = -e_2 t_2 - e_3 t_3$, holds pointwise and, together with *the parity formulas*, gives

$$p_3 \text{sn}(\varsigma - e_2 t_2 - e_3 t_3) = \frac{p_3 (\text{cn}(e_2 t_2 + e_3 t_3) \text{dn}(e_2 t_2 + e_3 t_3) \text{sn}(\varsigma) - \text{sn}(e_2 t_2 + e_3 t_3) \text{cn}(\varsigma) \text{dn}(\varsigma))}{1 + \xi_{23} k^2 x_1 h_{231}^2 t^2}$$

Using the given hypotheses and formulas, we get

$$p_3 \text{sn}(\varsigma - e_2 t_2 - e_3 t_3) = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2) (|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{|x_1 x_3|} (1 + \xi_{23} k^2 x_1 h_{231}^2 t^2)} \quad \blacklozenge$$

which is the third formula of $\star$.

The identity $\blacklozenge$ is verified separately on each sign region, with $h_{231}, h_{232}, h_{233}, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

In case (a), both values of ξ_{23} occur, because the sign of $z_2 z_3$

is not constrained; in case (b), $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$

and $g_i^2 - 1 = \frac{1}{z_i}$, so $z_2 z_3 > 0$. The eight sign regions cover both cases.

$x_1 > 0, x_3 > 0, y_2 > 0$, hence $\xi_{23} = 1, \text{sgn}(u z_2) = 1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 > 0, x_3 > 0, y_2 < 0$, hence $\xi_{23} = 1, \text{sgn}(u z_2) = -1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 > 0, x_3 < 0, y_2 > 0$, hence $\xi_{23} = -1, \text{sgn}(u z_2) = 1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 > 0, x_3 < 0, y_2 < 0$, hence $\xi_{23} = -1, \text{sgn}(u z_2) = -1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 < 0, x_3 > 0, y_2 > 0$, hence $\xi_{23} = -1, \text{sgn}(u z_2) = -1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 < 0, x_3 > 0, y_2 < 0$, hence $\xi_{23} = -1, \text{sgn}(u z_2) = 1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 < 0, x_3 < 0, y_2 > 0$, hence $\xi_{23} = 1, \text{sgn}(u z_2) = -1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 < 0, x_3 < 0, y_2 < 0$, hence $\xi_{23} = 1, \text{sgn}(u z_2) = 1$	$\blacklozenge \rightarrow 0 == 0$	✓

The derivation: the formula for U in case (c) for $u > 0$ **Verified****Given hypotheses and formulas, in addition to those above:**

$$\xi_{23} = \text{sgn}(x_1 x_3), \quad \text{sgn}(u z_2) = \text{sgn}(x_1 y_2)$$

$$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{p_1^2 p_3^2 q_2^4 \xi_{23}}}{p_1 p_3 q_2^2}$$

$$t_2 + t_3 = 2mK + iK', \quad m \in \mathbb{Z}, \quad (-1)^m = S_{23}, \quad e_2 = e_3$$

$$\text{sn}(v + 2mK) = (-1)^m \text{sn}(v), \quad \text{sn}(v \pm iK') = \frac{1}{k \text{sn}(v)} \quad \text{the shift formulas}$$

$$\text{sn}(\varsigma) = \frac{t}{p_1}, \quad t \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $e_2 = e_3$ and $(-1)^{m e_2} = (-1)^m$, *the shift formulas* give

$$p_3 \text{sn}(\varsigma - e_2(t_2 + t_3)) = \frac{(-1)^m p_3}{k \text{sn}(\varsigma)} = \frac{S_{23} p_1 p_3}{k t}$$

Using the given hypotheses and formulas, we get

$$p_3 \text{sn}(\varsigma - e_2(t_2 + t_3)) = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2) \text{sgn}(x_1)}{k \sqrt{|x_1 x_3|} t} \quad \blacktriangle$$

which is the fourth formula of $\star$.

The identity $\blacktriangle$

is verified separately on each of the four sign regions realizable in case (c), with $\varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and $g_i^2 - 1 = \frac{1}{z_i}$, so $z_2 z_3 > 0$.

$$x_1 > 0, \quad x_3 > 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \text{sgn}(u z_2) = 1 \quad \blacktriangle \rightarrow 0 == 0 \quad \checkmark$$

$$x_1 > 0, \quad x_3 > 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \text{sgn}(u z_2) = -1 \quad \blacktriangle \rightarrow 0 == 0 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \text{sgn}(u z_2) = -1 \quad \blacktriangle \rightarrow 0 == 0 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \text{sgn}(u z_2) = 1 \quad \blacktriangle \rightarrow 0 == 0 \quad \checkmark$$

CASE 1: ALL CLAIMS VERIFIED

Section 7

Results of Section 7 are used in the proof of Lemma 12: Case 2

```
In[ ]:= (* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[UU, XX, YY, ZZ, tt, wConfigs, sTriples, dataOf, lhsOf, derOf,
  assumOf, derCheck, resultsAll, statusChip, smallChip, lblZ, lblY, lblX];
```

```

statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
(*Case 2:  $u < 0$ . We write  $u = -UU$  with  $UU > 0$ ,
so  $M = 1 + UU$ ,  $k' = 1/\sqrt{1 + UU}$ ,  $k = k' \sqrt{UU}$ ,
 $g_l = 1/\sqrt{1 + 1/z_l}$ . Sign regions: two  $z_l$ -windows and the sign of  $y_l$ ;
the sign of  $x_l$  is forced by  $u x_l y_l z_l > 0$ ,
which for  $u < 0$  means  $x_l y_l z_l < 0$ . The window  $z_l > 0$  carries  $1 + u z_l > 0$ , that is,
 $1 - UU z_l > 0$ . On each region,  $p_l$  and  $q_l$  take their explicit radical values;
the identity is checked there for all 8 sign combinations of  $e'$ ,  $e_0$ ,  $\epsilon_{s_l}$ .)
lblZ[1] = TraditionalForm[HoldForm[Subscript[z, l] > 0]];
lblZ[2] = TraditionalForm[HoldForm[Subscript[z, l] < -1]];
lblY[1] = TraditionalForm[HoldForm[Subscript[y, l] > 0]];
lblY[-1] = TraditionalForm[HoldForm[Subscript[y, l] < 0]];
lblX[1] = TraditionalForm[HoldForm[Subscript[x, l] > 0]];
lblX[-1] = TraditionalForm[HoldForm[Subscript[x, l] < 0]];
wConfigs = {{Row[{lblZ[1], " ", lblY[1], " hence ", lblX[-1]}], ZZ, 1, -1},
  {Row[{lblZ[1], " ", lblY[-1], " hence ", lblX[1]}], ZZ, -1, 1},
  {Row[{lblZ[2], " ", lblY[1], " hence ", lblX[1]}], -1 - ZZ, 1, 1},
  {Row[{lblZ[2], " ", lblY[-1], " hence ", lblX[-1]}], -1 - ZZ, -1, -1}};
sTriples = Tuples[{1, -1}, 3];
dataOf[{lab_, zl_, sy_, sx_}] :=
Module[{yl = sy YY, x1 = sx XX, kk = Sqrt[UU] / Sqrt[1 + UU],
  kp = 1 / Sqrt[1 + UU], gl, p1, ql, Dd}, gl = 1 / Sqrt[1 + 1 / zl];
p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
ql = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
Dd = -(1 - x1 tt^2) (1 + UU x1 tt^2);
{zl, yl, x1, kk, kp, gl, p1, ql, Dd}];
lhsOf[{zl_, yl_, x1_, kk_, kp_, gl_, p1_, ql_, Dd_}, ep_, e0_, eps_] :=
Module[{snl, cnl, RR = Sqrt[x1 Dd]},
  snl = I eps Sqrt[(gl^2 - 1) p1^2 ql^2] / (kk p1 ql);
  cnl = eps Sqrt[(gl^2 - kp^2) p1^2 ql^2] / (kk p1 ql);
  ((1 + zl) ql cnl (tt / p1) + ep (1 + zl) ql snl gl (e0 I kp RR / (XX p1))) /
  (zl + x1 tt^2)];
derOf[{zl_, yl_, x1_, kk_, kp_, gl_, p1_, ql_, Dd_}, ep_, e0_, eps_] :=
Module[{uu = -UU},
  eps (tt Sqrt[uu x1 yl (1 + zl) (1 + uu zl)] - ep e0 Sqrt[uu yl zl Dd]) /
  (yl uu (zl + x1 tt^2))];
assumOf[{lab_, zl_, sy_, sx_}] := UU > 0 && XX > 0 && YY > 0 && ZZ > 0 &&
  Element[tt, Reals] && sx XX (- (1 - sx XX tt^2) (1 + UU sx XX tt^2)) ≥ 0 &&
  1 - UU zl > 0 && zl + sx XX tt^2 ≠ 0;
derCheck[w_, {ep_, e0_, eps_}] := Module[{d = dataOf[w]},

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TimeConstrained[FullSimplify[lhsOf[d, ep, e0, eps] - derOf[d, ep, e0, eps] == 0,
  assumOf[w]], 180, False] === True];
(* =====*)
(*PART 2:THE GIVEN HYPOTHESES AND THE FORMULAS TO TEST*)
(* =====*)
Print[Panel[Column[{Style[Row[{"Given hypotheses, Case 2  (" ,
  TraditionalForm[HoldForm[u < 0]], " ) :"}], Bold, 14],
Style[Row[{TraditionalForm[HoldForm[u < 0]], " ,  ",
  TraditionalForm[HoldForm[k' == 1 / Sqrt[1 - u]]], " ,  ", TraditionalForm[
  HoldForm[k == k' Sqrt[-u]]], " ,  ", TraditionalForm[HoldForm[k]],
  " ,  ", TraditionalForm[HoldForm[k']], " ∈ (0, 1)"}], 14],
Style[Row[{"For  ", TraditionalForm[HoldForm[i]], " = 1, 2, 3, 4 :"}],
Bold, 14], Style[Row[{TraditionalForm[HoldForm[Subscript[x, i]]],
  " ,  ", TraditionalForm[HoldForm[Subscript[y, i]]], " ,  ",
  TraditionalForm[HoldForm[Subscript[z, i]]], " ∈ ℝ",
  " ,  ", TraditionalForm[HoldForm[Subscript[x, i]]],
  " ,  ", TraditionalForm[HoldForm[Subscript[y, i]]],
  " ,  ", TraditionalForm[HoldForm[Subscript[z, i]]],
  " ≠ -1, 0,  ", TraditionalForm[HoldForm[Subscript[g, i] =
  dn[Subscript[t, i] == 1 / Sqrt[1 + 1 / Subscript[z, i]]]]], 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, i]]], " ,  ",
  TraditionalForm[HoldForm[Subscript[q, i]]], " ∈ ",
  Subscript["ℝ", "> 0"], " ∪ i ", Subscript["ℝ", "> 0"],
  " ,  ", TraditionalForm[HoldForm[Subscript[e, 0]]],
  " ,  ", TraditionalForm[HoldForm[Derivative[1][e]]],
  " ,  ", TraditionalForm[HoldForm[Subscript[e, i]]],
  " ,  ", TraditionalForm[HoldForm[Subscript[ε, i]]], " ,  ",
  TraditionalForm[HoldForm[Subscript[ξ, 23]]], " ∈ {±1}"}], 14],
Style[Row[{TraditionalForm[HoldForm[u Subscript[x, i] ×
  Subscript[y, i] × Subscript[z, i] > 0]], " ,  ",
  TraditionalForm[HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]],
  " ,  ", TraditionalForm[HoldForm[1 + u Subscript[z, i] > 0]]}], 14],
Style[Row[{TraditionalForm[HoldForm[D[Subscript[x, 1], t] =
  -(1 - Subscript[x, 1] t^2) (1 - u Subscript[x, 1] t^2)]],
  " ,  ", TraditionalForm[HoldForm[R == Sqrt[Subscript[x, 1] ×
  D[Subscript[x, 1], t]]]], " ,  ", TraditionalForm[
  HoldForm[Subscript[x, 1] × D[Subscript[x, 1], t] ≥ 0]]}],
14], "", Style["Formulas:", Bold, 14],
Style[Row[{TraditionalForm[HoldForm[cn[v + w] == (cn[v] × cn[w] -
  sn[v] × sn[w] × dn[v] × dn[w]) / (1 - k^2 sn[v]^2 × sn[w]^2)]],
  "  ", Style["the second addition formula", Italic, 13,
  RGBColor[0.10, 0.30, 0.70]]}], 14], Style[
Row[{TraditionalForm[HoldForm[{sn, cn, dn}[-v] == {-sn[v], cn[v], dn[v]}]],
  "  ", Style["the parity formulas", Italic,
  13, RGBColor[0.75, 0.35, 0.05]]}], 14], "",
Style[Row[{"We want to test these formulas  ", Style["*", 14], " :"}],
Bold, 14],

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Style[TraditionalForm[HoldForm[Subscript[W, 2] == Subscript[ε, 2]
  (t Sqrt[u Subscript[x, 1] × Subscript[y, 2] (1 + Subscript[z, 2]) (1 +
    u Subscript[z, 2])] + Subscript[e, 0] × Subscript[e, 2] Sqrt[
    u Subscript[y, 2] × Subscript[z, 2] × D[Subscript[x, 1], t])] /
  (Subscript[y, 2] u (Subscript[z, 2] + Subscript[x, 1] t^2))]], 15],
Style[TraditionalForm[HoldForm[Subscript[W, 1] == Subscript[ε, 1]
  (t Sqrt[u Subscript[x, 1] × Subscript[y, 1] (1 + Subscript[z, 1]) (1 +
    u Subscript[z, 1])] - Subscript[e, 0] × Subscript[e, 1] Sqrt[
    u Subscript[y, 1] × Subscript[z, 1] × D[Subscript[x, 1], t])] /
  (Subscript[y, 1] u (Subscript[z, 1] + Subscript[x, 1] t^2))]], 15],
Style[Row[{TraditionalForm[HoldForm[U == Subscript[ε, 2] × Subscript[ε, 3]
  Sign[u Subscript[z, 2]] (Abs[Subscript[x, 1]] Subscript[h, 232]
  t + Subscript[e, 0] k' Subscript[h, 231] × Subscript[h, 233]
  Sqrt[Subscript[x, 1] × D[Subscript[x, 1], t]]) /
  (Sqrt[Abs[Subscript[x, 1] × Subscript[x, 3]])
  (Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
  Subscript[x, 1] × Subsuperscript[h, 231, 2] t^2))]], "    ",
Style["in cases (a) and (b)", Italic, 13, GrayLevel[0.35]]}], 15],
Style[Row[{TraditionalForm[HoldForm[U == Subscript[ε, 2] ×
  Subscript[ε, 3] Sign[u Subscript[z, 2]] Subscript[e, 0] ×
  Subscript[e, 2] k' Sqrt[Subscript[x, 1] × D[Subscript[x, 1], t]] /
  (k Sqrt[Abs[Subscript[x, 1] × Subscript[x, 3]])
  (1 - Subscript[x, 1] t^2))]], "    ",
Style["in case (c)", Italic, 13, GrayLevel[0.35]]}], 15]],
Spacings → 1.1], Background → RGBColor[0.99, 0.96,
0.88], FrameMargins → 14]]];

(* =====*)
(*PART 3:THE PARAMETER NORMALIZATION*)
(* =====*)
resultsAll = {};
ClearAll[nCheck];
nCheck[sx_] :=
Module[{x1 = sx XX, p1, Dd, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
Dd = -(1 - x1 tt^2) (1 + UU x1 tt^2);
assum = UU > 0 && XX > 0 && Element[tt, Reals];
TimeConstrained[
FullSimplify[p1^2 (1 - x1 tt^2) (1 / (1 + UU) + UU / (1 + UU) x1 tt^2) -
(- (1 / (1 + UU)) x1 Dd / x1^2) == 0, assum], 180, False] == True];
Module[{grid, ok}, grid = nCheck /@ {1, -1};
ok = And @@ grid;
AppendTo[resultsAll, ok];
Print[
Panel[Column[{Row[{Style["The parameter normalization for ", Bold, 15,
  RGBColor[0.75, 0.35, 0.05]], Style[TraditionalForm[HoldForm[u < 0]],
  Bold, 15, RGBColor[0.75, 0.35, 0.05]], Spacer[20], statusChip[ok]}],
Style["Given hypotheses and formulas, in addition to those above:",
  Bold, 13], Style[Row[{TraditionalForm[HoldForm[sn[v]^2 + cn[v]^2 == 1]],

```

```

", " ", TraditionalForm[HoldForm[dn[v]^2 == k'^2 + k^2 cn[v]^2]],
" ", Style["the Pythagorean relations", Italic, 13,
  RGBColor[0.75, 0.35, 0.05]]], 13], Style[Row[{TraditionalForm[
  HoldForm[Subscript[x, 1] × Subsuperscript[p, 1, 2] == 1]], " ", " ",
  TraditionalForm[HoldForm[cn[ς] = t / Subscript[p, 1]]], " ", " ",
  TraditionalForm[HoldForm[cn[ς]^2 == Subscript[x, 1] t^2]]}],
13], "", Style["Verification:", Bold, 13],
Style[Row[{"By ", Style["the Pythagorean relations",
  Italic, 13, RGBColor[0.75, 0.35, 0.05]]], " and ",
  TraditionalForm[HoldForm[k^2 == -u k'^2]], " we have"}], 13],
Style[TraditionalForm[HoldForm[(Subscript[p, 1] × sn[ς] × dn[ς])^2 ==
  Subsuperscript[p, 1, 2] (1 - cn[ς]^2) (k'^2 + k^2 cn[ς]^2) == - (k'^2
  Subscript[x, 1] × D[Subscript[x, 1], t] / Subsuperscript[x, 1, 2]) ==
  (I k' R / Abs[Subscript[x, 1]])^2]], 14],
Style[Row[{"By ", Style["the parity formulas", Italic, 13, RGBColor[
  0.75, 0.35, 0.05]]], " replacing ", TraditionalForm[HoldForm[ς]],
" by ", TraditionalForm[HoldForm[-ς]], " preserves ",
  TraditionalForm[HoldForm[cn[ς]]], " and reverses the sign of ",
  TraditionalForm[HoldForm[sn[ς] × dn[ς]]], " ."}], 13],
Style[Row[{"Thus, we may fix ", Style[TraditionalForm[
  HoldForm[Subscript[p, 1] × sn[ς] × dn[ς] ==
  -Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]],
  RGBColor[0.75, 0.35, 0.05]]], " so that the two values of ",
  TraditionalForm[HoldForm[Subscript[e, 0]]],
" correspond to the two admissible choices of ",
  TraditionalForm[HoldForm[ς]], " ."}], 13],
Style[Row[{"The middle equality of the displayed chain, that is, ",
  TraditionalForm[HoldForm[Subsuperscript[p, 1, 2]
  (1 - cn[ς]^2) (k'^2 + k^2 cn[ς]^2) == - (k'^2 Subscript[x,
  1] × D[Subscript[x, 1], t] / Subsuperscript[x, 1, 2])]],
" , is verified on each sign region:"}], 13],
Grid[{{Style[lblX[1], 12, GrayLevel[0.3]], smallChip[grid[[1]]]},
  {Style[lblX[-1], 12, GrayLevel[0.3]], smallChip[grid[[2]]]}},
  Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.75, 0.35, 0.05], 0.93],
FrameMargins → 12]]];];

(* =====*)
(*PART 4:THE DERIVATION AND ITS VERIFICATION*)
(* =====*)
Module[{grid, regionOK, ok},
  grid = Table[derCheck[w, s], {w, wConfigs}, {s, sTriples}];
  regionOK = And@@@ grid;
  ok = And@@ regionOK;
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["The derivation: the formulas for ", Bold, 15,
      RGBColor[0.10, 0.30, 0.70]]], Style[TraditionalForm[

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```

HoldForm[Subscript[W, 2]]], Bold, 15, RGBColor[0.10, 0.30, 0.70]],
Style[" and ", Bold, 15, RGBColor[0.10, 0.30, 0.70]],
Style[TraditionalForm[HoldForm[Subscript[W, 1]]], Bold,
15, RGBColor[0.10, 0.30, 0.70]], Style[" for ", Bold, 15,
RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[u < 0]],
Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok]]],
Style["Given hypotheses and formulas, in addition to those above:",
Bold, 13], Style[
Row[{"For ", TraditionalForm[HoldForm[l]], " = 1, 2 :"}], Bold, 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[x, 1] ×
Subsuperscript[p, 1, 2] = 1]], ", ", TraditionalForm[
HoldForm[Subscript[y, l] × Subsuperscript[q, l, 2] = 1]], ", ",
TraditionalForm[HoldForm[Subscript[p, l] = Subscript[p, 1]]]}], 13],
Style[TraditionalForm[HoldForm[sn[Subscript[t, l]] =
I Subscript[ε, l] Sqrt[(Subsuperscript[g, l, 2] - 1)
Subsuperscript[p, l, 2] × Subsuperscript[q, l, 2]] /
(k Subscript[p, l] × Subscript[q, l])]], 14],
Style[TraditionalForm[HoldForm[cn[Subscript[t, l]] =
Subscript[ε, l] Sqrt[(Subsuperscript[g, l, 2] - k'^2)
Subsuperscript[p, l, 2] × Subsuperscript[q, l, 2]] /
(k Subscript[p, l] × Subscript[q, l])]], 14],
Style[Row[{TraditionalForm[HoldForm[cn[ς] = t / Subscript[p, 1]]],
", ", TraditionalForm[HoldForm[Subscript[p, 1] × sn[ς] × dn[ς] =
-Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]}], 14],
Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[Derivative[1][e] ×
Subscript[t, l]] = {Derivative[1][e] × sn[Subscript[t, l]],
cn[Subscript[t, l]], dn[Subscript[t, l]]}],
" ", Style["by the parity formulas", Italic, 13,
RGBColor[0.75, 0.35, 0.05]]}], 14], Style[Row[{TraditionalForm[
HoldForm[sn[Subscript[t, l]]^2 = (1 - Subsuperscript[g, l, 2]) / k^2 =
-((1 - u) / (u (1 + Subscript[z, l])))], ", ",
TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, l]]^2 × sn[ς]^2 =
(Subscript[z, l] + Subscript[x, 1] t^2) / (1 + Subscript[z, l]) ≠ 0]],
", by the choice of ", TraditionalForm[HoldForm[t]]}],
14], "", Style["Verification:", Bold, 13],
Style[Row[{"Since ", TraditionalForm[HoldForm[± Subscript[t, l]]],
" and ", TraditionalForm[HoldForm[ς]],
" are not poles and ", TraditionalForm[
HoldForm[1 - k^2 sn[Subscript[t, l]]^2 × sn[ς]^2 ≠ 0]], ", ",
Style["the second addition formula", Italic, 13, RGBColor[0.10,
0.30, 0.70]], ", at ", TraditionalForm[HoldForm[v = ς]], ", ",
TraditionalForm[HoldForm[w = Derivative[1][e] × Subscript[t, l]]],
", holds pointwise and, together with ",
Style["the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]],
", gives"}], 13], Style[TraditionalForm[
HoldForm[Subscript[q, l] × cn[ς + Derivative[1][e] × Subscript[t, l]] =
((1 + Subscript[z, l]) Subscript[q, l] × cn[Subscript[t, l]] × cn[ς] -

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Derivative[1][e] (1 + Subscript[z, l]) Subscript[q, l] ×
  sn[Subscript[t, l]] × dn[Subscript[t, l]] × sn[ϕ] × dn[ϕ] /
(Subscript[z, l] + Subscript[x, 1] t^2)], 14],
Style["Using the given formulas, we get", 13],
Style[Row[{TraditionalForm[HoldForm[
  Subscript[q, l] × cn[ϕ + Derivative[1][e] × Subscript[t, l]] =
  Subscript[ε, l] (t Sqrt[u Subscript[x, 1] × Subscript[y, l]
    (1 + Subscript[z, l]) (1 + u Subscript[z, l])] -
  Derivative[1][e] × Subscript[e, 0] Sqrt[u Subscript[
    y, l] × Subscript[z, l] × D[Subscript[x, 1], t]]) /
  (Subscript[y, l] u (Subscript[z, l] + Subscript[x, 1] t^2))]],
  " ", Style["★", 15, RGBColor[0.10, 0.30, 0.70]]], 14],
Style[Row[{"Setting ", TraditionalForm[HoldForm[l = 2]], " ",
  TraditionalForm[HoldForm[Derivative[1][e] = -Subscript[e, 2]]],
  " in ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]]],
  " gives the first formula of ", Style["★", 13], " ."}], 13],
Style[Row[{"Setting ", TraditionalForm[HoldForm[l = 1]], " ",
  TraditionalForm[HoldForm[Derivative[1][e] = Subscript[e, 1]]],
  " in ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]]],
  " gives the second formula of ", Style["★", 13], " ."}], 13], "",
Style[Row[{"The identity ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]],
  " is verified separately on each sign region:"}], 13],
Grid[Table[{Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{Style["★", 12, RGBColor[0.10, 0.30, 0.70]],
    " for all 8 sign combinations of ",
    TraditionalForm[HoldForm[Derivative[1][e]]], " ",
    TraditionalForm[HoldForm[Subscript[e, 0]]], " ", TraditionalForm[
      HoldForm[Subscript[ε, l]], " ∈ {±1} → 0 == 0"}],
    12, GrayLevel[0.3]], smallChip[regionOK[[iw]]], {iw, 4}],
  Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.10, 0.30, 0.70], 0.93],
FrameMargins →
12]]];];

(* =====*)
(*PART 5:U-FORMULA IN CASES (a) AND (b)*)
(* =====*)
ClearAll[X3, RR, h1, h2, h3, ee0, ep2, ep3,
  kk, kp, w3Configs, thCheck, lblX3, lblY2, lblXi, lblUZ];
kk = Sqrt[UU] / Sqrt[1 + UU]; kp = 1 / Sqrt[1 + UU];
lblX3[1] = TraditionalForm[HoldForm[Subscript[x, 3] > 0]];
lblX3[-1] = TraditionalForm[HoldForm[Subscript[x, 3] < 0]];
lblY2[1] = TraditionalForm[HoldForm[Subscript[y, 2] > 0]];
lblY2[-1] = TraditionalForm[HoldForm[Subscript[y, 2] < 0]];
lblXi[1] = TraditionalForm[HoldForm[Subscript[ξ, 23] = 1]];
lblXi[-1] = TraditionalForm[HoldForm[Subscript[ξ, 23] = -1]];
lblUZ[1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] = 1]];
lblUZ[-1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] = -1]];

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w3Configs =
  Table[{Row[{lblX[sx], " ", " ", lblX3[s3], " ", " ", lblY2[sy], " ", hence " ",
    lblXi[sx s3], " ", " ", lblUZ[sx sy]]}, sx, s3, sy],
    {sx, {1, -1}}, {s3, {1, -1}}, {sy, {1, -1}}];
w3Configs = Flatten[w3Configs, 2];
thCheck[{lab_, sx_, s3_, sy_}] :=
  Module[{x1 = sx XX, xi = sx s3, suz = sx sy, p1, p3, q2, S23, denom,
    lhs, rhs, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
  p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
  q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
  S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
  denom = h3^2 - xi kk^2 x1 h1^2 tt^2;
  lhs = p3 (S23 h2 (tt / p1) + I S23 h1 h3 (-ee0 I kp RR / (XX p1))) / denom;
  rhs = ep2 ep3 suz (XX h2 tt + ee0 kp h1 h3 RR) / (Sqrt[XX X3] denom);
  assum = XX > 0 && X3 > 0 && YY > 0 && UU > 0 && RR ≥ 0 &&
    Element[tt | h1 | h2 | h3 | ee0 | ep2 | ep3, Reals] && denom ≠ 0;
  TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] == True];
Module[{grid, ok}, grid = thCheck /@ w3Configs;
ok = And @@ grid;
AppendTo[resultsAll, ok];
Print[
  Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
    RGBColor[0.55, 0.25, 0.70]], Style[TraditionalForm[HoldForm[U]],
    Bold, 15, RGBColor[0.55, 0.25, 0.70]], Style[
    " in cases (a) and (b) for ", Bold, 15, RGBColor[0.55, 0.25, 0.70]],
    Style[TraditionalForm[HoldForm[u < 0]], Bold, 15,
    RGBColor[0.55, 0.25, 0.70]], Spacer[20], statusChip[ok]]},
  Style["Given hypotheses and formulas, in addition to those above:",
  Bold, 13], Style[Row[{TraditionalForm[HoldForm[Subscript[x, 3] ×
    Subsuperscript[p, 3, 2] == 1]], " ", " ", TraditionalForm[
    HoldForm[Subscript[y, 2] × Subsuperscript[q, 2, 2] == 1]], " ", " ",
    TraditionalForm[HoldForm[Subscript[p, 2] = Subscript[p, 1]]],
    " ", " ", TraditionalForm[HoldForm[Subscript[q, 3] = Subscript[q, 2]]],
    " ", " ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    " ", " ", TraditionalForm[HoldForm[Subscript[h, 232]]], " ", " ",
    TraditionalForm[HoldForm[Subscript[h, 233]]], " ∈ ℝ"}], 13],
  Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, 23] =
    Sign[Subscript[x, 1] × Subscript[x, 3]]],
    " ", " ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] =
    Sign[Subscript[x, 1] × Subscript[y, 2]]]]}], 13],
  Style[Row[{TraditionalForm[HoldForm[Subscript[S, 23] = Subscript[ε, 2] ×
    Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2] × Subsuperscript[
    p, 3, 2] × Subsuperscript[q, 2, 4] × Subscript[ξ, 23]] /
    (Subscript[p, 1] × Subscript[p, 3] × Subsuperscript[q, 2, 2])]],
    " ", " ", TraditionalForm[HoldForm[Subsuperscript[S, 23, 2] =
    Subscript[ξ, 23]]}], 13],
  Style[TraditionalForm[HoldForm[{sn, cn, dn}[Subscript[e, 2] ×

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Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]] ==
{I Subscript[S, 23] × Subscript[h, 231], Subscript[S, 23] ×
Subscript[h, 232], Subscript[h, 233]]}], 13],
Style[Row[{TraditionalForm[HoldForm[sn[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]] ==
I Subscript[S, 23] × Subscript[h, 231]]], ", ",
TraditionalForm[HoldForm[sn[Subscript[e, 2] × Subscript[t, 2] +
Subscript[e, 3] × Subscript[t, 3]]^2 ==
-Subscript[ξ, 23] × Subsuperscript[h, 231, 2]]], ", ",
TraditionalForm[HoldForm[sn[Subscript[e, 2] × Subscript[t, 2] +
Subscript[e, 3] × Subscript[t, 3]] × dn[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]] ==
I Subscript[S, 23] × Subscript[h, 231] × Subscript[h, 233]]],
", ", TraditionalForm[HoldForm[cn[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]] ==
Subscript[S, 23] × Subscript[h, 232]]]]}], 13],
Style[Row[{TraditionalForm[HoldForm[Subsuperscript[h, 233, 2] -
Subscript[ξ, 23] k^2 Subsuperscript[h, 231, 2] == 1]], " (by ",
TraditionalForm[HoldForm[dn[v]^2 + k^2 sn[v]^2 == 1]], " at ",
TraditionalForm[HoldForm[v == Subscript[e, 2] × Subscript[t, 2] +
Subscript[e, 3] × Subscript[t, 3]]], " )"]}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[x, 1] ×
Subsuperscript[p, 1, 2] == 1]], ", ",
TraditionalForm[HoldForm[cn[ς] == t / Subscript[p, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[p, 1] × sn[ς] × dn[ς] ==
-Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]]}], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ς]^2 == 1 - cn[ς]^2 ==
1 - Subscript[x, 1] t^2]], ", ",
TraditionalForm[HoldForm[1 - k^2 sn[ς]^2 × sn[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]]^2 ==
Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
Subscript[x, 1] × Subsuperscript[h, 231, 2] t^2 ≠ 0]],
", by the choice of ", TraditionalForm[HoldForm[t]]}], 13],
"", Style["Verification:", Bold, 13], Style[
Row[{"Since ", TraditionalForm[HoldForm[± (Subscript[e, 2] × Subscript[
t, 2] + Subscript[e, 3] × Subscript[t, 3])]], " and ",
TraditionalForm[HoldForm[ς]], " are not poles and ",
TraditionalForm[HoldForm[1 - k^2 sn[ς]^2 × sn[Subscript[e, 2] ×
Subscript[t, 2] + Subscript[e, 3] × Subscript[t, 3]]^2 ≠ 0]],
", ", Style["the second addition formula", Italic, 13,
RGBColor[0.10, 0.30, 0.70]],
", at ", TraditionalForm[HoldForm[v = ς]], ", ",
TraditionalForm[HoldForm[w = -Subscript[e, 2] × Subscript[t, 2] -
Subscript[e, 3] × Subscript[t, 3]]],
", holds pointwise and, together with ", Style[
"the parity formulas", Italic, 13,
RGBColor[0.75, 0.35, 0.05]], ", gives"}], 13],

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Style[TraditionalForm[HoldForm[Subscript[p, 3] × cn[ϕ - Subscript[e, 2] ×
    Subscript[t, 2] - Subscript[e, 3] × Subscript[t, 3]] ==
    (Subscript[p, 3] (cn[Subscript[e, 2] × Subscript[t, 2] +
        Subscript[e, 3] × Subscript[t, 3]] × cn[ϕ] +
        sn[Subscript[e, 2] × Subscript[t, 2] + Subscript[e, 3] ×
            Subscript[t, 3]] × dn[Subscript[e, 2] × Subscript[t, 2] +
            Subscript[e, 3] × Subscript[t, 3]] × sn[ϕ] × dn[ϕ])) /
    (Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
        Subscript[x, 1] × Subsuperscript[h, 231, 2] t^2)], 14],
Style["Using the given hypotheses and formulas, we get", 13],
Style[
    Row[{TraditionalForm[HoldForm[Subscript[p, 3] × cn[ϕ - Subscript[e, 2] ×
        Subscript[t, 2] - Subscript[e, 3] × Subscript[t, 3]] ==
        Subscript[ε, 2] × Subscript[ε, 3] Sign[u Subscript[z, 2]]
        (Abs[Subscript[x, 1]] Subscript[h, 232] t +
            Subscript[e, 0] k' Subscript[h, 231] × Subscript[h, 233]
            Sqrt[Subscript[x, 1] × D[Subscript[x, 1], t]]) /
        (Sqrt[Abs[Subscript[x, 1] × Subscript[x, 3]])
        (Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
            Subscript[x, 1] × Subsuperscript[h, 231, 2] t^2))]],
        " ", Style["♦", 14, RGBColor[0.55, 0.25, 0.70]]], 14],
Style[Row[{"which is the third formula of ", Style["*", 13], " ."}],
    13], "",
Style[Row[{"The identity ", Style["♦", 13, RGBColor[0.55, 0.25, 0.70]],
    " is verified separately on each sign region, with ",
    TraditionalForm[HoldForm[Subscript[h, 231]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 233]]],
    ", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
    ", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
    ", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
    " symbolic, that is, for all their real values at once,
    which covers all their sign choices:"}], 13],
Style[Row[{"In case (a), both values of ", TraditionalForm[
    HoldForm[Subscript[ξ, 23]]], " occur, because the sign of ",
    TraditionalForm[HoldForm[Subscript[z, 2] × Subscript[z, 3]]],
    " is not constrained; in case (b), ",
    TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
    ", because ", TraditionalForm[HoldForm[k^2 ==
        (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
    ", ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
        - (1 / (1 + Subscript[z, i]))]], ", and ", TraditionalForm[
        HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]], ", so ",
        TraditionalForm[HoldForm[Subscript[z, 2] × Subscript[z, 3] > 0]],
        " . The eight sign regions cover both cases."}],
    12, GrayLevel[0.35]], Grid[Table[
    {Style[w3Configs[[iw, 1]], 12, GrayLevel[0.3]], Style[

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Row[{Style["♦", 12, RGBColor[0.55, 0.25, 0.70]], " → 0 == 0"}],
12, GrayLevel[0.3]], smallChip[grid[[iw]]], {iw, 8}],
Alignment → Left, Spacings → {2, 0.7}]], Spacings → 1],
Background → Lighter[RGBColor[0.55, 0.25, 0.70],
0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 6:U-FORMULA IN CASE (c) FOR u<0*)
(* =====*)
ClearAll[cCheck, e0s, e2s];
cCheck[{lab_, sx_, s3_, sy_}] :=
Module[{x1 = sx XX, xi = sx s3, suz = sx sy, p1, p3, q2, S23, lhs, rhs, assum},
p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
lhs = S23 e2s I p3 (-e0s I kp RR / (XX p1)) / (kk (1 - x1 tt^2));
rhs = suz ep2 ep3 e0s e2s kp RR / (kk Sqrt[XX X3] (1 - x1 tt^2));
assum = XX > 0 && X3 > 0 && YY > 0 && UU > 0 &&
Element[tt | RR | e0s | e2s | ep2 | ep3, Reals] && 1 - x1 tt^2 ≠ 0;
TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] == True];
Module[{cConfigs, grid, ok}, cConfigs = Select[w3Configs, #[[2]] × #[[3]] == 1 &];
grid = cCheck /@ cConfigs;
ok = And@@grid;
AppendTo[resultsAll, ok];
Print[
Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[U]], Bold,
15, RGBColor[0.00, 0.50, 0.50]], Style[" in case (c) for ", Bold, 15,
RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u < 0]],
Bold, 15, RGBColor[0.00, 0.50, 0.50]], Spacer[20], statusChip[ok]}],
Style["Given hypotheses and formulas, in addition to those above:",
Bold, 13], Style[Row[{TraditionalForm[HoldForm[
Subscript[ξ, 23] == Sign[Subscript[x, 1] × Subscript[x, 3]]]],
", ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
Sign[Subscript[x, 1] × Subscript[y, 2]]]]]], 13],
Style[TraditionalForm[HoldForm[Subscript[S, 23] == Subscript[ε, 2] ×
Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2] × Subsuperscript[p, 3,
2] × Subsuperscript[q, 2, 4] × Subscript[ξ, 23]] / (Subscript[
p, 1] × Subscript[p, 3] × Subsuperscript[q, 2, 2])]], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[t, 2] + Subscript[t, 3] ==
2 K m + I K']], ", ", TraditionalForm[HoldForm[m ∈ ℤ]], ", ",
TraditionalForm[HoldForm[(-1)^m == Subscript[S, 23]]], ", ",
TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]]], 13],
Style[Row[{TraditionalForm[HoldForm[cn[v + 2 m K] == (-1)^m cn[v]]],
", ", TraditionalForm[
HoldForm[cn[v ≠ I K'] == ±I dn[v] / (k sn[v])]], ", ",
Style["the shift formulas", Italic, 13, RGBColor[0.00, 0.50, 0.50]]}],

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13], Style[Row[{TraditionalForm[HoldForm[
    Subscript[x, 1]  $\times$  Subsuperscript[p, 1, 2] == 1]], ", ",
    TraditionalForm[HoldForm[cn[ $\varsigma$ ] = t / Subscript[p, 1]]], ", ",
    TraditionalForm[HoldForm[Subscript[p, 1]  $\times$  sn[ $\varsigma$ ]  $\times$  dn[ $\varsigma$ ] =
        -Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ $\varsigma$ ] ^2 == 1 - cn[ $\varsigma$ ] ^2 ==
    1 - Subscript[x, 1] t^2]], ", ",
    TraditionalForm[HoldForm[1 - Subscript[x, 1] t^2  $\neq$  0]],
    ", by the choice of ", TraditionalForm[HoldForm[t]]], 13], ""],
Style["Verification:", Bold, 13], Style[Row[{"Since ", TraditionalForm[
    HoldForm[Subscript[e, 2] == Subscript[e, 3]]], " and ",
    TraditionalForm[HoldForm[(-1) ^ (m Subscript[e, 2]) == (-1) ^ m]],
    ", ", Style["the shift formulas", Italic, 13,
        RGBColor[0.00, 0.50, 0.50]], " give"}], 13],
Style[TraditionalForm[HoldForm[Subscript[p, 3]  $\times$ 
    cn[ $\varsigma$  - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
    (-1) ^ m Subscript[e, 2] I Subscript[p, 3]  $\times$  dn[ $\varsigma$ ] / (k sn[ $\varsigma$ )] ==
    Subscript[S, 23]  $\times$  Subscript[e, 2] I
    Subscript[p, 3]  $\times$  sn[ $\varsigma$ ]  $\times$  dn[ $\varsigma$ ] / (k sn[ $\varsigma$ ] ^2)]], 14],
Style["Using the given hypotheses and formulas, we get", 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3]  $\times$ 
    cn[ $\varsigma$  - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
    Subscript[ $\varepsilon$ , 2]  $\times$  Subscript[ $\varepsilon$ , 3] Sign[u Subscript[z, 2]]
    Subscript[e, 0]  $\times$  Subscript[e, 2] k' Sqrt[Subscript[x, 1]  $\times$ 
         $\mathcal{D}$ [Subscript[x, 1], t]] / (k Sqrt[Abs[Subscript[x, 1]  $\times$ 
            Subscript[x, 3]] (1 - Subscript[x, 1] t^2))]],
    " ", Style["▲", 14, RGBColor[0.00, 0.50, 0.50]]], 14],
Style[Row[{"which is the fourth formula of ", Style["*", 13], " ."}],
    13], ""],
Style[Row[{"The identity ", Style["▲", 13, RGBColor[0.00, 0.50, 0.50]],
    " is verified separately on each of the four sign
    regions realizable in case (c), with ",
    TraditionalForm[HoldForm[Subscript[e, 0]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
    TraditionalForm[HoldForm[Subscript[ $\varepsilon$ , 2]]], ", ", TraditionalForm[
    HoldForm[Subscript[ $\varepsilon$ , 3]]], ", ", TraditionalForm[HoldForm[R]],
    " symbolic, that is, for all their real values at once,
    which covers all their sign choices:"}], 13],
Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ $\xi$ , 23] == 1]],
    ", because ", TraditionalForm[HoldForm[k^2 ==
    (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
    ", ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
    - (1 / (1 + Subscript[z, i]))]], ", and ", TraditionalForm[
    HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]], ", so ",
    TraditionalForm[HoldForm[Subscript[z, 2]  $\times$  Subscript[z, 3] > 0]],
    " ."}], 12, GrayLevel[0.35]],
Grid[Table[{Style[cConfigs[iw, 1]], 12, GrayLevel[0.3]], Style[

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Row[{Style["▲", 12, RGBColor[0.00, 0.50, 0.50]], " → 0 == 0"}],
12, GrayLevel[0.3]], smallChip[grid[[iw]]], {iw, 4}],
Alignment → Left, Spacings → {2, 0.7}]], Spacings → 1],
Background → Lighter[RGBColor[0.00, 0.50, 0.50], 0.93],
FrameMargins →
12]]];];
(* =====*)
(*PART 7:SUMMARY*)
(* =====*)
Print[Framed[Style[If[And@@resultsAll, "CASE 2: ALL CLAIMS VERIFIED",
"SOME CLAIMS FAILED"], Bold, White, 16], Background →
If[And@@resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]]];

```

Given hypotheses, Case 2 ($u < 0$) :

$$u < 0, \quad k' = \frac{1}{\sqrt{1-u}}, \quad k = k' \sqrt{-u}, \quad k, k' \in (0, 1)$$

For $i = 1, 2, 3, 4$:

$$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0, \quad g_i = \operatorname{dn}(t_i) = \frac{1}{\sqrt{1+\frac{1}{z_i}}}$$

$$p_i, q_i \in \mathbb{R}_{>0} \cup i\mathbb{R}_{>0}, \quad e_0, e', e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$$

$$u x_i y_i z_i > 0, \quad z_i (1 + z_i) > 0, \quad 1 + u z_i > 0$$

$$\mathcal{D}(x_1, t) = -(1 - x_1 t^2)(1 - u x_1 t^2), \quad R = \sqrt{x_1 \mathcal{D}(x_1, t)}, \quad x_1 \mathcal{D}(x_1, t) \geq 0$$

Formulas:

$$\operatorname{cn}(v + w) = \frac{\operatorname{cn}(v) \operatorname{cn}(w) - \operatorname{sn}(v) \operatorname{sn}(w) \operatorname{dn}(v) \operatorname{dn}(w)}{1 - k^2 \operatorname{sn}(v)^2 \operatorname{sn}(w)^2} \quad \text{the second addition formula}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(-v) = \{-\operatorname{sn}(v), \operatorname{cn}(v), \operatorname{dn}(v)\} \quad \text{the parity formulas}$$

We want to test these formulas ★ :

$$W_2 = \frac{\varepsilon_2 (t \sqrt{u x_1 y_2 (1+z_2)(1+u z_2)} + e_0 e_2 \sqrt{u y_2 z_2 \mathcal{D}(x_1, t)})}{y_2 u (z_2 + x_1 t^2)}$$

$$W_1 = \frac{\varepsilon_1 (t \sqrt{u x_1 y_1 (1+z_1)(1+u z_1)} - e_0 e_1 \sqrt{u y_1 z_1 \mathcal{D}(x_1, t)})}{y_1 u (z_1 + x_1 t^2)}$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{|x_1 x_3|} (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)} \quad \text{in cases (a) and (b)}$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)}}{k \sqrt{|x_1 x_3|} (1 - x_1 t^2)} \quad \text{in case (c)}$$

The parameter normalization for $u < 0$

Verified

Given hypotheses and formulas, in addition to those above:

$$\operatorname{sn}(v)^2 + \operatorname{cn}(v)^2 = 1, \quad \operatorname{dn}(v)^2 = (k')^2 + k^2 \operatorname{cn}(v)^2 \quad \text{the Pythagorean relations}$$

$$x_1 p_1^2 = 1, \quad \operatorname{cn}(\varsigma) = \frac{t}{p_1}, \quad \operatorname{cn}(\varsigma)^2 = x_1 t^2$$

Verification:

By *the Pythagorean relations* and $k^2 = -u (k')^2$, we have

$$(p_1 \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma))^2 = p_1^2 (1 - \operatorname{cn}(\varsigma)^2) ((k')^2 + k^2 \operatorname{cn}(\varsigma)^2) = -\frac{(k')^2 x_1 \mathcal{D}(x_1, t)}{x_1^2} = \left(\frac{i k' R}{|x_1|} \right)^2$$

By *the parity formulas*, replacing ς by $-\varsigma$ preserves $\operatorname{cn}(\varsigma)$ and reverses the sign of $\operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma)$.

Thus, we may fix $p_1 \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma) = -\frac{e_0 i k' R}{|x_1|}$, so that the two values of e_0 correspond to the two admissible choices of ς .

The middle equality of the displayed chain, that is,

$$p_1^2 (1 - \operatorname{cn}(\varsigma)^2) ((k')^2 + k^2 \operatorname{cn}(\varsigma)^2) = -\frac{(k')^2 x_1 \mathcal{D}(x_1, t)}{x_1^2}, \quad \text{is verified on each sign region:}$$

$$x_1 > 0 \quad \checkmark$$

$$x_1 < 0 \quad \checkmark$$

The derivation: the formulas for W_2 and W_1 for $u < 0$

Verified

Given hypotheses and formulas, in addition to those above:

For $l = 1, 2$:

$$x_1 p_1^2 = 1, \quad y_l q_l^2 = 1, \quad p_l = p_1$$

$$\operatorname{sn}(t_l) = \frac{i \varepsilon_l \sqrt{(g_l^2 - 1) p_l^2 q_l^2}}{k p_l q_l}$$

$$\operatorname{cn}(t_l) = \frac{\varepsilon_l \sqrt{(g_l^2 - (k')^2) p_l^2 q_l^2}}{k p_l q_l}$$

$$\operatorname{cn}(\zeta) = \frac{t}{p_1}, \quad p_1 \operatorname{sn}(\zeta) \operatorname{dn}(\zeta) = -\frac{e_0 i k' R}{|x_1|}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(e' t_l) = \{e' \operatorname{sn}(t_l), \operatorname{cn}(t_l), \operatorname{dn}(t_l)\} \quad \text{by the parity formulas}$$

$$\operatorname{sn}(t_l)^2 = \frac{1 - g_l^2}{k^2} = -\frac{1 - u}{u(1 + z_l)}, \quad 1 - k^2 \operatorname{sn}(t_l)^2 \operatorname{sn}(\zeta)^2 = \frac{z_l + x_1 t^2}{1 + z_l} \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $\pm t_l$ and ζ are not poles and $1 - k^2 \operatorname{sn}(t_l)^2 \operatorname{sn}(\zeta)^2 \neq 0$, the second addition formula, at $v = \zeta$, $w = e' t_l$, holds pointwise and, together with the parity formulas, gives

$$q_l \operatorname{cn}(\zeta + e' t_l) = \frac{(1 + z_l) q_l \operatorname{cn}(t_l) \operatorname{cn}(\zeta) - e' (1 + z_l) q_l \operatorname{sn}(t_l) \operatorname{dn}(t_l) \operatorname{sn}(\zeta) \operatorname{dn}(\zeta)}{z_l + x_1 t^2}$$

Using the given formulas, we get

$$q_l \operatorname{cn}(\zeta + e' t_l) = \frac{\varepsilon_l (t \sqrt{u x_1 y_l (1 + z_l) (1 + u z_l)} - e' e_0 \sqrt{u y_l z_l} \mathcal{D}(x_1, t))}{y_l u (z_l + x_1 t^2)} \quad \star$$

Setting $l = 2$, $e' = -e_2$ in $\star$ gives the first formula of $\star$.

Setting $l = 1$, $e' = e_1$ in $\star$ gives the second formula of $\star$.

The identity $\star$ is verified separately on each sign region:

$z_l > 0, y_l > 0$, hence $x_1 < 0$	$\star$ for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\} \rightarrow 0 == 0$	✓
$z_l > 0, y_l < 0$, hence $x_1 > 0$	$\star$ for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\} \rightarrow 0 == 0$	✓
$z_l < -1, y_l > 0$, hence $x_1 > 0$	$\star$ for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\} \rightarrow 0 == 0$	✓
$z_l < -1, y_l < 0$, hence $x_1 < 0$	$\star$ for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\} \rightarrow 0 == 0$	✓

The derivation: the formula for U in cases (a) and (b) for $u < 0$

Verified

Given hypotheses and formulas, in addition to those above:

$$x_3 p_3^2 = 1, \quad y_2 q_2^2 = 1, \quad p_2 = p_1, \quad q_3 = q_2, \quad h_{231}, h_{232}, h_{233} \in \mathbb{R}$$

$$\xi_{23} = \operatorname{sgn}(x_1 x_3), \quad \operatorname{sgn}(u z_2) = \operatorname{sgn}(x_1 y_2)$$

$$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{p_1^2 p_3^2 q_2^4 \xi_{23}}}{p_1 p_3 q_2^2}, \quad S_{23}^2 = \xi_{23}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(e_2 t_2 + e_3 t_3) = \{i S_{23} h_{231}, S_{23} h_{232}, h_{233}\}$$

$$\operatorname{sn}(e_2 t_2 + e_3 t_3) = i S_{23} h_{231}, \quad \operatorname{sn}(e_2 t_2 + e_3 t_3)^2 = -\xi_{23} h_{231}^2,$$

$$\operatorname{sn}(e_2 t_2 + e_3 t_3) \operatorname{dn}(e_2 t_2 + e_3 t_3) = i S_{23} h_{231} h_{233}, \quad \operatorname{cn}(e_2 t_2 + e_3 t_3) = S_{23} h_{232}$$

$$h_{233}^2 - \xi_{23} k^2 h_{231}^2 = 1 \quad (\text{by } \operatorname{dn}(v)^2 + k^2 \operatorname{sn}(v)^2 = 1 \text{ at } v = e_2 t_2 + e_3 t_3)$$

$$x_1 p_1^2 = 1, \quad \operatorname{cn}(\varsigma) = \frac{t}{p_1}, \quad p_1 \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma) = -\frac{e_0 i k' R}{|x_1|}$$

$$\operatorname{sn}(\varsigma)^2 = 1 - \operatorname{cn}(\varsigma)^2 = 1 - x_1 t^2,$$

$$1 - k^2 \operatorname{sn}(\varsigma)^2 \operatorname{sn}(e_2 t_2 + e_3 t_3)^2 = h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2 \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $\pm(e_2 t_2 + e_3 t_3)$ and ς are not poles and $1 - k^2 \operatorname{sn}(\varsigma)^2 \operatorname{sn}(e_2 t_2 + e_3 t_3)^2 \neq 0$,
the second addition formula, at $v = \varsigma$, $w = -e_2 t_2 - e_3 t_3$

, holds pointwise and, together with the parity formulas, gives

$$p_3 \operatorname{cn}(\varsigma - e_2 t_2 - e_3 t_3) = \frac{p_3 (\operatorname{cn}(e_2 t_2 + e_3 t_3) \operatorname{cn}(\varsigma) + \operatorname{sn}(e_2 t_2 + e_3 t_3) \operatorname{dn}(e_2 t_2 + e_3 t_3) \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma))}{h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2}$$

Using the given hypotheses and formulas, we get

$$p_3 \operatorname{cn}(\varsigma - e_2 t_2 - e_3 t_3) = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{|x_1 x_3|} (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)} \quad \blacklozenge$$

which is the third formula of $\star$.

The identity $\blacklozenge$ is verified separately on each sign region, with $h_{231}, h_{232}, h_{233}, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

In case (a), both values of ξ_{23} occur, because the sign of $z_2 z_3$

is not constrained; in case (b), $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$,

$g_i^2 - 1 = -\frac{1}{1+z_i}$, and $z_i(1+z_i) > 0$, so $z_2 z_3 > 0$. The eight sign regions cover both cases.

$x_1 > 0, x_3 > 0, y_2 > 0$, hence $\xi_{23} = 1, \operatorname{sgn}(u z_2) = 1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 > 0, x_3 > 0, y_2 < 0$, hence $\xi_{23} = 1, \operatorname{sgn}(u z_2) = -1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 > 0, x_3 < 0, y_2 > 0$, hence $\xi_{23} = -1, \operatorname{sgn}(u z_2) = 1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 > 0, x_3 < 0, y_2 < 0$, hence $\xi_{23} = -1, \operatorname{sgn}(u z_2) = -1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 < 0, x_3 > 0, y_2 > 0$, hence $\xi_{23} = -1, \operatorname{sgn}(u z_2) = -1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 < 0, x_3 > 0, y_2 < 0$, hence $\xi_{23} = -1, \operatorname{sgn}(u z_2) = 1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 < 0, x_3 < 0, y_2 > 0$, hence $\xi_{23} = 1, \operatorname{sgn}(u z_2) = -1$	$\blacklozenge \rightarrow 0 == 0$	✓
$x_1 < 0, x_3 < 0, y_2 < 0$, hence $\xi_{23} = 1, \operatorname{sgn}(u z_2) = 1$	$\blacklozenge \rightarrow 0 == 0$	✓

The derivation: the formula for U in case (c) for $u < 0$ **Verified****Given hypotheses and formulas, in addition to those above:**

$$\xi_{23} = \operatorname{sgn}(x_1 x_3), \quad \operatorname{sgn}(u z_2) = \operatorname{sgn}(x_1 y_2)$$

$$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{p_1^2 p_3^2 q_2^4 \xi_{23}}}{p_1 p_3 q_2^2}$$

$$t_2 + t_3 = 2 K m + i K', \quad m \in \mathbb{Z}, \quad (-1)^m = S_{23}, \quad e_2 = e_3$$

$$\operatorname{cn}(v + 2 m K) = (-1)^m \operatorname{cn}(v), \quad \operatorname{cn}(v \mp i K') = \frac{\pm i \operatorname{dn}(v)}{k \operatorname{sn}(v)} \quad \text{the shift formulas}$$

$$x_1 p_1^2 = 1, \quad \operatorname{cn}(\varsigma) = \frac{t}{p_1}, \quad p_1 \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma) = -\frac{e_0 i k' R}{|x_1|}$$

$$\operatorname{sn}(\varsigma)^2 = 1 - \operatorname{cn}(\varsigma)^2 = 1 - x_1 t^2, \quad 1 - x_1 t^2 \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $e_2 = e_3$ and $(-1)^{m e_2} = (-1)^m$, *the shift formulas* give

$$p_3 \operatorname{cn}(\varsigma - e_2 (t_2 + t_3)) = \frac{(-1)^m e_2 i p_3 \operatorname{dn}(\varsigma)}{k \operatorname{sn}(\varsigma)} = \frac{S_{23} e_2 i p_3 \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma)}{k \operatorname{sn}(\varsigma)^2}$$

Using the given hypotheses and formulas, we get

$$p_3 \operatorname{cn}(\varsigma - e_2 (t_2 + t_3)) = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)}}{k \sqrt{|x_1 x_3|} (1 - x_1 t^2)} \quad \blacktriangle$$

which is the fourth formula of $\star$.

The identity $\blacktriangle$

is verified separately on each of the four sign regions realizable in case (c), with

$e_0, e_2, \varepsilon_2, \varepsilon_3, R$

symbolic, that is, for all their real values at once, which covers all their sign choices:

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$, $g_i^2 - 1 = -\frac{1}{1+z_i}$, and $z_i(1+z_i) > 0$, so $z_2 z_3 > 0$.

$$x_1 > 0, \quad x_3 > 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = 1 \quad \blacktriangle \rightarrow 0 == 0 \quad \checkmark$$

$$x_1 > 0, \quad x_3 > 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = -1 \quad \blacktriangle \rightarrow 0 == 0 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = -1 \quad \blacktriangle \rightarrow 0 == 0 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = 1 \quad \blacktriangle \rightarrow 0 == 0 \quad \checkmark$$

CASE 2: ALL CLAIMS VERIFIED

Section 8

Results of Section 8 are used in the proof of Theorem 2

In[1783]:=

```
(* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[UU, XX, X3, Z2, Z3, tt, RR, ee0, ep2, ep3, H1s, H2s, H3s, sX, statusChip,
```

```

smallChip, colGiven, colTest, lblZ2, lblZ3, lblX, zwin, wsign, mkA, mkAB,
mkAP, chkR, bOK, gdict, p6h, p6ok, p7h, p7ok, HAat, HBat, uLemPosat,
uLemNegat, uLemPosCat, uLemNegCat, uThmABat, uThmCat, resultsAll];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
colGiven = RGBColor[0.85, 0.45, 0.00];
colTest = RGBColor[0.10, 0.35, 0.75];
(*Every check below is a rational identity in a fixed set of radical atoms,
decided by Together;
no simplification search is involved.Sign regions:each of z2,
z3 lies in one of the two windows z>0 or z<-1,by z(1+z)>0;
we write x1=sx XX with XX>0,and the sign of x3 is forced by Sign[x1 x3]=Sign[
  z2 z3],the sign of y2 by Sign[y2]=Sign[u x1 z2].On each region the atoms
  sw2=Sqrt[|z2|],sv2=Sqrt[|1+z2|],sb2=Sqrt[1+u z2],etc.are real principal
  roots,because their radicands are positive there by the window facts;
each de-nesting step used below is certified by squaring,
which is a rational identity,and both sides are nonnegative
by construction.On each locus the parametrization is certified
by checking the corresponding guard equality rationally.*)
lblZ2[1] = TraditionalForm[HoldForm[Subscript[z, 2] > 0]];
lblZ2[2] = TraditionalForm[HoldForm[Subscript[z, 2] < -1]];
lblZ3[1] = TraditionalForm[HoldForm[Subscript[z, 3] > 0]];
lblZ3[2] = TraditionalForm[HoldForm[Subscript[z, 3] < -1]];
lblX[1] = TraditionalForm[HoldForm[Subscript[x, 1] > 0]];
lblX[-1] = TraditionalForm[HoldForm[Subscript[x, 1] < 0]];
zwin[1, Z_] := Z; zwin[2, Z_] := -1 - Z;
wsign[1] = 1; wsign[2] = -1;
sX = Sqrt[XX X3];
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
mkA[w2_, w3_, uv_] :=
  Module[{z2v = zwin[w2, Z2], z3v = zwin[w3, Z3], s2 = wsign[w2],
    s3 = wsign[w3]}, <|"z2" → z2v, "z3" → z3v, "s2" → s2, "s3" → s3, "xi" → s2 s3,
    "az2" → s2 z2v, "az3" → s3 z3v, "a12" → s2 (1 + z2v), "a13" → s3 (1 + z3v),
    "sw2" → Sqrt[s2 z2v], "sw3" → Sqrt[s3 z3v], "sv2" → Sqrt[s2 (1 + z2v)],
    "sv3" → Sqrt[s3 (1 + z3v)], "sb2" → Sqrt[1 + uv z2v], "sb3" → Sqrt[1 + uv z3v] |>];
mkAB[w2_, w3_] := Module[{at = mkA[w2, w3, 0]}, at["sb2"] = at["sv3"] / at["sw3"];
  at["sb3"] = at["sv2"] / at["sw2"]; at];
mkAP[w_] := Module[{at = mkA[w, w, 0], sS},
  sS = Sqrt[at["z2"] + at["z3"] + at["z2"] × at["z3"]];
  at["sS"] = sS; at["sb2"] = at["sw3"] × at["sv2"] / sS;
  at["sb3"] = at["sw2"] × at["sv3"] / sS; at];

```

```

bOK[at_, uv_] :=
  chkR[at["sb2"]^2, 1 + uv at["z2"]] && chkR[at["sb3"]^2, 1 + uv at["z3"]];
gdict[at_, 1] := {at["sv2"] / at["sw2"], at["sv3"] / at["sw3"],
  (1 + at["z2"]) / at["z2"], (1 + at["z3"]) / at["z3"]};
gdict[at_, 2] := {at["sw2"] / at["sv2"], at["sw3"] / at["sv3"],
  at["z2"] / (1 + at["z2"]), at["z3"] / (1 + at["z3"])};
p6h[at_, kk2_, mS_, e2_, e3_, reg_] :=
  Module[{g2, g3, sr1, sr2, sr3, sr4, sr5, d, xi = at["xi"], W},
    {g2, g3} = gdict[at, reg][[1 ;; 2]];
    If[reg == 1, W = at["sw2"] × at["sw3"];
      sr1 = at["sb3"] / W; sr2 = at["sb2"] / W;
      sr3 = at["sb2"] × at["sb3"] / W; sr4 = 1 / W;
      sr5 = at["sb2"] × at["sb3"] / (at["az2"] × at["az3"]),
      W = at["sv2"] × at["sv3"];
      sr1 = at["sb3"] / (mS W); sr2 = at["sb2"] / (mS W);
      sr3 = at["sb2"] × at["sb3"] / (mS^2 W); sr4 = 1 / W;
      sr5 = at["sb2"] × at["sb3"] / (mS^2 at["a12"] × at["a13"])]];
    d = kk2 - (g2^2 - 1) (g3^2 - 1);
    {(e2 g3 sr1 + e3 g2 sr2) / d,
      (sr3 + e2 e3 g2 g3 sr4) / d, (kk2 g2 g3 + e2 e3 xi sr5) / d}];
p6ok[at_, kk2_, kp2_, mS_, reg_] :=
  Module[{g2, g3, d2, d3, sr1, sr2, sr3, sr4, sr5, xi = at["xi"], W},
    {g2, g3, d2, d3} = gdict[at, reg];
    If[reg == 1, W = at["sw2"] × at["sw3"];
      sr1 = at["sb3"] / W; sr2 = at["sb2"] / W;
      sr3 = at["sb2"] × at["sb3"] / W; sr4 = 1 / W;
      sr5 = at["sb2"] × at["sb3"] / (at["az2"] × at["az3"]),
      W = at["sv2"] × at["sv3"];
      sr1 = at["sb3"] / (mS W); sr2 = at["sb2"] / (mS W);
      sr3 = at["sb2"] × at["sb3"] / (mS^2 W); sr4 = 1 / W;
      sr5 = at["sb2"] × at["sb3"] / (mS^2 at["a12"] × at["a13"])]];
    chkR[g2^2, d2] && chkR[g3^2, d3] && chkR[(g2^2 - 1) (g3^2 - kp2) xi, sr1^2] &&
      chkR[(g3^2 - 1) (g2^2 - kp2) xi, sr2^2] && chkR[(g2^2 - kp2) (g3^2 - kp2) xi,
        sr3^2] && chkR[(g2^2 - 1) (g3^2 - 1) xi, sr4^2] &&
      chkR[(g2^2 - 1) (g2^2 - kp2) (g3^2 - 1) (g3^2 - kp2), sr5^2];
p7h[at_, kkAt_, kp2_, e2_, reg_] :=
  Module[{g2, g3}, {g2, g3} = gdict[at, reg][[1 ;; 2]];
    {e2 (g2^2 - g3^2) / (2 kkAt g2 g3),
      (g2^2 g3^2 - kp2) / (2 kkAt g2 g3), (g2^2 + g3^2) / (2 g2 g3)}];
p7ok[at_, kkAt_, kk2_, reg_] :=
  Module[{g2, g3, d2, d3}, {g2, g3, d2, d3} = gdict[at, reg];
    chkR[g2^2, d2] && chkR[g3^2, d3] && chkR[kkAt^2, kk2];
HAat[at_, uv_, e2_, e3_] :=
  Module[{d = uv at["z2"] × at["z3"] - 1, sA2 = at["sw3"] × at["sv3"] × at["sb3"],
    sA3 = at["sw2"] × at["sv2"] × at["sb2"], sB = at["sb2"] × at["sb3"],
    sC = at["sw2"] × at["sv2"] × at["sw3"] × at["sv3"]},
    {(at["s2"] × at["z2"] sA2 + e2 e3 at["s3"] × at["z3"] sA3) / d,

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(at["xi"] × at["z2"] × at["z3"] sB + e2 e3 sC) / d,
(uv at["xi"] sC + e2 e3 sB) / d});
HBat[at_] := Module[{q = at["z2"] × at["z3"],
  s = at["z2"] + at["z3"], sC = at["sw2"] × at["sv2"] × at["sw3"] × at["sv3"]},
  {q (at["z3"] - at["z2"]) / (2 sC), q (s + 2) / (2 sC), (s + 2 q) at["xi"] / (2 sC)}];
uLemPosat[{h1_, h2_, h3_}, kk2_, xi_, sx_, suz_] :=
  ep2 ep3 suz (XX h2 h3 tt + ee0 h1 RR) / (sX (1 + xi kk2 sx XX h1^2 tt^2));
uLemNegat[{h1_, h2_, h3_}, kk2_, kpAt_, xi_, sx_, suz_] :=
  ep2 ep3 suz (XX h2 tt + ee0 kpAt h1 h3 RR) / (sX (h3^2 - xi kk2 sx XX h1^2 tt^2));
uLemPosCat[kkAt_, sx_, suz_] := ep2 ep3 suz sx / (sX kkAt tt);
uLemNegCat[kkAt_, kpAt_, sx_, suz_, e2_] :=
  ep2 ep3 suz ee0 e2 kpAt RR / (kkAt sX (1 - sx XX tt^2));
uThmABat[{H1_, H2_, H3_}, uv_, at_, sx_, e2_] :=
  ep2 ep3 at["s2"] (at["sw2"] × at["sw3"] / sX)
  (XX H2 H3 tt + ee0 e2 H1 RR) / (at["z2"] × at["z3"] + uv sx XX H1^2 tt^2);
uThmCat[at_, sy2_] := ep2 ep3 (at["sw2"] × at["sw3"] / sX) sy2 / tt;
resultsAll = {};
(* =====*)
(*PART 2:THE GIVEN HYPOTHESES AND THE FORMULAS TO TEST*)
(* =====*)
ClearAll[braceGrp];
braceGrp[items_List, sz_ : 13, col_ : GrayLevel[0]] := DisplayForm[
  RowBox[{GridBox[({ToBoxes[Style[#, sz, col]]} & /@ items), GridBoxAlignment →
    {"Columns" → {{Left}}}, GridBoxSpacings → {"Rows" → {{1.8}}},
    StyleBox["", FontColor → GrayLevel[0.55], SpanMaxSize → Infinity]}]];
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
  Style[Row[{TraditionalForm[HoldForm[u ≠ 0]], " ",
    TraditionalForm[HoldForm[u < 1]], " ", TraditionalForm[HoldForm[k]],
    " ", TraditionalForm[HoldForm[k']]], " ∈ (0, 1)"}], 14],
  Style[Row[{"For ", TraditionalForm[HoldForm[i]], " = 1, 2, 3, 4 :"}],
  Bold, 14], Style[Row[{TraditionalForm[HoldForm[Subscript[x, i]]],
    " ", TraditionalForm[HoldForm[Subscript[y, i]]], " ",
    TraditionalForm[HoldForm[Subscript[z, i]]], " ∈ ℝ",
    " ", TraditionalForm[HoldForm[Subscript[x, i]]],
    " ", TraditionalForm[HoldForm[Subscript[y, i]]], " ",
    TraditionalForm[HoldForm[Subscript[z, i]]], " ≠ -1, 0"}], 14],
  Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, 23] = Sign[Subscript[z,
    2] × Subscript[z, 3]] = Sign[Subscript[x, 1] × Subscript[x, 3]]}],
    " ", TraditionalForm[HoldForm[Sign[Subscript[y, 2]] =
    Sign[u Subscript[x, 1] × Subscript[z, 2]]}], 14],
  Style[Row[{TraditionalForm[HoldForm[u Subscript[x, i] ×
    Subscript[y, i] × Subscript[z, i] > 0]], " ",
    TraditionalForm[HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]],
    " ", TraditionalForm[HoldForm[1 + u Subscript[z, i] > 0]]}], 14],
  Style[Row[{TraditionalForm[HoldForm[Subscript[e, 0]]], " ",
    TraditionalForm[HoldForm[Subscript[e, i]]], " ",
    TraditionalForm[HoldForm[Subscript[ε, i]]], " ",

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TraditionalForm[HoldForm[Subscript[ξ, 23]]], " ∈ {±1}"]], 14],
Style[Row[{TraditionalForm[HoldForm[ $\partial$ [Subscript[x, 1], t] ==
- (1 - Subscript[x, 1] t^2) (1 - u Subscript[x, 1] t^2)]],
", ", TraditionalForm[HoldForm[R == Sqrt[Subscript[x, 1] ×
 $\partial$ [Subscript[x, 1], t]]]], ", ", TraditionalForm[
HoldForm[Subscript[x, 1] ×  $\partial$ [Subscript[x, 1], t] ≥ 0]]]], 14], ""],
Style[Row[{"(i) if ", TraditionalForm[HoldForm[u > 0]], " : ",
TraditionalForm[HoldForm[k = Sqrt[u]]], ", ", TraditionalForm[
HoldForm[k' = Sqrt[1 - u]]], ", ", TraditionalForm[HoldForm[
Subscript[g, i] = Sqrt[(1 + Subscript[z, i]) / Subscript[z, i]]]],
", ", TraditionalForm[HoldForm[k^2 - (Subsuperscript[g, 2, 2] - 1)
(Subsuperscript[g, 3, 2] - 1) = (u Subscript[z, 2] × Subscript[z, 3] -
1) / (Subscript[z, 2] × Subscript[z, 3])]]]], 14], ""],
Style[Row[{"(ii) if ", TraditionalForm[HoldForm[u < 0]], " : ",
TraditionalForm[HoldForm[k' = 1 / Sqrt[1 - u]]], ", ", TraditionalForm[
HoldForm[k = k' Sqrt[-u]]], ", ", TraditionalForm[HoldForm[
Subscript[g, i] = Sqrt[Subscript[z, i] / (1 + Subscript[z, i])]]],
", ", TraditionalForm[HoldForm[k^2 - (Subsuperscript[g, 2, 2] - 1)
(Subsuperscript[g, 3, 2] - 1) = - ((1 + u (Subscript[z, 2] +
Subscript[z, 3] + Subscript[z, 2] × Subscript[z, 3])) /
((1 - u) (1 + Subscript[z, 2]) (1 + Subscript[z, 3])))]]], 14],
""], Style[Row[{TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
", ", TraditionalForm[HoldForm[Subscript[h, 233]]],
", ", TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]], " ∈ ℝ"}], 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, 231] = 0]],
" if and only if ", TraditionalForm[
HoldForm[Subscript[g, 2] = Subscript[g, 3]]], " and ",
TraditionalForm[HoldForm[Subscript[e, 2] = -Subscript[e, 3]]], " ; ",
TraditionalForm[HoldForm[Subscript[h, 232] = 0]], " if and only if ",
TraditionalForm[HoldForm[Subscript[g, 2] × Subscript[g, 3] = k']],
" and ", TraditionalForm[
HoldForm[Subscript[e, 2] = -Subscript[e, 3]]], " ;"}], 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, 233] = 0]],
" if and only if ", TraditionalForm[
HoldForm[k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) =
k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)]],
" and ", TraditionalForm[HoldForm[Subscript[e, 2] =
Subscript[e, 3]]], " ; in particular, ",
TraditionalForm[HoldForm[Subscript[h, 233] ≠ 0]], " if ",
TraditionalForm[HoldForm[k^2 = (Subsuperscript[g, 2, 2] - 1)
(Subsuperscript[g, 3, 2] - 1)]], " ."}], 14],
""], Style[Row[{"the equalities ", TraditionalForm[
HoldForm[u Subscript[z, 2] × Subscript[z, 3] = 1]], " and ",
TraditionalForm[HoldForm[1 + u (Subscript[z, 2] + Subscript[z,

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3] + Subscript[z, 2] × Subscript[z, 3]) == 0]], " ",
Style["cannot hold simultaneously", Bold, 14], " ."}], 14],
"", Style["Given formulas and expressions:", Bold, 14],
Style[TraditionalForm[HoldForm[
U == HoldForm[Subscript[ε, 2] × Subscript[ε, 3] Sign[u Subscript[z, 2]] /
Sqrt[Abs[Subscript[x, 1] × Subscript[x, 3]]]] Piecewise[
{{(Abs[Subscript[x, 1]] Subscript[h, 232] × Subscript[h, 233] t +
Subscript[e, 0] × Subscript[h, 231] Sqrt[Subscript[x, 1] ×
D[Subscript[x, 1], t]]) / (1 + Subscript[ξ, 23] k^2
Subscript[x, 1] × Subsuperscript[h, 231, 2] t^2), u > 0},
{(Abs[Subscript[x, 1]] Subscript[h, 232] t + Subscript[e, 0]
k' Subscript[h, 231] × Subscript[h, 233] Sqrt[Subscript[x, 1] ×
D[Subscript[x, 1], t]]) / (Subsuperscript[h, 233, 2] -
Subscript[ξ, 23] k^2 Subscript[x, 1] × Subsuperscript[
h, 231, 2] t^2), u < 0}}]], 15, Bold, colGiven], "",
Grid[{{braceGrp[{TraditionalForm[HoldForm[Subscript[h, 231] ==
(Subscript[e, 2] × Subscript[g, 3] Sqrt[(Subsuperscript[g, 2, 2] -
1) (Subsuperscript[g, 3, 2] - Superscript[k', 2])
Subscript[ξ, 23]] + Subscript[e, 3] × Subscript[g, 2]
Sqrt[(Subsuperscript[g, 3, 2] - 1) (Subsuperscript[g, 2, 2] -
Superscript[k', 2]) Subscript[ξ, 23]]) / (k^2 -
(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))]],
TraditionalForm[HoldForm[Subscript[h, 232] ==
(Sqrt[(Subsuperscript[g, 2, 2] - Superscript[k', 2])
(Subsuperscript[g, 3, 2] - Superscript[k', 2]) Subscript[ξ,
23]] + Subscript[e, 2] × Subscript[e, 3] × Subscript[g, 2] ×
Subscript[g, 3] Sqrt[(Subsuperscript[g, 2, 2] - 1)
(Subsuperscript[g, 3, 2] - 1) Subscript[ξ, 23]]) / (k^2 -
(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))]],
TraditionalForm[HoldForm[Subscript[h, 233] == (k^2 Subscript[g, 2] ×
Subscript[g, 3] + Subscript[e, 2] × Subscript[e, 3] × Subscript[
ξ, 23] Sqrt[(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g,
2, 2] - Superscript[k', 2]) (Subsuperscript[g, 3, 2] - 1)
(Subsuperscript[g, 3, 2] - Superscript[k', 2]))] / (k^2 -
(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))]],
13, colGiven], Row[{Style["(a) ", Bold, 13], Style[
TraditionalForm[HoldForm[k^2 ≠ (Subsuperscript[g, 2, 2] - 1)
(Subsuperscript[g, 3, 2] - 1)], 13]}]}],
Alignment → {Left, Center}, Spacings → {1.5, 0}], "", Grid[
{{braceGrp[Row[{TraditionalForm[HoldForm[Subscript[h, 231] == Subscript[
e, 2] (Subsuperscript[g, 2, 2] - Subsuperscript[g, 3, 2]) /
(2 k Subscript[g, 2] × Subscript[g, 3])]],
", ", TraditionalForm[HoldForm[Subscript[h, 232] ==
(Subsuperscript[g, 2, 2] × Subsuperscript[g, 3, 2] - Superscript[
k', 2]) / (2 k Subscript[g, 2] × Subscript[g, 3])]],
", ", TraditionalForm[HoldForm[Subscript[h, 233] ==
(Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) /

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      (2 Subscript[g, 2] × Subscript[g, 3]]]]], 13, colGiven],
Row[{Style["(b)   ", Bold, 13], Style[TraditionalForm[HoldForm[
      k^2 = (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
13], Style["   ", 13], Style[TraditionalForm[
      HoldForm[Subscript[e, 2] = -Subscript[e, 3]], 13]]}],
Alignment → {Left, Center}, Spacings → {1.5, 0}], ""],
Grid[{{braceGrp[{Style[TraditionalForm[HoldForm[U = HoldForm[
      Subscript[ε, 2] × Subscript[ε, 3] Sign[u Subscript[z, 2]] /
      Sqrt[Abs[Subscript[x, 1] × Subscript[x, 3]]]] Piecewise[
      {{Sign[Subscript[x, 1]] / (k t), u > 0}, {Subscript[e, 0] ×
      Subscript[e, 2] k' Sqrt[Subscript[x, 1] × Subscript[x, 1],
      t]] / (k (1 - Subscript[x, 1] t^2)), u < 0}]]]], Bold]],
14, colGiven], Row[{Style["(c)   ", Bold, 13], Style[
      TraditionalForm[HoldForm[
      k^2 = (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
13], Style["   ", 13], Style[TraditionalForm[
      HoldForm[Subscript[e, 2] = Subscript[e, 3]], 13]]}],
Alignment → {Left, Center}, Spacings → {1.5, 0}], ""],
Style[Row[{"We want to test these formulas   ",
      Style["★", 14], " :"}], Bold, 14],
Style[Row[{"We show that the formulas for   ",
      TraditionalForm[HoldForm[U]],
      " below follow from the given hypotheses, formulas, and
      expressions above,"}], 12, GrayLevel[0.35]],
Style[Row[{"together with the following claim:   in each case,
      the denominator of the given formula and the denominator
      of the tested formula vanish for the same values of   ",
      TraditionalForm[HoldForm[t]], " ,   that is, one of them is
      zero if and only if the other is; hence the compared
      formulas are defined for exactly the same   ",
      TraditionalForm[HoldForm[t]], " ."}], 12, GrayLevel[0.35]],
"", Style[TraditionalForm[HoldForm[U = HoldForm[Subscript[ε, 2] ×
      Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2] ×
      Subscript[z, 3] / (Subscript[x, 1] × Subscript[x, 3])]]
      ((Abs[Subscript[x, 1]] Subscript[H, 2] × Subscript[H, 3] t +
      Subscript[e, 0] × Subscript[e, 2] × Subscript[H, 1] Sqrt[Subscript[
      x, 1] × Subscript[x, 1], t]]) / (Subscript[z, 2] × Subscript[
      z, 3] + u Subscript[x, 1] × Subsuperscript[H, 1, 2] t^2))]],
15, Bold, colTest], "", Grid[{{braceGrp[{TraditionalForm[
      HoldForm[Subscript[H, 1] = (Abs[Subscript[z, 2]] Sqrt[
      Subscript[z, 3] (1 + Subscript[z, 3]) (1 + u Subscript[z, 3])] +
      Subscript[e, 2] × Subscript[e, 3] Abs[Subscript[z, 3]] Sqrt[
      Subscript[z, 2] (1 + Subscript[z, 2]) (1 + u Subscript[z, 2])]] /
      (u Subscript[z, 2] × Subscript[z, 3] - 1)]], TraditionalForm[
      HoldForm[Subscript[H, 2] = (Abs[Subscript[z, 2] × Subscript[z, 3]]
      Sqrt[(1 + u Subscript[z, 2]) (1 + u Subscript[z, 3])] +
      Subscript[e, 2] × Subscript[e, 3] Sqrt[Subscript[z, 2] ×

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Subscript[z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3])) /
(u Subscript[z, 2] × Subscript[z, 3] - 1)]], TraditionalForm[
HoldForm[Subscript[H, 3] = (u Sqrt[Subscript[z, 2] × Subscript[z, 3]
(1 + Subscript[z, 2]) (1 + Subscript[z, 3])) Sign[Subscript[z,
2] × Subscript[z, 3]] + Subscript[e, 2] × Subscript[e, 3]
Sqrt[(1 + u Subscript[z, 2]) (1 + u Subscript[z, 3]))] /
(u Subscript[z, 2] × Subscript[z, 3] - 1)]], 13, colTest],
Row[{Style["(a) ", Bold, 13], Style[TraditionalForm[
HoldForm[u Subscript[z, 2] × Subscript[z, 3] ≠ 1]], 13]}]},
Alignment → {Left, Center}, Spacings → {1.5, 0}], "",
Grid[{{braceGrp[TraditionalForm[HoldForm[
Subscript[H, 1] = Subscript[z, 2] × Subscript[z, 3] (Subscript[z, 3] -
Subscript[z, 2]) / (2 Sqrt[Subscript[z, 2] × Subscript[z, 3]
(1 + Subscript[z, 2]) (1 + Subscript[z, 3]))]],
TraditionalForm[HoldForm[Subscript[H, 2] = Subscript[z, 2] ×
Subscript[z, 3] (Subscript[z, 2] + Subscript[z, 3] + 2) /
(2 Sqrt[Subscript[z, 2] × Subscript[z, 3] (1 + Subscript[z, 2])
(1 + Subscript[z, 3]))]], TraditionalForm[
HoldForm[Subscript[H, 3] = (Subscript[z, 2] + Subscript[z, 3] +
2 Subscript[z, 2] × Subscript[z, 3]) Sign[Subscript[z, 2] ×
Subscript[z, 3]] / (2 Sqrt[Subscript[z, 2] × Subscript[z, 3]
(1 + Subscript[z, 2]) (1 + Subscript[z, 3]))]]],
13, colTest], Row[{Style["(b) ", Bold, 13], Style[
TraditionalForm[HoldForm[u Subscript[z, 2] × Subscript[z, 3] = 1]],
13], Style["", 13], Style[TraditionalForm[
HoldForm[Subscript[e, 2] = -Subscript[e, 3]], 13]}]},
Alignment → {Left, Center}, Spacings → {1.5, 0}], "",
Grid[{{braceGrp[Style[TraditionalForm[HoldForm[
U = HoldForm[Subscript[ε, 2] × Subscript[ε, 3] Sqrt[Subscript[z,
2] × Subscript[z, 3] / (Subscript[x, 1] × Subscript[x, 3])] ×
HoldForm[Sign[Subscript[y, 2]] / t]], Bold]], 14, colTest],
Row[{Style["(c) ", Bold, 13], Style[TraditionalForm[
HoldForm[u Subscript[z, 2] × Subscript[z, 3] = 1]], 13],
Style["", 13], Style[TraditionalForm[
HoldForm[Subscript[e, 2] = Subscript[e, 3]], 13]}]},
Alignment → {Left, Center}, Spacings → {1.5, 0}],
Spacings → 1.1], Background →
RGBColor[
0.99,
0.96,
0.88],
FrameMargins → 14]];
(* =====*)
(*PART 3:CASE (a) FOR u>0*)
(* =====*)
ClearAll[regA, aCheckPos, aPreCk];
regA = Flatten[

```

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Table[{Row[{lblZ2[w2], " ", lblZ3[w3], " ", lblX[sx]}], w2, w3, sx},
  {w2, {1, 2}}, {w3, {1, 2}}, {sx, {1, -1}}, 2];
aCheckPos[{lab_, w2_, w3_, sx_}, {e2_, e3_}] := Module[{at = mkA[w2, w3, UU],
  xi, suz, W, h, H, brOK, numCk, denCk, fullCk}, xi = at["xi"];
  suz = at["s2"];
  W = at["sw2"] at["sw3"];
  h = p6h[at, UU, 1, e2, e3, 1]; H = HAat[at, UU, e2, e3];
  brOK = bOK[at, UU] && p6ok[at, UU, 1 - UU, 1, 1] &&
    chkR[h[[1]] W, xi e2 H[[1]] && chkR[h[[2]] W, xi H[[2]] && chkR[h[[3]] H[[3]]];
  numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
    xi W (XX (xi H2s / W) H3s tt + ee0 (xi e2 H1s / W) RR)];
  denCk = chkR[at["z2"] at["z3"] + UU sx XX H1s^2 tt^2,
    at["z2"] at["z3"] (1 + xi UU sx XX (xi e2 H1s / W)^2 tt^2)];
  fullCk = chkR[uLemPosat[{xi e2 H1s / W, xi H2s / W, H3s}, UU, xi, sx, suz],
    uThmABat[{H1s, H2s, H3s}, UU, at, sx, e2]];
  {brOK, numCk && denCk, fullCk}];
aPreCk[{lab_, w2_, w3_, sx_}] :=
  Module[{at = mkA[w2, w3, UU], xi, W}, xi = at["xi"];
  W = at["sw2"] at["sw3"];
  chkR[at["z2"] at["z3"] / (xi W sX), W / sX]];
Module[{grid, brRegion, idRegion, fullRegion, preAll, ok},
  grid = Table[aCheckPos[w, s], {w, regA}, {s, Tuples[{1, -1}, 2]};
  brRegion = Table[And@@grid[[iw, All, 1]], {iw, Length[regA]}];
  idRegion = Table[And@@grid[[iw, All, 2]], {iw, Length[regA]}];
  fullRegion = Table[And@@grid[[iw, All, 3]], {iw, Length[regA]}];
  preAll = And@@(aPreCk /@ regA);
  ok = (And@@brRegion) && (And@@idRegion) && (And@@fullRegion) && preAll;
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
      RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[U]], Bold,
      15, RGBColor[0.10, 0.30, 0.70]], Style[" in case (a) for ", Bold, 15,
      RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[u > 0]],
      Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok]}],
      Style[Row[{"By ", TraditionalForm[HoldForm[
        Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
        Sign[Subscript[x, 1] Subscript[x, 3]]], " : ",
        TraditionalForm[HoldForm[Subscript[ξ, 23] Subscript[z, 2]
          Subscript[z, 3] == Abs[Subscript[z, 2] Subscript[z, 3]]], " ",
        TraditionalForm[HoldForm[Subscript[ξ, 23] Subscript[x, 1]
          Subscript[x, 3] == Abs[Subscript[x, 1] Subscript[x, 3]]], " ."}],
      12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
      Style[Row[{"in this case ", TraditionalForm[HoldForm[
        k^2 ≠ (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
        " , that is, ", TraditionalForm[HoldForm[
          u Subscript[z, 2] Subscript[z, 3] ≠ 1]],
        " , and the following identities hold:"}], 13], Style[

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Row[{TraditionalForm[HoldForm[k == Sqrt[u]]], " ", TraditionalForm[
  HoldForm[Subsuperscript[g, i, 2] - 1 == 1 / Subscript[z, i]], " ",
  TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - Superscript[k', 2] ==
    (1 + u Subscript[z, i]) / Subscript[z, i]]],
  " ", TraditionalForm[HoldForm[Subscript[g, i] ==
    Sqrt[Subscript[z, i] (1 + Subscript[z, i]) / Abs[Subscript[z, i]]]],
  " ", Style["♦", 14, RGBColor[0.55, 0.33, 0.10]]], 13],
Style[Row[{"Hence, substituting the identities ", Style["♦", 13,
  RGBColor[0.55, 0.33, 0.10]], " and the equality ", TraditionalForm[
  HoldForm[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] ==
    Abs[Subscript[z, 2] Subscript[z, 3]]}], " into ", Style[
  Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    " ", TraditionalForm[HoldForm[Subscript[h, 232]]], " ",
    TraditionalForm[HoldForm[Subscript[h, 233]]]}], 13, colGiven],
  " ", we get, with ", Style[Row[{"the case-(a) ",
    TraditionalForm[HoldForm[Subscript[H, 1]]], " ",
    TraditionalForm[HoldForm[Subscript[H, 2]]], " ", TraditionalForm[
      HoldForm[Subscript[H, 3]]}], 13, colTest], " "}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, 231] ==
  Subscript[ξ, 23] Subscript[e, 2] Subscript[H, 1] / Sqrt[Abs[
    Subscript[z, 2] Subscript[z, 3]]]], " ", TraditionalForm[
  HoldForm[Subscript[h, 232] == Subscript[ξ, 23] Subscript[H, 2] /
    Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]]], " ",
  TraditionalForm[HoldForm[Subscript[h, 233] == Subscript[H, 3]]],
  " ", Style["♦", 14, RGBColor[0.00, 0.45, 0.25]]], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for all 4 sign combinations of ",
    TraditionalForm[HoldForm[Subscript[e, 2]]], " ",
    TraditionalForm[HoldForm[Subscript[e, 3]]], " → 0 == 0"}],
    12, GrayLevel[0.3]], smallChip[brRegion[[iw]]],
  {iw, Length[regA]}], Alignment → Left, Spacings → {2, 0.7}],
""], Style[Row[{"Thus, substituting the identities ",
  Style["♦", 13, RGBColor[0.00, 0.45, 0.25]],
  " into the numerator and the denominator of ",
  Style["the given U-formula", Italic, 13, colGiven], " we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
  HoldForm[Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
    Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]] == Subscript[ξ, 23]
    Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]] (Abs[Subscript[x, 1]]
    Subscript[h, 232] Subscript[h, 233] t + Subscript[e, 0]
    Subscript[h, 231] Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]])],
  14], Style["the denominator identity:", 13], Style[
  TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] +
    u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 ==
    Subscript[z, 2] Subscript[z, 3] (1 + Subscript[ξ, 23] k^2

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Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)]], 14],
Style[Row[{"which are verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for all 4 sign combinations of ",
TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[e, 3]]], " → 0 == 0"}],
12, GrayLevel[0.3]], smallChip[idRegion[[iw]]],
{iw, Length[regA]}], Alignment → Left, Spacings → {2, 0.7}],
"", Style["and, finally, the prefactor identity:", 13],
Style[TraditionalForm[
HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
HoldForm[Subscript[z, 2] Subscript[z, 3] / (Subscript[ξ, 23]
Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]])] = Sqrt[Subscript[
z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]], 14],
Style[Row[{"which holds by ", TraditionalForm[HoldForm[
Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] =
Abs[Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] =
Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"}], 13],
Style[Row[{"and is verified separately on each of the 8
sign regions (sign combinations of ",
TraditionalForm[HoldForm[Subscript[z, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[z, 3]]],
", ", TraditionalForm[HoldForm[Subscript[x, 1]]],
" as above) → 0 == 0 ", smallChip[preAll]]], 13], "",
Style[Row[{"Together, the numerator, denominator, and prefactor
identities, along with ", TraditionalForm[
HoldForm[Sign[u Subscript[z, 2]] = Sign[Subscript[z, 2]]],
", , give the formula ", Style["*", 13, colTest],
" in case (a):"}], 13], Style[
Row[{TraditionalForm[HoldForm[U = HoldForm[Subscript[ε, 2] Subscript[ε,
3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2] Subscript[z, 3] /
(Subscript[x, 1] Subscript[x, 3])] ((Abs[Subscript[x, 1]
Subscript[H, 2] Subscript[H, 3] t + Subscript[e, 0]
Subscript[e, 2] Subscript[H, 1] Sqrt[Subscript[x, 1]
D[Subscript[x, 1], t]) / (Subscript[z, 2] Subscript[z, 3] +
u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2))]],
" , Style["*", 14]]], 14, colTest], "", Style[
Row[{"The identity ", Style["*", 13, colTest],
" is also verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]],
", ", TraditionalForm[HoldForm[Subscript[e, 0]]],

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", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
" symbolic, that is, for all their real values at once,
  which covers all their sign choices:"]], 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]], Style[
  Row[{"for all 4 sign combinations of ",
    TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, 3]]], " → 0 == 0"}], 12,
    GrayLevel[0.3]], smallChip[fullRegion[[iw]]], {iw, Length[regA]}],
  Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.10, 0.30, 0.70],
  0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 4:CASES (b) AND (c) FOR u>0,ON u z2 z3=1*)
(* =====*)
ClearAll[regBpos, bCheckPos, bPreCk, cCheckPos];
regBpos =
  Flatten[Table[{Row[{lblZ2[w], ", ", lblZ3[w], ", ", lblX[sx]}], w, sx},
    {w, {1, 2}}, {sx, {1, -1}}], 1];
bCheckPos[{lab_, w_, sx_}, e2_] :=
  Module[{at = mkAB[w, w], uv, kkAt, W, d2, d3, h, H, brOK, numCk, denCk, fullCk},
    uv = 1 / (at["z2"] at["z3"]);
    W = at["sw2"] at["sw3"]; kkAt = 1 / W;
    {d2, d3} = gdict[at, 1][[3 ;; 4]];
    h = p7h[at, kkAt, 1 - uv, e2, 1]; H = Hbat[at];
    brOK = bOK[at, uv] && p7ok[at, kkAt, uv, 1] && chkR[kkAt^2, (d2 - 1) (d3 - 1)] &&
      chkR[h[[1]] W, e2 H[[1]]] && chkR[h[[2]] W, H[[2]]] && chkR[h[[3]] W, H[[3]]];
    numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
      W (XX (H2s / W) H3s tt + ee0 (e2 H1s / W) RR)];
    denCk = chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
      at["z2"] at["z3"] (1 + sx XX (e2 H1s / W)^2 kkAt^2 tt^2)];
    fullCk = chkR[uLemPosat[{e2 H1s / W, H2s / W, H3s}, uv, 1, sx, at["s2"]],
      uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]];
    {brOK, numCk && denCk, fullCk}];
bPreCk[{lab_, w_, sx_}] :=
  Module[{at = mkAB[w, w], W}, W = at["sw2"] at["sw3"];
    chkR[at["z2"] at["z3"] / (W sx), W / sx]];
cCheckPos[{lab_, w_, sx_}] :=
  Module[{at = mkAB[w, w], uv, kkAt, d2, d3}, uv = 1 / (at["z2"] at["z3"]);
    kkAt = 1 / (at["sw2"] at["sw3"]);
    {d2, d3} = gdict[at, 1][[3 ;; 4]];
    p7ok[at, kkAt, uv, 1] && chkR[kkAt^2, (d2 - 1) (d3 - 1)] &&
      chkR[uLemPosCat[kkAt, sx, at["s2"]], uThmCat[at, at["s2"] sx]]];
Module[{gridB, brRegion, idRegion, fullRegion, preAll, cRegion, ok},
  gridB = Table[bCheckPos[w, s], {w, regBpos}, {s, {1, -1}}];
  brRegion = Table[And@@gridB[[iw, All, 1]], {iw, Length[regBpos]}];
  idRegion = Table[And@@gridB[[iw, All, 2]], {iw, Length[regBpos]}];

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fullRegion = Table[And@@gridB[[iw, All, 3]], {iw, Length[regBpos]};
preAll = And@@ (bPreCk /@ regBpos);
cRegion = cCheckPos /@ regBpos;
ok = (And@@brRegion) && (And@@idRegion) &&
      (And@@fullRegion) && preAll && (And@@cRegion);
AppendTo[resultsAll, ok];
Print[
  Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
    RGBColor[0.55, 0.25, 0.70]], Style[TraditionalForm[HoldForm[U]],
    Bold, 15, RGBColor[0.55, 0.25, 0.70]], Style[
    " in cases (b) and (c) for ", Bold, 15, RGBColor[0.55, 0.25, 0.70]],
    Style[TraditionalForm[HoldForm[u > 0]], Bold, 15,
    RGBColor[0.55, 0.25, 0.70]], Spacer[20], statusChip[ok]]},
  Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
    ", because ", TraditionalForm[HoldForm[
    k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
    " and ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
    1 / Subscript[z, i]]], ", so ", TraditionalForm[HoldForm[
    Subscript[z, 2] Subscript[z, 3] > 0]], ", and the signs of ",
    TraditionalForm[HoldForm[Subscript[z, 2]]], " and ",
    TraditionalForm[HoldForm[Subscript[z, 3]]], " coincide."}],
  12, GrayLevel[0.35]], Style[Row[{"By ", TraditionalForm[
    HoldForm[Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
    Sign[Subscript[x, 1] Subscript[x, 3]] == 1]], " : ",
    TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] ==
    Abs[Subscript[z, 2] Subscript[z, 3]]], ", ",
    TraditionalForm[HoldForm[Subscript[x, 1] Subscript[x, 3] ==
    Abs[Subscript[x, 1] Subscript[x, 3]]], " ."}],
  12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
  Style[Row[{"in these cases ", TraditionalForm[HoldForm[
    k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)]],
    ", that is, ", TraditionalForm[HoldForm[
    u Subscript[z, 2] Subscript[z, 3] == 1]],
    ", and the following identities hold:"}], 13],
  Style[Row[{TraditionalForm[HoldForm[
    k == 1 / Sqrt[Subscript[z, 2] Subscript[z, 3]]],
    ", ", TraditionalForm[HoldForm[Subscript[g, i] ==
    Sqrt[Subscript[z, i] (1 + Subscript[z, i]) / Abs[Subscript[z, i]]],
    " ", Style["▲", 14, RGBColor[0.55, 0.33, 0.10]]}], 13],
  Style[Row[{"Hence, in case (b) ( ", TraditionalForm[
    HoldForm[Subscript[e, 2] == -Subscript[e, 3]]],
    " ), substituting the identities ",
    Style["▲", 13, RGBColor[0.55, 0.33, 0.10]], " and the equality ",
    TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] ==
    Abs[Subscript[z, 2] Subscript[z, 3]]], " into ", Style[
    Row[{"the case-(b) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",

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TraditionalForm[HoldForm[Subscript[h, 233]]]], 13, colGiven],
" , we get, with ", Style[Row[{"the case-(b) ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ", TraditionalForm[
HoldForm[Subscript[H, 3]]]], 13, colTest], " ,"}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, 231] == Subscript[e, 2]
Subscript[H, 1] / Sqrt[Subscript[z, 2] Subscript[z, 3]]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232] ==
Subscript[H, 2] / Sqrt[Subscript[z, 2] Subscript[z, 3]]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233] == Subscript[H, 3]]],
" ", Style["▲", 14, RGBColor[0.00, 0.45, 0.25]]]], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
" = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
smallChip[brRegion[[iw]]], {iw, Length[regBpos]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"Thus, substituting the identities ",
Style["▲", 13, RGBColor[0.00, 0.45, 0.25]],
" into the numerator and the denominator of ",
Style["the given U-formula", Italic, 13, colGiven], ", we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
HoldForm[Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]] ==
Sqrt[Subscript[z, 2] Subscript[z, 3]] (Abs[Subscript[x, 1]
Subscript[h, 232] Subscript[h, 233] t + Subscript[e, 0] Subscript[
h, 231] Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]])]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
HoldForm[Subscript[z, 2] Subscript[z, 3] + u Subscript[x, 1]
Subsuperscript[H, 1, 2] t^2 == Subscript[z, 2] Subscript[z, 3]
(1 + k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
" = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
smallChip[idRegion[[iw]]], {iw, Length[regBpos]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"and, finally, ", Style["the prefactor identity", Italic,
13, RGBColor[0.80, 0.20, 0.50]], " :"}, 13], Style[TraditionalForm[
HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
HoldForm[Subscript[z, 2] Subscript[z, 3] /
Sqrt[Subscript[z, 2] Subscript[z, 3]]] == Sqrt[Subscript[z, 2]
Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]], 14],

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Style[Row[{"which holds by ", TraditionalForm[HoldForm[Subscript[z, 2]
    Subscript[z, 3] == Abs[Subscript[z, 2] Subscript[z, 3]]],
    " and ", TraditionalForm[HoldForm[Subscript[x, 1] Subscript[x, 3] ==
    Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"}], 13],
Style[Row[{"and is verified separately on each of the 4
    sign regions (sign combinations of ",
    TraditionalForm[HoldForm[Subscript[z, 2]]], " ",
    TraditionalForm[HoldForm[Subscript[z, 3]]],
    " ", TraditionalForm[HoldForm[Subscript[x, 1]]],
    " as above) → 0 == 0 ", smallChip[preAll]]], 13], "",
Style[Row[{"Together, the numerator, denominator, and prefactor
    identities, along with ", TraditionalForm[
    HoldForm[Sign[u Subscript[z, 2]] == Sign[Subscript[z, 2]]],
    " , give the formula ", Style["*", 13, colTest],
    " in case (b):"}], 13], Style[
Row[{TraditionalForm[HoldForm[U == HoldForm[Subscript[ε, 2] Subscript[ε,
    3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2] Subscript[z, 3] /
    (Subscript[x, 1] Subscript[x, 3])] (Abs[Subscript[x, 1]]
    Subscript[H, 2] Subscript[H, 3] t + Subscript[e, 0]
    Subscript[e, 2] Subscript[H, 1] Sqrt[Subscript[x, 1]
    D[Subscript[x, 1], t]) / (Subscript[z, 2] Subscript[z, 3] +
    u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2)]],
    " ", Style["*", 14]]], 14, colTest], "", Style[
Row[{"The identity ", Style["*", 13, colTest],
    " is also verified separately on each sign region, with ",
    TraditionalForm[HoldForm[Subscript[H, 1]]],
    " ", TraditionalForm[HoldForm[Subscript[H, 2]]],
    " ", TraditionalForm[HoldForm[Subscript[H, 3]]],
    " ", TraditionalForm[HoldForm[Subscript[e, 0]]],
    " ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
    " ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
    " symbolic, that is, for all their real values at once,
    which covers all their sign choices:"}], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
    Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
        " = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
    smallChip[fullRegion[[iw]]], {iw, Length[regBpos]}],
    Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"In case (c) ( ", TraditionalForm[
    HoldForm[Subscript[e, 2] == Subscript[e, 3]]],
    " ), the ", Style["given", Italic, 13, colGiven],
    " and ", Style["tested", Italic, 13, colTest],
    " formulas are their replacements:"}], 13],
Style[Row[{Style[TraditionalForm[HoldForm[U ==
    HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]] /
    Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
    HoldForm[Sign[Subscript[x, 1]] / (k t)]]], 14, colGiven],

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" and ", Style[TraditionalForm[HoldForm[
  U == HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sqrt[Subscript[z, 2]
    Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
  HoldForm[Sign[Subscript[y, 2]] / t]], 14, colTest], " ,"], 14],
Style[Row[{"which coincide because ", TraditionalForm[
  HoldForm[Sign[Subscript[y, 2]] ==
    Sign[u Subscript[x, 1] Subscript[z, 2]]]], " ", TraditionalForm[
  HoldForm[k == 1 / Sqrt[Subscript[z, 2] Subscript[z, 3]]]],
" , and ", Style["the prefactor identity", Italic, 13,
  RGBColor[0.80, 0.20, 0.50]], " ; their denominators ",
TraditionalForm[HoldForm[k t]], " and ",
TraditionalForm[HoldForm[t]], " vanish for the same t because ",
TraditionalForm[HoldForm[k ≠ 0]], " ."]], 13],
Style[Row[{"The equality of the ", Style["given", Italic, 13, colGiven],
" and ", Style["tested", Italic, 13, colTest],
" formulas is verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[ε, 2]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"}], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
  Style["→ 0 == 0", 12, GrayLevel[0.3]],
  smallChip[cRegion[[iw]]]}, {iw, Length[regBpos]}],
  Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.55, 0.25, 0.70], 0.93],
FrameMargins →
12]]];];

(* =====*)
(*PART 5:CASE (a) FOR u<0*)
(* =====*)
ClearAll[aCheckNeg, aPreCkNeg, h233ConvCk,
  regZ, regMix, bCheckNegII, bPreCkII, cCheckNegII];
regZ = Flatten[Table[
  {Row[{lblZ2[w2], " ", " ", lblZ3[w3]}], w2, w3}, {w2, {1, 2}}, {w3, {1, 2}}, 1];
regMix = Flatten[Table[{Row[{lblZ2[w2], " ", " ", lblZ3[3 - w2], " ", " ", lblX[sx]}],
  w2, 3 - w2, sx}, {w2, {1, 2}}, {sx, {1, -1}}, 1];
h233ConvCk[{lab_, w2_, w3_}] :=
Module[{at = mkA[w2, w3, -UU], kk2 = UU / (1 + UU), d2, d3},
  {d2, d3} = gdict[at, 2][[3 ;; 4]];
  chkR[kk2 (d2 + d3) - (kk2 - (d2 - 1) (d3 - 1)),
  (1 - (-UU) at["z2"] at["z3"]) / ((1 + UU) (1 + at["z2"]) (1 + at["z3"]))];];
aCheckNeg[{lab_, w2_, w3_, sx_}, {e2_, e3_}] :=
Module[{at = mkA[w2, w3, -UU], uv = -UU, kk2, mS, kpAt, xi, suz,
  W, h, H, brh, brOK, numCk, denCk, fullCk}, kk2 = UU / (1 + UU);
  mS = Sqrt[1 + UU]; kpAt = 1 / mS; xi = at["xi"]; suz = -at["s2"];
  W = at["sw2"] at["sw3"];
  h = p6h[at, kk2, mS, e2, e3, 2]; H = HAat[at, uv, e2, e3];

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brh = {-xi e2 mS H1s / (W H3s), -xi H2s / (W H3s), 1 / H3s};
brOK = bOK[at, uv] && p6ok[at, kk2, 1 / (1 + UU), mS, 2] && chkR[h[[3]] H[[3]], 1] &&
  chkR[h[[1]] W H[[3]], -xi e2 mS H[[1]]] && chkR[h[[2]] W H[[3]], -xi H[[2]]];
numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
  -xi W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)];
denCk = chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
  at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - xi kk2 sx XX brh[[1]]^2 tt^2)];
fullCk = chkR[uLemNegat[brh, kk2, kpAt, xi, sx, suz],
  uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]];
{brOK, numCk && denCk, fullCk}];
aPreCkNeg[{lab_, w2_, w3_, sx_}] :=
Module[{at = mkA[w2, w3, -UU], xi, W}, xi = at["xi"];
W = at["sw2"] at["sw3"];
chkR[at["z2"] at["z3"] / (xi W sX), W / sX]];
bCheckNegII[{lab_, w2_, w3_, sx_}, e2_] :=
Module[{at = mkAB[w2, w3], uv, kk2, mS, kpAt, xi, suz, W, h, H,
  brh, brOK, numCk, denCk, fullCk}, uv = 1 / (at["z2"] at["z3"]);
kk2 = -uv / (1 - uv); mS = Sqrt[1 - uv]; kpAt = 1 / mS; xi = at["xi"];
suz = -at["s2"]; W = at["sw2"] at["sw3"];
h = p6h[at, kk2, mS, e2, -e2, 2]; H = HBat[at];
brh = {-xi e2 mS H1s / (W H3s), -xi H2s / (W H3s), 1 / H3s};
brOK = bOK[at, uv] && p6ok[at, kk2, 1 / (1 - uv), mS, 2] && chkR[h[[3]] H[[3]], 1] &&
  chkR[h[[1]] W H[[3]], -xi e2 mS H[[1]]] && chkR[h[[2]] W H[[3]], -xi H[[2]]];
numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
  -xi W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)];
denCk = chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
  at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - xi kk2 sx XX brh[[1]]^2 tt^2)];
fullCk = chkR[uLemNegat[brh, kk2, kpAt, xi, sx, suz],
  uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]];
{brOK, numCk && denCk, fullCk}];
bPreCkII[{lab_, w2_, w3_, sx_}] :=
Module[{at = mkAB[w2, w3], xi, W}, xi = at["xi"];
W = at["sw2"] at["sw3"];
chkR[at["z2"] at["z3"] / (xi W sX), W / sX]];
cCheckNegII[{lab_, w2_, w3_, sx_}, e2_] :=
Module[{at = mkAB[w2, w3], uv, kk2, mS, kpAt, xi, suz, W, h},
  uv = 1 / (at["z2"] at["z3"]);
  kk2 = -uv / (1 - uv);
  mS = Sqrt[1 - uv];
  kpAt = 1 / mS; xi = at["xi"]; suz = -at["s2"];
  W = at["sw2"] at["sw3"];
  h = p6h[at, kk2, mS, e2, e2, 2];
  bOK[at, uv] && p6ok[at, kk2, 1 / (1 - uv), mS, 2] && chkR[h[[3]], 0] &&
    chkR[h[[1]], e2 mS W] && chkR[h[[2]], W] && chkR[kk2 mS^2 W^2, 1] &&
    chkR[h[[3]]^2 - xi kk2 sx XX h[[1]]^2 tt^2, sx XX tt^2] &&
    chkR[uLemNegat[h, kk2, kpAt, xi, sx, suz], uThmCat[at, suz sx]]];
Module[{grid, brRegion, idRegion, fullRegion, preAll, convAll, gridB,

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brRegionB, idRegionB, fullRegionB, preAllB, gridC, cRegion, ok},
grid = Table[aCheckNeg[w, s], {w, regA}, {s, Tuples[{1, -1}, 2]};
brRegion = Table[And@@grid[[iw, All, 1]], {iw, Length[regA]};
idRegion = Table[And@@grid[[iw, All, 2]], {iw, Length[regA]};
fullRegion = Table[And@@grid[[iw, All, 3]], {iw, Length[regA]};
preAll = And@@ (aPreCkNeg /@ regA);
convAll = And@@ (h233ConvCk /@ regZ);
gridB = Table[bCheckNegII[w, s], {w, regMix}, {s, {1, -1}}];
brRegionB = Table[And@@gridB[[iw, All, 1]], {iw, Length[regMix]};
idRegionB = Table[And@@gridB[[iw, All, 2]], {iw, Length[regMix]};
fullRegionB = Table[And@@gridB[[iw, All, 3]], {iw, Length[regMix]};
preAllB = And@@ (bPreCkII /@ regMix);
gridC = Table[cCheckNegII[w, s], {w, regMix}, {s, {1, -1}}];
cRegion = Table[And@@gridC[[iw]], {iw, Length[regMix]};
ok = (And@@brRegion) && (And@@idRegion) &&
      (And@@fullRegion) && preAll && convAll && (And@@brRegionB) &&
      (And@@idRegionB) && (And@@fullRegionB) && preAllB && (And@@cRegion);
AppendTo[resultsAll, ok];
Print[
Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
  RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u]], Bold,
  15, RGBColor[0.00, 0.50, 0.50]], Style[" in case (a) for ", Bold, 15,
  RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u < 0]],
  Bold, 15, RGBColor[0.00, 0.50, 0.50]], Spacer[20], statusChip[ok]}],
Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[h, 233] == 0]],
" if and only if ", TraditionalForm[
  HoldForm[k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) ==
  k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
" and ", TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[
  e, 3]]], " . When ", TraditionalForm[HoldForm[u < 0]], " ,"}],
12, GrayLevel[0.35]], Style[TraditionalForm[HoldForm[
  k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) -
  (k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)) ==
  (1 - u Subscript[z, 2] Subscript[z, 3]) /
  ((1 - u) (1 + Subscript[z, 2]) (1 + Subscript[z, 3]))]], 12,
GrayLevel[0.35]], Style[Row[{"which is verified separately on
  each of the 4 sign regions (sign combinations of ",
  TraditionalForm[HoldForm[Subscript[z, 2]]], " , ",
  TraditionalForm[HoldForm[Subscript[z, 3]]], " ) → 0 == 0 ",
smallChip[convAll], " , so ", TraditionalForm[
  HoldForm[k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) ==
  k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
" if and only if ", TraditionalForm[HoldForm[u Subscript[z, 2]
  Subscript[z, 3] == 1]], " ."}], 12, GrayLevel[0.35]],
Style[Row[{"Thus, when ", TraditionalForm[HoldForm[u < 0]],
" , ", TraditionalForm[HoldForm[Subscript[h, 233] == 0]],
" if and only if ", TraditionalForm[

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HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], " and ",
TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]],
" . Therefore, in cases (i) and (ii) below, ", TraditionalForm[
HoldForm[Subscript[h, 233] != 0]], " , and the identity ",
TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] == 1]],
" gives ", TraditionalForm[HoldForm[Subscript[H, 3] != 0]],
" , so the divisions by ", TraditionalForm[
HoldForm[Subscript[H, 3]]], " are legitimate there."]],
12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
Style[Row[{"in this case ", TraditionalForm[HoldForm[
k^2 != (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)]},
" , that is, ", TraditionalForm[HoldForm[
1 + u (Subscript[z, 2] + Subscript[z, 3] +
Subscript[z, 2] Subscript[z, 3]) != 0]], " ;"}], 13],
Style[Row[{"we consider three possible cases: (i) ",
TraditionalForm[HoldForm[u Subscript[z, 2] Subscript[z, 3] != 1]],
" , (ii) ", TraditionalForm[
HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], " and ",
TraditionalForm[HoldForm[Subscript[e, 2] == -Subscript[e, 3]]],
" , (iii) ", TraditionalForm[
HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], " and ",
TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]], " ."}],
13], "", Style[Row[{"(i) ", TraditionalForm[
HoldForm[u Subscript[z, 2] Subscript[z, 3] != 1]], " :"}],
Bold, 13], Style["The following identities hold:", 13],
Style[Row[{TraditionalForm[HoldForm[k == Sqrt[u / (u - 1)]]],
" , ", TraditionalForm[HoldForm[k' == 1 / Sqrt[1 - u]]],
" , ", TraditionalForm[HoldForm[
Subsuperscript[g, i, 2] - 1 == -(1 / (1 + Subscript[z, i]))]], " , ",
TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - Superscript[k', 2] ==
-((1 + u Subscript[z, i]) / ((1 - u) (1 + Subscript[z, i])))]],
" , ", TraditionalForm[HoldForm[Subscript[g, i] == Sqrt[
Subscript[z, i] (1 + Subscript[z, i]) / Abs[1 + Subscript[z, i]]]],
" , ", Style["●", 14, RGBColor[0.55, 0.33, 0.10]]}], 13],
Style[Row[{TraditionalForm[HoldForm[u^2 Subscript[z, 2]
Subscript[z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) -
(1 + u Subscript[z, 2]) (1 + u Subscript[z, 3]) ==
(u Subscript[z, 2] Subscript[z, 3] - 1) (1 + u (Subscript[z, 2] +
Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]))]],
" , ", Style["●", 14, RGBColor[0.55, 0.05, 0.10]]}], 13],
Style[Row[{"and, by ", TraditionalForm[HoldForm[
Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
Sign[Subscript[x, 1] Subscript[x, 3]]], " and ",
TraditionalForm[HoldForm[Sign[1 + Subscript[z, i]] ==
Sign[Subscript[z, i]]], " : ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] ==
Abs[Subscript[z, 2] Subscript[z, 3]]], " , ", TraditionalForm[

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HoldForm[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] ==
  Abs[Subscript[x, 1] Subscript[x, 3]]], ", ", TraditionalForm[
  HoldForm[Subscript[ξ, 23] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
    Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]]], " ."]], 13],
Style[Row[{"Hence, substituting the identities ",
  Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
  " and the equalities ", TraditionalForm[
    HoldForm[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] ==
      Abs[Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
        HoldForm[Subscript[ξ, 23] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
          Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]]], " into ", Style[
          Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
            ", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
              TraditionalForm[HoldForm[Subscript[h, 233]]]]], 13, colGiven],
    " , and using the identity ", Style["●", 13,
      RGBColor[0.55, 0.05, 0.10]], " , we get, with ", Style[
        Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[H, 1]]],
          ", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
            ", ", TraditionalForm[HoldForm[Subscript[H, 3]]]]],
          13, colTest], " ,"}], 13], Style[
  Row[{TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] = 1]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 231] =
      -Subscript[ξ, 23] Subscript[e, 2] Sqrt[1 - u] Subscript[H, 1] /
        (Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232] =
      -Subscript[ξ, 23] Subscript[H, 2] /
        (Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
    " ", Style["●", 14, RGBColor[0.00, 0.45, 0.25]]], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for all 4 sign combinations of ",
    TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, 3]]], " → 0 == 0"}],
    12, GrayLevel[0.3]], smallChip[brRegion[[iw]]]],
  {iw, Length[regA]}], Alignment → Left, Spacings → {2, 0.7}],
""], Style[Row[{"Thus, substituting the identities ",
  Style["●", 13, RGBColor[0.00, 0.45, 0.25]],
  " into the numerator and the denominator of ",
  Style["the given U-formula", Italic, 13, colGiven], ", we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
  HoldForm[Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
    Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]] =
    -Subscript[ξ, 23] Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]
    Subsuperscript[H, 3, 2] (Abs[Subscript[x, 1]] Subscript[h, 232] t +
    Subscript[e, 0] k' Subscript[h, 231] Subscript[h, 233]
    Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]])]], 14],

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Style["the denominator identity:", 13], Style[TraditionalForm[
  HoldForm[Subscript[z, 2] Subscript[z, 3] +
    u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 ==
    Subscript[z, 2] Subscript[z, 3] Subsuperscript[H, 3, 2]
    (Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
      Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
  TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
  TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
  TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for all 4 sign combinations of ",
    TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, 3]]], " → 0 == 0"}],
    12, GrayLevel[0.3]], smallChip[idRegion[[iw]]],
  {iw, Length[regA]}], Alignment → Left, Spacings → {2, 0.7}],
"", Style["and, finally, the prefactor identity:", 13],
Style[TraditionalForm[
  HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
    HoldForm[Subscript[z, 2] Subscript[z, 3] / (Subscript[ξ, 23]
      Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]])] == Sqrt[Subscript[
        z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]], 14],
Style[Row[{"which holds by ", TraditionalForm[HoldForm[
  Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] ==
  Abs[Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
  HoldForm[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] ==
  Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"}], 13],
Style[Row[{"and is verified separately on each of the 8
  sign regions (sign combinations of ",
  TraditionalForm[HoldForm[Subscript[z, 2]]], ", ",
  TraditionalForm[HoldForm[Subscript[z, 3]]],
  ", ", TraditionalForm[HoldForm[Subscript[x, 1]]],
  " as above) → 0 == 0 ", smallChip[preAll]]], 13], "",
Style[Row[{"Together, the numerator, denominator, and prefactor
  identities, along with ", TraditionalForm[
  HoldForm[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]],
  " and the cancellation of the factor ",
  TraditionalForm[HoldForm[Subsuperscript[H, 3, 2]]],
  " ( ", TraditionalForm[HoldForm[Subscript[H, 3] ≠ 0]],
  " , as noted above), give the formula ",
  Style["*", 13, colTest], " in case (a):"}], 13],
Style[Row[{TraditionalForm[HoldForm[U == HoldForm[Subscript[ε,
  2] Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
  Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
  ((Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
  Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
  Subscript[x, 1] D[Subscript[x, 1], t]]) / (Subscript[z, 2]

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Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2]
t^2))]], "      ", Style["*", 14]], 14, colTest],
"", Style[Row[{"The identity ", Style["*", 13, colTest],
" is also verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]],
", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"}], 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]], Style[
Row[{"for all 4 sign combinations of ",
TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[e, 3]]], " → 0 == 0"}],
12, GrayLevel[0.3]], smallChip[fullRegion[[iw]]],
{iw, Length[regA]}], Alignment → Left, Spacings → {2, 0.7}],
"", Style[Row[{"(ii) ", TraditionalForm[
HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], " and ",
TraditionalForm[HoldForm[Subscript[e, 2] == -Subscript[e, 3]]], " :"}],
Bold, 13], Style[Row[{"Here ", TraditionalForm[
HoldForm[Subscript[ξ, 23] == -1]], ", because ",
TraditionalForm[HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]],
" and ", TraditionalForm[HoldForm[u < 0]], ", so ",
TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] < 0]],
", and the signs of ", TraditionalForm[HoldForm[Subscript[z, 2]]],
" and ", TraditionalForm[HoldForm[Subscript[z, 3]]],
" are opposite; the given expressions remain ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233]]]}], 12, colGiven],
", and the expressions to test are given by ", Style[
Row[{"the case-(b) ", TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]]}], 12, colTest], " ."}],
12, GrayLevel[0.35]], Style[Row[{TraditionalForm[HoldForm[
1 + u Subscript[z, 2] == (1 + Subscript[z, 3]) / Subscript[z, 3]],
", ", TraditionalForm[HoldForm[1 + u Subscript[z, 3] ==
(1 + Subscript[z, 2]) / Subscript[z, 2]]],
"      ", Style["●", 14, RGBColor[0.40, 0.15, 0.60]]}], 13],
Style[Row[{"Hence, substituting the identities ",
Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
" and ", Style["●", 13, RGBColor[0.40, 0.15, 0.60]],
" and the equalities ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] ==
Abs[Subscript[z, 2] Subscript[z, 3]]}], " and ", TraditionalForm[

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HoldForm[Subscript[ξ, 23] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]], " into ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233]]]}], 13, colGiven],
" , we get, with ", Style[Row[{"the case-(b) ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ", TraditionalForm[
HoldForm[Subscript[H, 3]]]}], 13, colTest], " ,"}], 13], Style[
Row[TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] = 1]],
", ", TraditionalForm[HoldForm[Subscript[h, 231] =
-Subscript[ξ, 23] Subscript[e, 2] Sqrt[1 - u] Subscript[H, 1] /
(Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
", ", TraditionalForm[HoldForm[Subscript[h, 232] =
-Subscript[ξ, 23] Subscript[H, 2] /
(Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
" ", Style["●", 14, RGBColor[0.75, 0.10, 0.40]]}], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regMix[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
" = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
smallChip[brRegionB[[iw]]]}, {iw, Length[regMix]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"Thus, substituting the identities ",
Style["●", 13, RGBColor[0.75, 0.10, 0.40]],
" into the numerator and the denominator of ",
Style["the given U-formula", Italic, 13, colGiven], ", we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
HoldForm[Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]] =
-Subscript[ξ, 23] Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]
Subsuperscript[H, 3, 2] (Abs[Subscript[x, 1] Subscript[h, 232] t +
Subscript[e, 0] k' Subscript[h, 231] Subscript[h, 233]
Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]])]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
HoldForm[Subscript[z, 2] Subscript[z, 3] +
u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 ==
Subscript[z, 2] Subscript[z, 3] Subsuperscript[H, 3, 2]
(Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regMix[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],

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" = ±1 → 0 == 0"]], 12, GrayLevel[0.3]],
smallChip[idRegionB[[iw]]], {iw, Length[regMix]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style["and, finally, the prefactor identity:", 13],
Style[TraditionalForm[
HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]]
HoldForm[Subscript[z, 2] Subscript[z, 3] / (Subscript[ξ, 23]
Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]])] = Sqrt[Subscript[
z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]], 14],
Style[Row[{"which holds by ", TraditionalForm[HoldForm[
Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] =
Abs[Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] =
Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"}], 13],
Style[Row[{"and is verified separately on each of the 4
sign regions (sign combinations of ",
TraditionalForm[HoldForm[Subscript[z, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[z, 3]]],
", ", TraditionalForm[HoldForm[Subscript[x, 1]]],
" as above) → 0 == 0 ", smallChip[preAllB]}], 13], "",
Style[Row[{"Together, the numerator, denominator, and prefactor
identities, along with ", TraditionalForm[
HoldForm[Sign[u Subscript[z, 2]] = -Sign[Subscript[z, 2]]],
" and the cancellation of the factor ",
TraditionalForm[HoldForm[Subsuperscript[H, 3, 2]]],
" ( ", TraditionalForm[HoldForm[Subscript[H, 3] ≠ 0]],
" , as noted above), give the formula ",
Style["*", 13, colTest], " in case (b):"}], 13],
Style[Row[TraditionalForm[HoldForm[U = HoldForm[Subscript[ε,
2] Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
((Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
Subscript[x, 1] D[Subscript[x, 1], t]) / (Subscript[z, 2]
Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2]
t^2))]], " ", Style["*", 14]}], 14, colTest],
"", Style[Row[{"The identity ", Style["*", 13, colTest],
" is also verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]],
", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"}], 13],
Grid[Table[{Style[regMix[[iw], 1]], 12, GrayLevel[0.3]], Style[

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Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
  " =  $\pm 1 \rightarrow 0 == 0$ "}], 12, GrayLevel[0.3]],
smallChip[fullRegionB[[iw]]], {iw, Length[regMix]]},
Alignment  $\rightarrow$  Left, Spacings  $\rightarrow$  {2, 0.7}], "",
Style[Row[{"(iii) ", TraditionalForm[
  HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], " and ",
  TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]], " :"}],
Bold, 13], Style[Row[{"As in case (ii), ",
  TraditionalForm[HoldForm[Subscript[ $\xi$ , 23] == -1]],
" ; the given expressions remain ", Style[
  Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    " ", TraditionalForm[HoldForm[Subscript[h, 232]]],
    " ", TraditionalForm[HoldForm[Subscript[h, 233]]]}],
12, colGiven], " , and the expression to test is ",
Style["the case-(c) U-formula", Italic, 12, colTest],
" ."}], 12, GrayLevel[0.35]],
Style[Row[{"Hence, substituting the identities ",
  Style["●", 13, RGBColor[0.55, 0.33, 0.10]], " and ",
  Style["●", 13, RGBColor[0.40, 0.15, 0.60]], " into ", Style[
  Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    " ", TraditionalForm[HoldForm[Subscript[h, 232]]],
    " ", TraditionalForm[HoldForm[Subscript[h, 233]]]}],
13, colGiven], " , we get"}], 13], Style[
Row[{TraditionalForm[HoldForm[Subscript[h, 233] == 0]], " , ",
  TraditionalForm[HoldForm[k Subscript[h, 231] == Subscript[e, 2]]],
" , ", TraditionalForm[HoldForm[Subscript[h, 232] == Sqrt[
  Abs[Subscript[z, 2] Subscript[z, 3]]]], " , ", TraditionalForm[
  HoldForm[Subsuperscript[h, 233, 2] - Subscript[ $\xi$ , 23] k^2 Subscript[
    x, 1] Subsuperscript[h, 231, 2] t^2 == Subscript[x, 1] t^2]],
" ", Style["●", 14, RGBColor[0.00, 0.55, 0.65]]}], 13],
Style[Row[{"Thus, by the identities ", Style["●", 13,
  RGBColor[0.00, 0.55, 0.65]], " , the ",
  TraditionalForm[HoldForm[Subscript[e, 0]]], "-term of ",
Style["the given U-formula", Italic, 13, colGiven],
" vanishes and it reduces to"}], 13], Style[TraditionalForm[
  HoldForm[U == HoldForm[Subscript[ $\varepsilon$ , 2] Subscript[ $\varepsilon$ , 3] Sqrt[Subscript[
    z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
  HoldForm[Sign[Subscript[y, 2] / t]]], 14, colTest],
Style[Row[{"which is ", Style["the tested case-(c) formula",
  Italic, 13, colTest], " ; the given denominator reduces to ",
  TraditionalForm[HoldForm[Subscript[x, 1] t^2]],
" , which vanishes for the same t as the tested denominator ",
  TraditionalForm[HoldForm[t]], " , namely ",
  TraditionalForm[HoldForm[t == 0]], " . The identities ",
Style["●", 13, RGBColor[0.00, 0.55, 0.65]],
" and the equality of the ", Style["given", Italic, 13, colGiven],
" and ", Style["tested", Italic, 13, colTest],

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" formulas are verified separately on each sign region:"]], 13],
Grid[Table[{Style[regMix[[iw], 1]], 12, GrayLevel[0.3]],
Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
" = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
smallChip[cRegion[[iw]]], {iw, Length[regMix]}],
Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.00, 0.50, 0.50],
0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 6:CASES (b) AND (c) FOR u<0,ON 1+u(z2+z3+z2 z3)=0*)
(* =====*)
ClearAll[pbCheck, pbPreCk, pcCheck];
pbCheck[{lab_, w_, sx_}, e2_] :=
Module[{at = mkAP[w], uv, kk2, mS, kpAt, kkAt, suz, W, d2, d3, h, H, brh, brOK,
numCk, denCk, fullCk}, uv = -1 / (at["z2"] + at["z3"] + at["z2"] at["z3"]);
kk2 = -uv / (1 - uv); mS = at["sv2"] at["sv3"] / at["sS"];
kpAt = 1 / mS; kkAt = 1 / (at["sv2"] at["sv3"]); suz = -at["s2"];
W = at["sw2"] at["sw3"];
{d2, d3} = gdict[at, 2][[3 ;; 4]];
h = p7h[at, kkAt, 1 / (1 - uv), e2, 2]; H = HAat[at, uv, e2, -e2];
brh = {-e2 mS H1s / (W H3s), -H2s / (W H3s), 1 / H3s};
brOK = bOK[at, uv] && p7ok[at, kkAt, kk2, 2] && chkR[kk2, (d2 - 1) (d3 - 1)] &&
chkR[mS^2, 1 - uv] && chkR[kpAt, kkAt at["sS"]] && chkR[h[[3]] H[[3]], 1] &&
chkR[h[[1]] W H[[3]], -e2 mS H[[1]] && chkR[h[[2]] W H[[3]], -H[[2]]];
numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
-W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)];
denCk = chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - kk2 sx XX brh[[1]]^2 tt^2)];
fullCk = chkR[uLemNegat[brh, kk2, kpAt, 1, sx, suz],
uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]];
{brOK, numCk && denCk, fullCk}];
pbPreCk[{lab_, w_, sx_}] := Module[{at = mkAP[w], W}, W = at["sw2"] at["sw3"];
chkR[at["z2"] at["z3"] / (W sx), W / sx]];
pcCheck[{lab_, w_, sx_}, e2_] :=
Module[{at = mkAP[w], uv, kk2, mS, kpAt, kkAt, suz, W, d2, d3, H, H1v, H2v},
uv = -1 / (at["z2"] + at["z3"] + at["z2"] at["z3"]);
kk2 = -uv / (1 - uv); mS = at["sv2"] at["sv3"] / at["sS"];
kpAt = 1 / mS; kkAt = 1 / (at["sv2"] at["sv3"]); suz = -at["s2"];
W = at["sw2"] at["sw3"];
{d2, d3} = gdict[at, 2][[3 ;; 4]];
H = HAat[at, uv, e2, e2];
H1v = -W at["sS"];
H2v = -at["sw2"] at["sv2"] at["sw3"] at["sv3"];
{bOK[at, uv] && chkR[kkAt^2, kk2] && chkR[kk2, (d2 - 1) (d3 - 1)] &&
chkR[mS^2, 1 - uv] && chkR[kpAt, kkAt at["sS"]] &&
chkR[H[[3]], 0] && chkR[H[[1]], H1v] && chkR[H[[2]], H2v],
chkR[ee0 e2 kpAt RR, -(kkAt / W) ee0 e2 H1v RR] &&

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chkR[kkAt (1 - sx XX tt^2), (kkAt / (at["z2"] at["z3"]))
  (at["z2"] at["z3"] + uv sx XX H1v^2 tt^2)], chkR[uThmABat[
  {H1v, H2v, 0}, uv, at, sx, e2], uLemNegCat[kkAt, kpAt, sx, suz, e2]]];
Module[{gridB, brRegion, idRegion, fullRegion,
  preAll, gridC, cBrRegion, cIdRegion, cRegion, ok},
  gridB = Table[pbCheck[w, s], {w, regBpos}, {s, {1, -1}}];
  brRegion = Table[And@@gridB[[iw, All, 1]], {iw, Length[regBpos]}];
  idRegion = Table[And@@gridB[[iw, All, 2]], {iw, Length[regBpos]}];
  fullRegion = Table[And@@gridB[[iw, All, 3]], {iw, Length[regBpos]}];
  preAll = And@@(pbPreCk /@ regBpos);
  gridC = Table[pcCheck[w, s], {w, regBpos}, {s, {1, -1}}];
  cBrRegion = Table[And@@gridC[[iw, All, 1]], {iw, Length[regBpos]}];
  cIdRegion = Table[And@@gridC[[iw, All, 2]], {iw, Length[regBpos]}];
  cRegion = Table[And@@gridC[[iw, All, 3]], {iw, Length[regBpos]}];
  ok = (And@@brRegion) && (And@@idRegion) && (And@@fullRegion) &&
    preAll && (And@@cBrRegion) && (And@@cIdRegion) && (And@@cRegion);
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
      RGBColor[0.45, 0.45, 0.10]], Style[TraditionalForm[HoldForm[U]],
      Bold, 15, RGBColor[0.45, 0.45, 0.10]], Style[
      " in cases (b) and (c) for ", Bold, 15, RGBColor[0.45, 0.45, 0.10]],
      Style[TraditionalForm[HoldForm[u < 0]], Bold, 15,
      RGBColor[0.45, 0.45, 0.10]], Spacer[20], statusChip[ok]]},
    Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
      ", because ", TraditionalForm[HoldForm[k^2 ==
        (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
      " and ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
        - (1 / (1 + Subscript[z, i]))]], ", so ", TraditionalForm[
        HoldForm[(1 + Subscript[z, 2]) (1 + Subscript[z, 3]) > 0]],
      " , and, by ", TraditionalForm[HoldForm[Sign[1 + Subscript[z, i]] ==
        Sign[Subscript[z, i]]]], ", the signs of ",
      TraditionalForm[HoldForm[Subscript[z, 2]]], " and ",
      TraditionalForm[HoldForm[Subscript[z, 3]]], " coincide."}],
    12, GrayLevel[0.35]], Style[Row[{"By ", TraditionalForm[
      HoldForm[Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
        Sign[Subscript[x, 1] Subscript[x, 3]] == 1]], " : ",
      TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] ==
        Abs[Subscript[z, 2] Subscript[z, 3]]]], ", ",
      TraditionalForm[HoldForm[Subscript[x, 1] Subscript[x, 3] ==
        Abs[Subscript[x, 1] Subscript[x, 3]]]], ", ",
      TraditionalForm[HoldForm[(1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
        Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]]], " ."}],
    12, GrayLevel[0.35]], Style[Row[{"Moreover, ",
      TraditionalForm[HoldForm[u Subscript[z, 2] Subscript[z, 3] ≠ 1]],
      " , because the equalities ",
      TraditionalForm[HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]],

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" and ", TraditionalForm[HoldForm[1 + u (Subscript[z, 2] +
      Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]) == 0]],
" cannot hold simultaneously; hence the expressions
to test are given by ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[H, 1]]],
    ", ", TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
    TraditionalForm[HoldForm[Subscript[H, 3]]]}], 12, colTest],
" . Also ", TraditionalForm[HoldForm[Subscript[h, 233] != 0]],
" (as stated in the hypotheses), and,
in case (b), the identity ",
TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] == 1]],
" gives ", TraditionalForm[HoldForm[Subscript[H, 3] != 0]],
" , so the divisions by ", TraditionalForm[
HoldForm[Subscript[H, 3]]], " are legitimate there."}],
12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
Style[Row[{"in these cases ", TraditionalForm[HoldForm[
k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)]],
" , that is, ", TraditionalForm[HoldForm[1 + u (Subscript[z, 2] +
      Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]) == 0]],
" , and the following identities hold:"}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[g, i] == Sqrt[
      Subscript[z, i] (1 + Subscript[z, i])] / Abs[1 + Subscript[z, i]]]],
", ", TraditionalForm[HoldForm[1 + u Subscript[z, 2] ==
      Subscript[z, 3] (1 + Subscript[z, 2]) / (Subscript[z, 2] +
      Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3])]], ", ",
TraditionalForm[HoldForm[1 + u Subscript[z, 3] == Subscript[z, 2]
      (1 + Subscript[z, 3]) / (Subscript[z, 2] + Subscript[z, 3] +
      Subscript[z, 2] Subscript[z, 3])]], ", ", TraditionalForm[
HoldForm[k == 1 / Sqrt[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]]],
", ", TraditionalForm[HoldForm[k' == k Sqrt[Subscript[z,
      2] + Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]]]],
" ", Style["■", 14, RGBColor[0.55, 0.33, 0.10]]}], 13],
Style[Row[{"Hence, in case (b) ( ", TraditionalForm[
HoldForm[Subscript[e, 2] == -Subscript[e, 3]]],
" ), substituting the identities ",
Style["■", 13, RGBColor[0.55, 0.33, 0.10]], " and the equality ",
TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] ==
      Abs[Subscript[z, 2] Subscript[z, 3]]]], " into ", Style[
Row[{"the case-(b) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
    TraditionalForm[HoldForm[Subscript[h, 233]]]}], 13, colGiven],
" , we get, with ", Style[Row[{"the case-(a) ",
      TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
      TraditionalForm[HoldForm[Subscript[H, 2]]], ", ", TraditionalForm[
HoldForm[Subscript[H, 3]]]}], 13, colTest], ", "}], 13], Style[
Row[TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] == 1]],
", ", TraditionalForm[HoldForm[

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Subscript[h, 231] == -Subscript[e, 2] Sqrt[1 - u] Subscript[H, 1] /
  (Sqrt[Subscript[z, 2] Subscript[z, 3] Subscript[H, 3]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232] == -Subscript[H, 2] /
  (Sqrt[Subscript[z, 2] Subscript[z, 3] Subscript[H, 3]])],
" ", Style["■", 14, RGBColor[0.00, 0.45, 0.25]]], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
    " = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
  smallChip[brRegion[[iw]]], {iw, Length[regBpos]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"Thus, substituting the identities ",
  Style["■", 13, RGBColor[0.00, 0.45, 0.25]],
  " into the numerator and the denominator of ",
  Style["the given U-formula", Italic, 13, colGiven], ", we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
  HoldForm[Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
    Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]] ==
    -Sqrt[Subscript[z, 2] Subscript[z, 3] Subsuperscript[H, 3, 2]
    (Abs[Subscript[x, 1] Subscript[h, 232] t +
      Subscript[e, 0] k' Subscript[h, 231] Subscript[h, 233]
      Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]])]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
  HoldForm[Subscript[z, 2] Subscript[z, 3] +
    u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 == Subscript[z, 2]
    Subscript[z, 3] Subsuperscript[H, 3, 2] (Subsuperscript[h, 233, 2] -
    k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
  TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
  TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
  TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
    " = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
  smallChip[idRegion[[iw]]], {iw, Length[regBpos]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"and, finally, ", Style["the prefactor identity", Italic,
  13, RGBColor[0.80, 0.20, 0.50]], " :"}], 13], Style[TraditionalForm[
  HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
    HoldForm[Subscript[z, 2] Subscript[z, 3] /
      Sqrt[Subscript[z, 2] Subscript[z, 3]] == Sqrt[Subscript[z, 2]
      Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]], 14],
Style[Row[{"which holds by ", TraditionalForm[
  HoldForm[Subscript[z, 2] Subscript[z, 3] ==
    Abs[Subscript[z, 2] Subscript[z, 3]]], " and ",
  TraditionalForm[HoldForm[Subscript[x, 1] Subscript[x, 3] ==

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Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"]], 13],
Style[Row[{"and is verified separately on each of the 4
sign regions (sign combinations of ",
TraditionalForm[HoldForm[Subscript[z, 2]]], " ",
TraditionalForm[HoldForm[Subscript[z, 3]]],
", ", TraditionalForm[HoldForm[Subscript[x, 1]]],
" as above) → 0 == 0 ", smallChip[preAll]]], 13], "",
Style[Row[{"Together, the numerator, denominator, and prefactor
identities, along with ", TraditionalForm[
HoldForm[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]]],
" and the cancellation of the factor ",
TraditionalForm[HoldForm[Subsuperscript[H, 3, 2]]],
" ( ", TraditionalForm[HoldForm[Subscript[H, 3] ≠ 0]],
", as noted above), give the formula ",
Style["*", 13, colTest], " in case (a):"]], 13],
Style[Row[TraditionalForm[HoldForm[U == HoldForm[Subscript[ε, 2]
Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
((Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
Subscript[x, 1] D[Subscript[x, 1], t]) / (Subscript[z, 2]
Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2]
t^2))]], " ", Style["*", 14]]], 14, colTest],
"", Style[Row[{"The identity ", Style["*", 13, colTest],
" is also verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]],
", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"]], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
" = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
smallChip[fullRegion[[iw]]], {iw, Length[regBpos]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"In case (c) ( ", TraditionalForm[
HoldForm[Subscript[e, 2] == Subscript[e, 3]]],
" ), the expressions to test remain ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]]]],
12, colTest], " , and the given expression is ",
Style["the case-(c) U-formula", Italic, 12, colGiven], " :"]],
12, GrayLevel[0.35]], Style[TraditionalForm[HoldForm[

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U = HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]] /
  Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]] .
  HoldForm[Subscript[e, 0] Subscript[e, 2] k' Sqrt[Subscript[x, 1]
    D[Subscript[x, 1], t]] / (k (1 - Subscript[x, 1] t^2))]],
14, colGiven], Style[Row[{"Substituting the identities ",
  Style["■", 13, RGBColor[0.55, 0.33, 0.10]], " into ", Style[
  Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[H, 1]]],
    "", TraditionalForm[HoldForm[Subscript[H, 2]]],
    "", TraditionalForm[HoldForm[Subscript[H, 3]]]}],
  13, colTest], " , we get"]], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[H, 3] == 0]],
  "", TraditionalForm[HoldForm[Subscript[H, 1] ==
    -Sqrt[Subscript[z, 2] Subscript[z, 3] (Subscript[z, 2] +
      Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3])]]], "",
  TraditionalForm[HoldForm[Subscript[H, 2] == -Sqrt[Subscript[z, 2]
    Subscript[z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3])]]],
  " ", Style["■", 14, RGBColor[0.40, 0.15, 0.60]]}], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
    " = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
  smallChip[cBrRegion[[iw]]], {iw, Length[regBpos]}],
  Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"Thus, substituting the identities ",
  Style["■", 13, RGBColor[0.40, 0.15, 0.60]], " and the equality ",
  TraditionalForm[HoldForm[k' == k Sqrt[Subscript[z, 2] +
    Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]]],
  " into the numerator and the denominator of ", Style[
  "the given case-(c) U-formula", Italic, 13, colGiven], " , we get"}],
13], Style["the numerator identity:", 13],
Style[TraditionalForm[HoldForm[Subscript[e, 0] Subscript[e, 2]
  k' Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]] ==
  -(k / Sqrt[Subscript[z, 2] Subscript[z, 3]])
  (Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
    Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]])]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
  HoldForm[k (1 - Subscript[x, 1] t^2) ==
    (k / (Subscript[z, 2] Subscript[z, 3])) (Subscript[z, 2] Subscript[z,
    3] + u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
  TraditionalForm[HoldForm[Subscript[H, 1]]], "",
  TraditionalForm[HoldForm[Subscript[H, 2]]], "",
  TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
    " = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],

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    smallChip[cIdRegion[[iw]]], {iw, Length[regBpos]]},
    Alignment → Left, Spacings → {2, 0.7}], "",
    Style[Row[{"Together, the numerator, denominator, and ",
      Style["prefactor", Italic, 13, RGBColor[0.80, 0.20, 0.50]],
      " identities, along with ", TraditionalForm[
        HoldForm[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]],
      " , give the formula ", Style["*", 13, colTest],
      " in case (a):"}], 13],
    Style[Row[{TraditionalForm[HoldForm[U == HoldForm[Subscript[ε, 2]
      Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
      Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
      ((Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
      Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
      Subscript[x, 1] D[Subscript[x, 1], t]] / (Subscript[z, 2]
      Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2]
      t^2))]], " ", Style["*", 14]], 14, colTest],
    Style[Row[{"By the denominator identity, the denominators ",
      TraditionalForm[HoldForm[k (1 - Subscript[x, 1] t^2)]], " and ",
      TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] +
      u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2]],
      " vanish for the same t, because ", TraditionalForm[
      HoldForm[k]], " ", TraditionalForm[
      HoldForm[Subscript[z, 2] Subscript[z, 3] ≠ 0]], " ."}], 13],
    Style[Row[{"The identity ", Style["*", 13, colTest],
      " is also verified separately on each sign region, with ",
      TraditionalForm[HoldForm[Subscript[e, 0]]],
      " ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
      " ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
      " symbolic, that is, for all their real values at once,
      which covers all their sign choices:"}], 13],
    Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
      Style[Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]],
        " = ±1 → 0 == 0"}], 12, GrayLevel[0.3]],
      smallChip[cRegion[[iw]]], {iw, Length[regBpos]]},
      Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
    Background → Lighter[RGBColor[0.45, 0.45, 0.10],
      0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 7:SUMMARY*)
(* =====*)
Print[Framed[Style[If[And@@resultsAll, "ALL CLAIMS VERIFIED",
  "SOME CLAIMS FAILED"], Bold, White, 16], Background →
  If[And@@resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]]];

```

Given hypotheses:

$u \neq 0, \quad u < 1, \quad k, k' \in (0, 1)$

For $i = 1, 2, 3, 4$:

$$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0$$

$$\xi_{23} = \text{sgn}(z_2 z_3) = \text{sgn}(x_1 x_3), \quad \text{sgn}(y_2) = \text{sgn}(u x_1 z_2)$$

$$u x_i y_i z_i > 0, \quad z_i (1 + z_i) > 0, \quad 1 + u z_i > 0$$

$$e_0, e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$$

$$\mathcal{D}(x_1, t) = -(1 - x_1 t^2)(1 - u x_1 t^2), \quad R = \sqrt{x_1 \mathcal{D}(x_1, t)}, \quad x_1 \mathcal{D}(x_1, t) \geq 0$$

$$(i) \text{ if } u > 0 : \quad k = \sqrt{u}, \quad k' = \sqrt{1-u}, \quad g_i = \sqrt{\frac{1+z_i}{z_i}}, \quad k^2 - (g_2^2 - 1)(g_3^2 - 1) = \frac{u z_2 z_3 - 1}{z_2 z_3}$$

$$(ii) \text{ if } u < 0 : \quad k' = \frac{1}{\sqrt{1-u}}, \quad k = k' \sqrt{-u},$$

$$g_i = \sqrt{\frac{z_i}{1+z_i}}, \quad k^2 - (g_2^2 - 1)(g_3^2 - 1) = -\frac{1+u(z_2+z_3+z_2 z_3)}{(1-u)(1+z_2)(1+z_3)}$$

$$h_{231}, h_{232}, h_{233}, H_1, H_2, H_3 \in \mathbb{R}$$

$$h_{231} = 0 \text{ if and only if } g_2 = g_3 \text{ and } e_2 = -e_3$$

$$; \quad h_{232} = 0 \text{ if and only if } g_2 g_3 = k' \text{ and } e_2 = -e_3 ;$$

$$h_{233} = 0 \text{ if and only if } k^2 (g_2^2 + g_3^2) = k^2 - (g_2^2 - 1)(g_3^2 - 1)$$

$$\text{and } e_2 = e_3 ; \text{ in particular, } h_{233} \neq 0 \text{ if } k^2 = (g_2^2 - 1)(g_3^2 - 1) .$$

the equalities $u z_2 z_3 = 1$ and

$$1 + u(z_2 + z_3 + z_2 z_3) = 0 \text{ cannot hold simultaneously .}$$

Given formulas and expressions:

$$U = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2)}{\sqrt{|x_1 x_3|}} \left(\left\{ \begin{array}{ll} \frac{|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)}}{1 + \xi_{23} k^2 x_1 h_{231}^2 t^2} & u > 0 \\ \frac{|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)}}{h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2} & u < 0 \end{array} \right\} \right)$$

$$\left. \begin{aligned} h_{231} &= \frac{e_2 g_3 \sqrt{(g_2^2 - 1)(g_3^2 - k'^2)} \xi_{23} + e_3 g_2 \sqrt{(g_3^2 - 1)(g_2^2 - k'^2)} \xi_{23}}{k^2 - (g_2^2 - 1)(g_3^2 - 1)} \\ h_{232} &= \frac{\sqrt{(g_2^2 - k'^2)(g_3^2 - k'^2)} \xi_{23} + e_2 e_3 g_2 g_3 \sqrt{(g_2^2 - 1)(g_3^2 - 1)} \xi_{23}}{k^2 - (g_2^2 - 1)(g_3^2 - 1)} \\ h_{233} &= \frac{k^2 g_2 g_3 + e_2 e_3 \xi_{23} \sqrt{(g_2^2 - 1)(g_2^2 - k'^2)(g_3^2 - 1)(g_3^2 - k'^2)}}{k^2 - (g_2^2 - 1)(g_3^2 - 1)} \end{aligned} \right\} \quad (a) \quad k^2 \neq (g_2^2 - 1)(g_3^2 - 1)$$

$$h_{231} = \frac{e_2 (g_2^2 - g_3^2)}{2 k g_2 g_3}, \quad h_{232} = \frac{g_2^2 g_3^2 - k'^2}{2 k g_2 g_3}, \quad h_{233} = \frac{g_2^2 + g_3^2}{2 g_2 g_3} \quad (b) \quad k^2 = (g_2^2 - 1)(g_3^2 - 1), \quad e_2 = -e_3$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2)}{\sqrt{|x_1 x_3|}} \left(\left\{ \begin{array}{ll} \frac{\text{sgn}(x_1)}{k t} & u > 0 \\ \frac{e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)}}{k (1 - x_1 t^2)} & u < 0 \end{array} \right\} \right) \quad (c) \quad k^2 = (g_2^2 - 1)(g_3^2 - 1), \quad e_2 = e_3$$

We want to test these formulas ★ :

We show that the formulas for U

below follow from the given hypotheses, formulas, and expressions above,

together with the following claim: in each case, the denominator of the given formula

and the denominator of the tested formula vanish for the same values of

t , that is, one of them is zero if and only if the other is; hence the

compared formulas are defined for exactly the same t .

$$U = \frac{\left(\varepsilon_2 \varepsilon_3 \operatorname{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}} \right) \left(|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} \right)}{z_2 z_3 + u x_1 H_1^2 t^2}$$

$$\left. \begin{aligned} H_1 &= \frac{|z_2| \sqrt{z_3(1+z_3)(1+u z_3)} + e_2 e_3 |z_3| \sqrt{z_2(1+z_2)(1+u z_2)}}{u z_2 z_3 - 1} \\ H_2 &= \frac{|z_2 z_3| \sqrt{(1+u z_2)(1+u z_3)} + e_2 e_3 \sqrt{z_2 z_3(1+z_2)(1+z_3)}}{u z_2 z_3 - 1} \\ H_3 &= \frac{u \sqrt{z_2 z_3(1+z_2)(1+z_3)} \operatorname{sgn}(z_2 z_3) + e_2 e_3 \sqrt{(1+u z_2)(1+u z_3)}}{u z_2 z_3 - 1} \end{aligned} \right\} \quad (\text{a}) \quad u z_2 z_3 \neq 1$$

$$\left. \begin{aligned} H_1 &= \frac{z_2 z_3 (z_3 - z_2)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \\ H_2 &= \frac{z_2 z_3 (z_2 + z_3 + 2)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \\ H_3 &= \frac{(z_2 + z_3 + 2 z_2 z_3) \operatorname{sgn}(z_2 z_3)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \end{aligned} \right\} \quad (\text{b}) \quad u z_2 z_3 = 1, \quad e_2 = -e_3$$

$$U = \left(\varepsilon_2 \varepsilon_3 \sqrt{\frac{z_2 z_3}{x_1 x_3}} \right) \frac{\operatorname{sgn}(y_2)}{t} \quad (\text{c}) \quad u z_2 z_3 = 1, \quad e_2 = e_3$$

The derivation: the formula for U in case (a) for $u > 0$

Verified

By $\xi_{23} = \operatorname{sgn}(z_2 z_3) = \operatorname{sgn}(x_1 x_3)$: $\xi_{23} z_2 z_3 = |z_2 z_3|$, $\xi_{23} x_1 x_3 = |x_1 x_3|$.

Verification:

in this case $k^2 \neq (g_2^2 - 1)(g_3^2 - 1)$, that is, $u z_2 z_3 \neq 1$, and the following identities hold:

$$k = \sqrt{u}, \quad g_i^2 - 1 = \frac{1}{z_i}, \quad g_i^2 - k^2 = \frac{1+u z_i}{z_i}, \quad g_i = \frac{\sqrt{z_i(1+z_i)}}{|z_i|} \quad \blacklozenge$$

Hence, substituting the identities $\blacklozenge$ and the equality $\xi_{23} z_2 z_3 = |z_2 z_3|$

into the case-(a) $h_{231}, h_{232}, h_{233}$, we get, with the case-(a) H_1, H_2, H_3 ,

$$h_{231} = \frac{\xi_{23} e_2 H_1}{\sqrt{|z_2 z_3|}}, \quad h_{232} = \frac{\xi_{23} H_2}{\sqrt{|z_2 z_3|}}, \quad h_{233} = H_3 \quad \blacklozenge$$

which are verified separately on each sign region:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 = 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 = 0$	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 = 0$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 = 0$	✓

$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓

Thus, substituting the identities ♦

into the numerator and the denominator of *the given U-formula*, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = \xi_{23} \sqrt{|z_2 z_3|} (|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 (1 + \xi_{23} k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\xi_{23} \sqrt{|z_2 z_3|}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23} x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 8 sign regions (sign combinations of z_2, z_3, x_1 as above) $\rightarrow 0 == 0$ ✓

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u z_2) = \text{sgn}(z_2)$, give the formula ★ in case (a):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓

The derivation: the formula for U in cases (b) and (c) for $u > 0$

Verified

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and

$g_i^2 - 1 = \frac{1}{z_i}$, so $z_2 z_3 > 0$, and the signs of z_2 and z_3 coincide.

By $\xi_{23} = \text{sgn}(z_2 z_3) = \text{sgn}(x_1 x_3) = 1$: $z_2 z_3 = |z_2 z_3|$, $x_1 x_3 = |x_1 x_3|$.

Verification:

in these cases $k^2 = (g_2^2 - 1)(g_3^2 - 1)$, that is, $u z_2 z_3 = 1$, and the following identities hold:

$$k = \frac{1}{\sqrt{z_2 z_3}}, \quad g_i = \frac{\sqrt{z_i(1+z_i)}}{|z_i|} \quad \blacktriangle$$

Hence, in case (b) ($e_2 = -e_3$), substituting the identities $\blacktriangle$ and the equality

$z_2 z_3 = |z_2 z_3|$ into the case-(b) $h_{231}, h_{232}, h_{233}$, we get, with the case-(b) H_1, H_2, H_3 ,

$$h_{231} = \frac{e_2 H_1}{\sqrt{z_2 z_3}}, \quad h_{232} = \frac{H_2}{\sqrt{z_2 z_3}}, \quad h_{233} = H_3 \quad \blacktriangle$$

which are verified separately on each sign region:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓

Thus, substituting the identities $\blacktriangle$

into the numerator and the denominator of the given U -formula, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = \sqrt{z_2 z_3} (|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 (1 + k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓

and, finally, the prefactor identity :

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\sqrt{z_2 z_3}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $z_2 z_3 = |z_2 z_3|$ and $x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 4 sign regions (sign combinations of z_2, z_3, x_1 as above) $\rightarrow 0 == 0$ ✓

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u z_2) = \text{sgn}(z_2)$, give the formula $\star$ in case (b):

$$U = \frac{(e_2 e_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}} (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)}))}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity $\star$ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, e_2, e_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$$\begin{array}{llll} z_2 > 0, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \end{array}$$

In case (c) ($e_2 = e_3$), the *given* and *tested* formulas are their replacements:

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2)}{\sqrt{|x_1 x_3|}} \frac{\operatorname{sgn}(x_1)}{k t} \quad \text{and} \quad U = \left(\varepsilon_2 \varepsilon_3 \sqrt{\frac{z_2 z_3}{x_1 x_3}} \right) \frac{\operatorname{sgn}(y_2)}{t},$$

which coincide because $\operatorname{sgn}(y_2) = \operatorname{sgn}(u x_1 z_2)$, $k = \frac{1}{\sqrt{z_2 z_3}}$, and *the prefactor identity*; their denominators $k t$ and t vanish for the same t because $k \neq 0$.

The equality of the *given* and *tested*

formulas is verified separately on each sign region, with $\varepsilon_2, \varepsilon_3$

symbolic, that is, for all their real values at once, which covers all their sign choices:

$$\begin{array}{llll} z_2 > 0, z_3 > 0, x_1 > 0 & \rightarrow 0 == 0 & \checkmark \\ z_2 > 0, z_3 > 0, x_1 < 0 & \rightarrow 0 == 0 & \checkmark \\ z_2 < -1, z_3 < -1, x_1 > 0 & \rightarrow 0 == 0 & \checkmark \\ z_2 < -1, z_3 < -1, x_1 < 0 & \rightarrow 0 == 0 & \checkmark \end{array}$$

The derivation: the formula for U in case (a) for $u < 0$

Verified

Here $h_{233} = 0$ if and only if $k^2 (g_2^2 + g_3^2) = k^2 - (g_2^2 - 1)(g_3^2 - 1)$ and $e_2 = e_3$. When $u < 0$,

$$k^2 (g_2^2 + g_3^2) - (k^2 - (g_2^2 - 1)(g_3^2 - 1)) = \frac{1 - u z_2 z_3}{(1 - u)(1 + z_2)(1 + z_3)}$$

which is verified separately on each of the 4 sign regions (sign combinations of z_2, z_3) $\rightarrow 0 == 0$ $\checkmark$, so $k^2 (g_2^2 + g_3^2) = k^2 - (g_2^2 - 1)(g_3^2 - 1)$ if and only if $u z_2 z_3 = 1$.

Thus, when $u < 0$, $h_{233} = 0$ if and only if $u z_2 z_3 = 1$ and

$e_2 = e_3$. Therefore, in cases (i) and (ii) below, $h_{233} \neq 0$, and the identity $h_{233} H_3 = 1$ gives $H_3 \neq 0$, so the divisions by H_3 are legitimate there.

Verification:

in this case $k^2 \neq (g_2^2 - 1)(g_3^2 - 1)$, that is, $1 + u(z_2 + z_3 + z_2 z_3) \neq 0$;

we consider three possible cases: (i) $u z_2 z_3 \neq 1$

, (ii) $u z_2 z_3 = 1$ and $e_2 = -e_3$, (iii) $u z_2 z_3 = 1$ and $e_2 = e_3$.

(i) $u z_2 z_3 \neq 1$:

The following identities hold:

$$k = \sqrt{\frac{u}{u-1}}, \quad k' = \frac{1}{\sqrt{1-u}}, \quad g_i^2 - 1 = -\frac{1}{1+z_i}, \quad g_i^2 - k'^2 = -\frac{1+u z_i}{(1-u)(1+z_i)}, \quad g_i = \frac{\sqrt{z_i(1+z_i)}}{|1+z_i|} \quad \bullet$$

$$u^2 z_2 z_3 (1+z_2)(1+z_3) - (1+u z_2)(1+u z_3) = (u z_2 z_3 - 1)(1+u(z_2 + z_3 + z_2 z_3)) \quad \bullet$$

and, by $\xi_{23} = \operatorname{sgn}(z_2 z_3) = \operatorname{sgn}(x_1 x_3)$ and $\operatorname{sgn}(1+z_i) = \operatorname{sgn}(z_i)$:

$$\xi_{23} z_2 z_3 = |z_2 z_3|, \quad \xi_{23} x_1 x_3 = |x_1 x_3|, \quad \xi_{23} (1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|.$$

Hence, substituting the identities $\bullet$ and the equalities $\xi_{23} z_2 z_3 = |z_2 z_3|$

and $\xi_{23}(1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|$ into the case-(a) $h_{231}, h_{232}, h_{233}$, and using the identity $\bullet$, we get, with the case-(a) H_1, H_2, H_3 ,

$$h_{233} H_3 = 1, \quad h_{231} = -\frac{\xi_{23} e_2 \sqrt{1-u} H_1}{\sqrt{|z_2 z_3|} H_3}, \quad h_{232} = -\frac{\xi_{23} H_2}{\sqrt{|z_2 z_3|} H_3} \quad \bullet$$

which are verified separately on each sign region:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓

Thus, substituting the identities $\bullet$

into the numerator and the denominator of the given U -formula, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = -\xi_{23} \sqrt{|z_2 z_3|} H_3^2 (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 H_3^2 (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\xi_{23} \sqrt{|z_2 z_3|}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23} x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 8 sign regions (sign combinations of z_2, z_3, x_1 as above) $\rightarrow 0 == 0$ ✓

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u z_2) = -\text{sgn}(z_2)$ and the cancellation of the factor H_3^2 ($H_3 \neq 0$, as noted above), give the formula $\star$ in case (a):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity $\star$ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$

symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of $e_2, e_3 \rightarrow 0 == 0$	✓

(ii) $u z_2 z_3 = 1$ and $e_2 = -e_3$:

Here $\xi_{23} = -1$, because $u z_2 z_3 = 1$ and $u < 0$, so $z_2 z_3 < 0$, and the signs of z_2 and z_3 are opposite; the given expressions remain **the case-(a) $h_{231}, h_{232}, h_{233}$** , and the expressions to test are given by **the case-(b) H_1, H_2, H_3** .

$$1 + u z_2 = \frac{1+z_3}{z_3}, \quad 1 + u z_3 = \frac{1+z_2}{z_2} \quad \bullet$$

Hence, substituting the identities $\bullet$ and $\bullet$ and the equalities

$\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23} (1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|$ into **the case-(a) $h_{231}, h_{232}, h_{233}$** , we get, with **the case-(b) H_1, H_2, H_3** ,

$$h_{233} H_3 = 1, \quad h_{231} = -\frac{\xi_{23} e_2 \sqrt{1-u} H_1}{\sqrt{|z_2 z_3|} H_3}, \quad h_{232} = -\frac{\xi_{23} H_2}{\sqrt{|z_2 z_3|} H_3} \quad \bullet$$

which are verified separately on each sign region:

$z_2 > 0, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓

Thus, substituting the identities $\bullet$

into the numerator and the denominator of **the given U-formula**, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = -\xi_{23} \sqrt{|z_2 z_3|} H_3^2 (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 H_3^2 (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\xi_{23} \sqrt{|z_2 z_3|}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23} x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 4 sign regions (sign combinations of z_2, z_3, x_1 as above) $\rightarrow 0 == 0$ ✓

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u z_2) = -\text{sgn}(z_2)$ and the cancellation of the factor H_3^2 ($H_3 \neq 0$, as noted above), give the formula ★ in case (b):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$$\begin{array}{llll} z_2 > 0, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 > 0, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 < -1, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 < -1, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \end{array}$$

(iii) $u z_2 z_3 = 1$ and $e_2 = e_3$:

As in case (ii), $\xi_{23} = -1$; the given expressions remain

the case-(a) $h_{231}, h_{232}, h_{233}$, and the expression to test is the case-(c) U -formula .

Hence, substituting the identities ● and ● into the case-(a) $h_{231}, h_{232}, h_{233}$, we get

$$h_{233} = 0, \quad k h_{231} = e_2, \quad h_{232} = \sqrt{|z_2 z_3|}, \quad h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2 = x_1 t^2 \quad \bullet$$

Thus, by the identities ● , the e_0 -term of the given U -formula vanishes and it reduces to

$$U = (\varepsilon_2 \varepsilon_3 \sqrt{\frac{z_2 z_3}{x_1 x_3}}) \frac{\text{sgn}(y_2)}{t}$$

which is the tested case-(c) formula ; the given denominator reduces to

$x_1 t^2$, which vanishes for the same t as the tested denominator

t , namely $t = 0$. The identities ● and the equality of the given

and tested formulas are verified separately on each sign region:

$$\begin{array}{llll} z_2 > 0, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 > 0, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 < -1, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \\ z_2 < -1, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 & \checkmark \end{array}$$

The derivation: the formula for U

in cases (b) and (c) for $u < 0$

Verified

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and $g_i^2 - 1 = -\frac{1}{1+z_i}$, so

$(1+z_2)(1+z_3) > 0$, and, by $\text{sgn}(1+z_i) = \text{sgn}(z_i)$, the signs of z_2 and z_3 coincide.

By $\xi_{23} = \text{sgn}(z_2 z_3) = \text{sgn}(x_1 x_3) = 1$: $z_2 z_3 = |z_2 z_3|$
 , $x_1 x_3 = |x_1 x_3|$, $(1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|$.

Moreover, $u z_2 z_3 \neq 1$, because the equalities $u z_2 z_3 = 1$ and $1 + u(z_2 + z_3 + z_2 z_3) = 0$ cannot hold simultaneously; hence the expressions to test are given by

the case-(a) H_1, H_2, H_3 . Also $h_{233} \neq 0$

(as stated in the hypotheses), and, in case (b), the identity $h_{233} H_3 = 1$ gives $H_3 \neq 0$, so the divisions by H_3 are legitimate there.

Verification:

in these cases $k^2 = (g_2^2 - 1)(g_3^2 - 1)$, that is,

$1 + u(z_2 + z_3 + z_2 z_3) = 0$, and the following identities hold:

$$g_i = \frac{\sqrt{z_i(1+z_i)}}{|1+z_i|}, \quad 1 + u z_2 = \frac{z_3(1+z_2)}{z_2+z_3+z_2 z_3}, \quad 1 + u z_3 = \frac{z_2(1+z_3)}{z_2+z_3+z_2 z_3}$$

$$, \quad k = \frac{1}{\sqrt{(1+z_2)(1+z_3)}}, \quad k' = k \sqrt{z_2 + z_3 + z_2 z_3} \quad \blacksquare$$

Hence, in case (b) ($e_2 = -e_3$), substituting the identities $\blacksquare$ and the equality

$z_2 z_3 = |z_2 z_3|$ into the case-(b) $h_{231}, h_{232}, h_{233}$, we get, with the case-(a) H_1, H_2, H_3 ,

$$h_{233} H_3 = 1, \quad h_{231} = -\frac{e_2 \sqrt{1-u} H_1}{\sqrt{z_2 z_3} H_3}, \quad h_{232} = -\frac{H_2}{\sqrt{z_2 z_3} H_3} \quad \blacksquare$$

which are verified separately on each sign region:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓

Thus, substituting the identities $\blacksquare$

into the numerator and the denominator of the given U-formula, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} =$$

$$- \sqrt{z_2 z_3} H_3^2 (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 H_3^2 (h_{233}^2 - k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓

and, finally, the prefactor identity :

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\sqrt{z_2 z_3}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $z_2 z_3 = |z_2 z_3|$ and $x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 4 sign regions (sign combinations of z_2, z_3, x_1 as above) $\rightarrow 0 == 0$ ✓

Together, the numerator, denominator, and prefactor identities, along with

$\text{sgn}(u z_2) = -\text{sgn}(z_2)$ and the cancellation of the factor H_3^2

($H_3 \neq 0$, as noted above), give the formula $\star$ in case (a):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity $\star$ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1 \rightarrow 0 == 0$	✓
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$$\begin{array}{ll} z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \end{array}$$

In case (c) ($e_2 = e_3$), the expressions to test remain

the case-(a) H_1, H_2, H_3 , and the given expression is the case-(c) U-formula :

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2)}{\sqrt{|x_1 x_3|}} \cdot \frac{e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)}}{k(1 - x_1 t^2)}$$

Substituting the identities ■ into the case-(a) H_1, H_2, H_3 , we get

$$H_3 = 0, H_1 = -\sqrt{z_2 z_3 (z_2 + z_3 + z_2 z_3)}, H_2 = -\sqrt{z_2 z_3 (1 + z_2)(1 + z_3)} \quad \blacksquare$$

which are verified separately on each sign region:

$$\begin{array}{ll} z_2 > 0, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \end{array}$$

Thus, substituting the identities ■ and the equality $k' = k \sqrt{z_2 + z_3 + z_2 z_3}$

into the numerator and the denominator of the given case-(c) U-formula, we get the numerator identity:

$$e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)} = -\frac{k(|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{z_2 z_3}}$$

the denominator identity:

$$k(1 - x_1 t^2) = \frac{k(z_2 z_3 + u x_1 H_1^2 t^2)}{z_2 z_3}$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$$\begin{array}{ll} z_2 > 0, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \end{array}$$

Together, the numerator, denominator, and prefactor

identities, along with $\operatorname{sgn}(u z_2) = -\operatorname{sgn}(z_2)$, give the formula ★ in case (a):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \operatorname{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

By the denominator identity, the denominators $k(1 - x_1 t^2)$

and $z_2 z_3 + u x_1 H_1^2 t^2$ vanish for the same t , because $k, z_2 z_3 \neq 0$.

The identity ★ is also verified separately on each sign region, with $e_0, \varepsilon_2, \varepsilon_3$

symbolic, that is, for all their real values at once, which covers all their sign choices:

$$\begin{array}{ll} z_2 > 0, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \\ z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 \rightarrow 0 == 0 \quad \checkmark \end{array}$$

ALL CLAIMS VERIFIED